

UTF Commands

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Contents

1	Module Index	1
1.1	Commands	1
2	Module Documentation	5
2.1	blackbox	5
2.1.1	Detailed Description	5
2.2	converged_flow	7
2.2.1	Detailed Description	7
2.3	set_app_var	8
2.4	set_sram_trace	9
2.4.1	Detailed Description	9
2.5	clocks	10
2.5.1	Detailed Description	10
2.6	keep_rtl_read	11
2.6.1	Detailed Description	11
2.7	keep_rtl_write	12
2.7.1	Detailed Description	12
2.8	optimization	13
2.8.1	Detailed Description	13
2.9	power_profile	16
2.9.1	Detailed Description	16
2.10	set_global_false_path	17
2.10.1	Detailed Description	17
2.11	set_power_estimation	18
2.11.1	Detailed Description	18
2.12	synplifypro_synthesis	19
2.12.1	Detailed Description	19
2.13	synthesis	20
2.13.1	Detailed Description	20
2.14	system_tasks	23
2.14.1	Detailed Description	23

2.15	zcei	24
2.15.1	Detailed Description	24
2.16	scemi	25
2.16.1	Detailed Description	25
2.17	zgate	26
2.17.1	Detailed Description	26
2.18	zinject	28
2.18.1	Detailed Description	28
2.19	profile	31
2.19.1	Detailed Description	31
2.20	memory_preferences	32
2.20.1	Detailed Description	32
2.21	assertion_synthesis	33
2.21.1	Detailed Description	33
2.22	ztopbuild	34
2.22.1	Detailed Description	34
2.23	design	35
2.24	dpi_synthesis	36
2.24.1	Detailed Description	36
2.25	defmapping	37
2.25.1	Detailed Description	37
2.26	create_new_target	38
2.26.1	Detailed Description	38
2.27	pbuild	39
2.27.1	Detailed Description	39
2.28	UCVLibExt	40
2.29	UCVcsVerdiFlowNoFSDB	41
2.30	architecture_file	42
2.31	clock_boundary_marker	43
2.31.1	Detailed Description	43
2.32	clock_config	45
2.33	clock_delay	47
2.34	clock_frequency	49
2.35	cluster_constraint	50
2.36	clustering	53
2.37	code_coverage	56
2.38	compile	57
2.39	coverage	59
2.40	debug	60
2.41	design_size	64

2.42	disk_space_checker	66
2.43	eco	67
2.44	encrypted_mod	69
2.45	enhanced_write_access	70
2.46	fault_emulation	71
2.47	force	72
2.48	fpga	75
2.49	fusa	77
2.50	global_db	78
2.51	grid_cmd	79
2.52	grid_engine	80
2.53	grid_task_association	81
2.54	load_edif	82
2.55	memories	83
2.56	modular_compile	86
2.57	move_initial_to_sw	87
2.58	nfs_settings	88
2.59	performance	89
2.60	post_compile_command	90
2.61	pre_compile_command	91
2.62	probe_config	92
2.63	probe_signals	93
2.64	read_classification	94
2.65	read_sdc	95
2.66	reg_init	96
2.67	run_manager	97
2.68	scheduling	98
2.69	set_bus_interface	99
	2.69.1 Detailed Description	99
2.70	set_dualedge	100
2.71	set_false_path_from	101
2.72	set_false_path_to	102
2.73	set_hwtop	103
2.74	set_perf_flow	104
2.75	set_power_profile	105
2.76	set_sw_control_signal	106
2.77	set_trigger	107
2.78	set_zcei_partition	108
2.79	signature	109
2.80	simxl	110

2.81 simxl_allow_concurrent	111
2.82 simxl_enable_zdpi	112
2.83 simxl_move_tf_to_tb	113
2.84 simxl_move_to_tb	114
2.85 simxl_set_hwtop	115
2.86 simxl_xlbram	116
2.87 smart_zice_config	117
2.88 synthesis_preferences	118
2.89 timing_analysis	119
2.90 timing_constraint	121
2.91 transparent_latch	122
2.92 tristate	123
2.93 vcs_exec_command	124
2.94 verilog_force_release	125
2.95 wire_resolution	126
2.96 xtors	128
2.97 zemi3	129
2.98 zemi4	132
2.99 zforce	135
2.100zpar	137

Index

138

Chapter 1

Module Index

1.1 Commands

Here is a list of all modules:

UCVLibExt	40
UCVcsVerdiFlowNoFSDB	41
architecture_file	42
assertion_synthesis	33
blackbox	5
clock_boundary_marker	43
clock_config	45
clock_delay	47
clock_frequency	49
clocks	10
cluster_constraint	50
clustering	53
code_coverage	56
compile	57
converged_flow	7
coverage	59
create_new_target	38
debug	60
defmapping	37
design	35
design_size	64
disk_space_checker	66
dpi_synthesis	36
eco	67
encrypted_mod	69
enhanced_write_access	70
fault_emulation	71
force	72
fpga	75
fusa	77
global_db	78
grid_cmd	79
grid_engine	80
grid_task_association	81
keep_rtl_read	11
keep_rtl_write	12
load_edif	82
memories	83

memory_preferences	32
modular_compile	86
move_initial_to_sw	87
nfs_settings	88
optimization	13
pbuild	39
performance	89
post_compile_command	90
power_profile	16
pre_compile_command	91
probe_config	92
probe_signals	93
profile	31
read_classification	94
read_sdc	95
reg_init	96
run_manager	97
scemi	25
scheduling	98
set_app_var	8
set_bus_interface	99
set_dualedge	100
set_false_path_from	101
set_false_path_to	102
set_global_false_path	17
set_hwtop	103
set_perf_flow	104
set_power_estimation	18
set_power_profile	105
set_sram_trace	9
set_sw_control_signal	106
set_trigger	107
set_zcei_partition	108
signature	109
simxl	110
simxl_allow_concurrent	111
simxl_enable_zdpi	112
simxl_move_tf_to_tb	113
simxl_move_to_tb	114
simxl_set_hwtop	115
simxl_xlbram	116
smart_zice_config	117
synplifypro_synthesis	19
synthesis	20
synthesis_preferences	118
system_tasks	23
timing_analysis	119
timing_constraint	121
transparent_latch	122
tristate	123
vcs_exec_command	124
verilog_force_release	125
wire_resolution	126
xtors	128
zcei	24
zemi3	129
zemi4	132
zforce	135

zgate	26
zinject	28
zpar	137
ztopbuild	34

Chapter 2

Module Documentation

2.1 blackbox

Functions

- **blackbox** args

2.1.1 Detailed Description

```
blackbox <ALGO> <MODULE> [-fnmatch]
        [-object_not_found [fatal|warning]] [-port <list>]
        [-reg_init [0|1]] [-value [0|1|NONE|REG|Z]]
```

COMMAND DESCRIPTION:

Specifies the behavior of blackboxes in the design.

MANDATORY ARGUMENTS:

- **<ALGO>** Specifies the processing to apply on a blackbox module.
Details about allowed values:
define: Specifies a module as a known blackbox.
drive: Specifies a value that will drive the outputs/inouts of the blackbox. You may assign different values to different ports with the '-port' option.
remove: [Will be DEPRECATED] Deletes all instantiations of the blackbox module given in an argument. This option is deprecated. Please use 'blackbox drive bbox -value none' instead 'blackbox remove' command.
fatal: Reports fatal error if unexpected blackboxes are found. On the next release -algo <ALGO> may replace <ALGO>
- **<MODULE>** Blackbox module name. To search a blackbox module into a specific EDIF library, use the following syntax: <edif_library_name>@<blackbox_module_name>. Both edif_library_name and blackbox_module_name can contain wildcards. Type: <MODULE> is a string.
On the next release -module <MODULE> may replace <MODULE>
unexpected: Use 'unexpected' as module name with blackbox fatal command to report fatal error if unexpected blackboxes are found

OPTIONAL ARGUMENTS:

- **-fnmatch** Indicates that object pathnames can contains wildcards. This option applies to both module name and port names.

- `-object_not_found [fatal|warning]` Specifies the error policy when objects cannot be found.
Available policies:
fatal: Default value. A fatal error is displayed.
warning: A warning is displayed instead of an error.
Default value: fatal.
- `-port <list>` Specifies a list of ports or port vectors upon which to apply the command. If not specified, the command is applied to all output and inouts ports. The `fnmatch` option can be also used to match port names.
- `-reg_init [0|1]` Specifies the register initial value [0|1].
Default value is 0 (only needed when `-value REG`).
Ex: `drive -value REG -reg_init 1`.
Available values are '0' or '1'.
- `-value <VALUE>` Specifies the driver value [0|1|NONE|REG|Z]. This option is meaningful only for drive, disconnect or remove commands. Ex: `drive -value 1`.
Available values:
0: A constant 0 will be inserted as a driver.
1: A constant 1 will be inserted as a driver.
NONE: No driver is inserted. This is mostly useful for the 'blackbox drive' command.
REG: A register writable at runtime will be inserted as a driver.
Z: A disabled tristate signal will be inserted as a driver. This signal can be controlled at runtime (enable and value are writable).

COMMAND USAGE EXAMPLES:

- Example 1:
- `blackbox drive bbox -value REG -reg_init 1`
Drives all the outputs and inouts of the blackbox module with registers initialized to 1. Their names can be found in the `zSelectProbes` hierarchy of the blackbox.
- Example 2:
- `blackbox drive work@BB30 -port {P*[1[0-9]\]} -value REG -reg_init 1 -fnmatch`
Drives ports matching wildcards in blackbox module BB30 from the EDIF library 'work' with a register value. Register initial value will be 1.
- Example 3:
- `blackbox fatal unexpected`
Report fatal error if unexpected blackboxes are encountered in the design

2.2 converged_flow

Functions

- **converged_flow** args

2.2.1 Detailed Description

is no longer supported - please specify a Taurus architecture file to enable the flow.

2.3 set_app_var

- **name** <string> #name of variable in optionsDB
- **value** <string> #value of a variable in optionsDB
- **-lock** hard|soft<enum> #lock type over name

Description

This command is used to add an option to OptionsDB.

Synopsis

```
set_app_var <name> <value> [-lock <hard|soft>]
```

Option(s)

- **name** <string> Name of option to be added to OptionsDB.
- **value** <string> Value of option to be added to OptionsDB.
- **-lock** <hard|soft> Whether recurrence of the option in same utf file should cause error/warning.
 - **hard**: Abort if same option has been defined previously with this flag
 - **soft**: Warn and ignore if same option has been defined previously with this flag

Constraint(s)

- At least one of the following options should be used: name
- At least one of the following options should be used: value

2.4 set_sram_trace

Functions

- **set_sram_trace** args

2.4.1 Detailed Description

set_sram_trace options

- `-clock driverClk|compositeClk|<clock rtlname>` Specify the clock type.

Note

The command should be used only in Taurus flow, otherwise ignored.

2.5 clocks

Functions

- **clocks** args

2.5.1 Detailed Description

Clock handling options

- `-skew_offset NONE | ALL` enable global skew offset
- `-skew_offset_synchronous <bool>` sync/async skew offset
- `-skew_offset_delay_in_ns <int>` async skew offset delay, but not currently in ns. zCui uses 24ns steps.
- `-localize_clock_tree NONE | FPGA | ZCORE` Localize clock tree
- `-glitches NOFILTER | FILTER | DELAY_ENABLES [-detect_clock_delay]` Filter glitches

Note

`-detect_clock_delay` is only valid with `with -glitches DELAY_ENABLES`

- `-direct_routing <bool>` Direct routing
- `-add_internal_clk #clock_signal` Specify internal clock path
- `-add_internal_data #data_signal` Specify internal data path
- `-clear_internal_clk_data <bool>` Clears all paths previously declared using "-add_internal_clk" and "-add_internal_data" options
- `-advanced_command_file #filename` Specify advanced clock handling commands

2.6 keep_rtl_read

Functions

- **keep_rtl_read** args

2.6.1 Detailed Description

Enable read access.

2.7 keep_rtl_write

Functions

- **keep_rtl_write** args

2.7.1 Detailed Description

Enable write access.

2.8 optimization

Functions

- **optimization** args

2.8.1 Detailed Description

- `-aggressive|-maxopt|-max_opt [-module <module list>][-instance <instance list>]` Aggressive optimization if `-module` or `-instance` are present the optimization directive is per the instance/module list (regex patterns can be used e.g. all modules which begin with 'a' `-module {a.*}` if `-module` or `-instance` are not present this is a global directive (legacy zFast:zview)
- `-normal|-keep_registers [-module <module list>][-instance <instance list>]` Medium optimization (legacy zFast:zview) if `-module` or `-instance` are present the optimization directive is per the instance/module list if `-module` or `-instance` are not present this is a global directive (legacy zFast:zview) (regex patterns can be used e.g. all modules which begin with 'a' `-module {a.*}`)
- `-no|-noopt|-no_opt` No optimization (legacy zFast:zview) if `-module` or `-instance` are present the optimization directive is per the instance/module list if `-module` or `-instance` are not present this is a global directive (legacy zFast:zview) (regex patterns can be used e.g. all modules which begin with 'a' `-module {a.*}`)
- `-pass_opt_levels <bool>` Specify whether WLS (Word Level Synthesis) should use the optimization levels specified by `-aggressive`, `-normal`, `-no_opt`, etc. Default is true.
- `-dsp_mult_threshold <int>` Synthesize DSP for mults when total bits of args are greater than nb (legacy zFast:DSPMulThreshold)
- `-dsp_limit <int>` Limit the maximum number of DSPs in any module (legacy Compile:DSPLimit). Acceptable values are ≥ 0 . When set to 0, this gives the default, which is no limit. Note that this is not exactly a limit for the modules. Rather, this is the maximum number of DSPs in any module, but after all of the inlining has happened into the modules. If you have used "optimization -top_down_roots <module_list>", it is a bit different. Call each module in the <module_list> a root module. Each root module is synthesized together with a sub-hierarchy below it. The DSP limit, in this case, is the limit on the number of DSPs in each root module together with that root module's own sub-hierarchy.
- `-remove_unconnected_blackboxes <bool>` Specify removal of unconnected blackboxes
- `-latch_clocks <bool>` Enable clock analysis on mem instances and latches.
- `-boundary_opt <bool>` Deprecated. Boundary optimization: performs ports merging for ports with same/opposite phase and cross-modules constant propagation. WLS flow option. Currently default false. Cannot be combined with the newer `-qor_opt_enable` or `-qor_opt_disable` options.
- `-number_of_threads <int>` Specify number of threads for WLS (Word Level Synthesis). Default is 16.
- `-qor_opt_enable <list of sub-options>` The sub-options are `INLINE`, `BOUNDARY`, `THREADED_GLOBAL`, and `ALL`. These enable the specified WLS (Word Level Synthesis) global optimizations. Default is currently for the global optimizations to be off.
- `-qor_opt_disable <list of sub-options>` The sub-options are `INLINE`, `BOUNDARY`, `THREADED_GLOBAL`, and `ALL`. These disable the specified WLS (Word Level Synthesis) global optimizations. Default is currently for the global optimizations to be off.
- `-inline <module_list>` Performs module inlining for a list of modules (this can get also pattern: `*` matches everything, `?` matches any single character, `[seq]` matches any character in seq, `[!seq]` matches any character not in seq>]
- `-auto_inline_limit <limit>` auto inline all modules with cost $\leq$ limit (Legacy zFast:Compile:AutoInseparableLimit)

- `-auto_inline_port_size_threshold <Threshold>` auto inline all modules with total port size $\geq$ Threshold (Legacy zFast:Compile:AutoInseparablePortSizeThreshold)
- `-print_auto_inline_cost <bool>` Print inline cost for `auto_inline_limit`.
- `-auto_inline_cost_refine <bool>` auto inling optimization in calculating inlining cost.
- `-auto_inline_params <seq_modules>=<bool>`. Enable/disable auto inling parameters. `seq_modules` enables inlining of sequential blocks. Default value is false.

Note

Avoid counting simple assignments (buffers) towards inlining cost.

- `-cell_size <int> [-module <module list>][-instance <instance list>]` Specifies the maximum cell size that PLayer will create.
- `-lut_costs <list> [-module <module list>][-instance <instance list>]` Specifies the relative costs of the different sized LUTs. This is a list of non-negative LUT costs giving the relative costs for LUT2 up through LUT6. if `-module` or `-instance` are present the optimization directive is per the instance/module list if `-module` or `-instance` are not present this is a global directive. (legacy Compile:PLY:LUTcost)

For example, with the following UTF commands

```
optimization -lut_costs {10 10 10 50 100}
```

```
optimization -lut_costs {10 12 15 20 30} -module {my_hw_top}
```

```
optimization -lut_costs {10 10 12 15 20} -module {dut}
```

The list {10 10 10 50 100} will be the global default, while the list {10 12 15 20 30} will be used for module `my_hw_top`, and the list {10 10 12 15 20} will be used for module `dut`.

- `-enhanced_loop_xform <bool> [-module <module list>]` Specifies whether or not to use WLS's enhanced loop transformation. Default is true. Specify false to turn off this transformation. If `-module` is present the optimization directive is per the module list. If `-module` is not present this is a global directive.
- `-enc_mux_threshold <int> -instance <list> [-regex]` Controls technology mapping of encoded muxes in terms of the min. number of selector pins in an encoded mux (e.g. a vector indexing operation or a case statement) required for the mux to be mapped as FPGA fabric mux vs dissolved into LUTs. This is intended to be used by CAEs during bringup rather than the end user.
- `-expand_dsp_mult_threshold <int> -instance <list> [-regex]` Specifies the max size of multiplier ops that do not use DSP
- `-drop_dead_logic <int> -instance <list> [-regex]` Eliminate stateful logic that doesn't contribute to functionality of a module (at the expense of losing visibility on removed logic). True when using full optimization
- `-share_level <0|2> -module <list> [-regex]` (legacy zFast: Compile:ShareLevel:PerModule=(modulename 0|2))
- `-inline_disable <module_list>` disable inline on list of modules

Note

(this can get also pattern: * matches everything, ? matches any single character, [seq] matches any character in seq, ![seq] matches any character not in seq>]

- `-never_inline <module_list>` set list of modules to be never inlined

Note

(this can get also pattern: * matches everything, ? matches any single character, [seq] matches any character in seq, ![seq] matches any character not in seq>]

- `-never_inline_disable <module_list>` disable never inline on list of modules

Note

(this can get also pattern: * matches everything, ? matches any single character, [seq] matches any character in seq, [!seq] matches any character not in seq>]

- `-top_down_roots <module_list>` List of modules as top for limited top-down approach (Legacy Compile-:TopDownRoots)
- `-celldefine <bool>` treat module in 'celldefine block as UDP for cost computation when auto inlining.

Note

`-celldefine` option should only be used with `-auto_inline_limit` or `-auto_inline_port_size_threshold`. Default value is t.

- `-area_effort <DEFAULT|LOW|MEDIUM|HIGH>`
- `-tree <module>`. This specifies as that this module and all below will be optimized and mapped together.

Note

For Synplify Mapping, optimizations work better when the complete module tree is optimized together.

- `-techmapping <REGULAR|TDS>`. This option is used to specify Timing Driven Synthesis using Synplify Mapping Technology.

For example:

```
optimization -tree {critical_module} -techmapping {TDS}
```

This will run technology mapping using Synplify Timing Driven Synthesis on critical_module and the tree below.

Note

`-techmapping` should only be used with `-tree`. Default value is REGULAR.

- `-techmapping_constraints_file <constraints for this module>`

Note

`-techmapping_constraints_file` should only be used with `-techmapping {TDS}`. This should contain FDC constraints for this module tree.

- `-techmapping_advanced_command_file <additional option to be passed to synplify>`

Note

`-techmapping_advanced_command_file` should only be used with `-techmapping {TDS}`. This should contain additional options for Synplify Timing Driven Synthesis.

2.9 power_profile

Functions

- **power_profile** args

2.9.1 Detailed Description

Enable power profile and debug dumping. The debug dumping signals can be specified by \$dumpvars calls in design, and the power profiled signals are specified by power weighted profile, and all these power profiled signals are also debug signals.

`power_profile -weight_profile <file> [-optimize CAPACITY|BALANCED|PERFORMANCE]`

Parameters

args	
	• <code>-weight_profile <file></code> Specify the power weight file generated by Spyglass to enable power profile and debug dumping. All the signals used by power profile are also debug dump signals. • <code>-optimize CAPACITY BALANCED PERFORMANCE</code> Specify the optimization strategy for debug dump. CAPACITY: allocate the largest number of debug resources. BALANCED: allocate the resource to balanced performance and capacity (default). PERFORMANCE: allocate the resource for best performance. • <code>-smash_hierarchy NONE CRITNET MEMORY BOTH</code> Specify to smash the instrumented memory or critical net instance hierarchy. NONE: do not smash any instrumented power profile instances (default). CRITNET: smash all the instrumented critical net power profile instances. MEMORY: smash all the instrumented memory power profile instances. BOTH: smash both critical net and memory power profile instances. • <code>-missing_bucket_signal_threshold <int></code> Specify the tolerance threshold for missing bucket signals. Default is 0. • <code>-missing_critical_net_threshold <int></code> Specify the tolerance threshold for missing critical nets. Default is 0. • <code>-autodrop_extreme_small_memory_weight <bool></code> Enable the automatic dropping for extreme small memory weights in IP which can impact accuracy. Default is false.

2.10 set_global_false_path

Functions

- **set_global_false_path** args

2.10.1 Detailed Description

Specifies global false path constraints to a given node

Usage:

```
set_global_false_path [-match=<string>]
```

This command is used to specify global false path constraints on each port and wire names that match a given pattern.

This command supports Unix wildcards. The '-match' option is used to specify a match pattern.

- `-match=<string>` Matching pattern using the form `Ins_pattern/Port_pattern`
- `-case_sens=<bool>` Matching pattern is case sensitive
- `-verbose=<bool>` Displays the name of false paths found

2.11 set_power_estimation

Functions

- **set_power_estimation** args

2.11.1 Detailed Description

```
set_power_estimation [-fnmatch] [-object_not_found [fatal|warning]] [-rtlname_file <filename>]
                    [-type [dynamic|fwc]] [-wire_file <filename>]
```

COMMAND DESCRIPTION:

Configures the behavior of the Power Estimation feature.

OPTIONAL ARGUMENTS:

- `-fnmatch` Indicates that object pathnames can contain wildcards.
- `-object_not_found [fatal|warning]` Specifies the error policy when objects cannot be found.
- Available policies:
`fatal`: Default value. A fatal error is displayed.
`warning`: A warning is displayed instead of an error.
- `-rtlname_file <filename>` Specifies the file containing the names of wires used for Power Estimation (one bit per line).
- `-type [dynamic|fwc]` Specifies the capture mechanism used by Power Estimation. If not specified, fast- waveform-capture will be preferred.
- Details about allowed values:
`dynamic`: Power Estimation will be performed via dynamic-probes.
`fwc`: Power Estimation will be done through fast waveform captures.
- `-wire_file <filename>` Specifies the file containing the EDIF names of wires used for Power Estimation (one bit per line).

COMMAND USAGE EXAMPLES:

- Example 1:

```
- set_power_estimation -type fwc -rtlname_file ../../src/power_est.txt
  Power estimation will be performed via the signals found
  in file ../../src/power_est.txt
```

2.12 synplifypro_synthesis

Functions

- **synplifypro_synthesis** args

2.12.1 Detailed Description

Specify synplifypro synthesis groups

- `#name | +tree #name | -tree #name` Specify hierarchy to be synthesized with synplifypro
- `-groupname #group_name` synthesis group name
- `-options {name=value ... nameN=valueN}` Synthesis options

2.13 synthesis

Functions

- **synthesis** args

2.13.1 Detailed Description

Synthesis options

- `-advanced_command_file <filename>` Specify legacy zFast command file which will effect DUT and all Xtors.
- `-advanced_command_file_dut_only <filename>` Specify legacy zFast command file will effect DUT only.
- `-advanced_command {<zFast_attribute>}` Specify any legacy zFast command
- `-blackbox <module names to blackbox>` The module names or unit names specified are forced as blackboxes (after elaboration stage). Hierarchical paths to module instances are not accepted.
- `-full_blackbox <module names to blackbox>` The module names or unit names specified are to be forced as blackboxes from elaboration stage itself. Hierarchical paths to module instances are not accepted.
- `-dont_make_bbox <module names to NOT blackbox>` The module names or unit names specified are not to be treated as blackboxes. Hierarchical paths to module instances are not accepted.
- `-blackbox_path <instance paths to blackbox> -preserve_ports` The hierarchical instance paths specified are forced as blackboxes (after elaboration stage); The `-preserve_ports` option can be added if you want to protect the ports of the blackboxed instances
- `-ignore_unique_case <bool>` Ignore unique keyword in unique case statement (legacy Compile:IgnoreUniqueCase)
- `-check_utf_vs_advanced_command_file <bool>` In WLS flow, do some consistency checks between UTF commands and synthesis advanced commands.
- `-wls_option {<wls_option>}` Specify one wls option.

Note

`-wls_option` and `-wls_option_file` will be applied in the order in which the two of them are found in the UTF file.

- `-wls_option_file <filename>` Specify additional wls option file

Note

`-wls_option_file` : option file can be set once throughout the utf file

- `-simon_option_file <filename>` Deprecated. Specify additional simon option file.

Note

`-simon_option_file` : option file can be set once throughout the utf file
Instead, use the newer option `-wls_option_file`.

- `-generate_db_for_fmcheck <bool>` Generate db in WLS (Word Level Synthesis) flow for fm-check
- `-generate_db_for_fmcheck_modules <module_names>` Generate db in WLS (Word Level Synthesis) flow for specified units for fmcheck.

- `-generate_db_for_fmcheck_module_file <module_file>` Generate db in WLS (Word Level Synthesis) flow for the units specified in the module file for fmcheck.
- `-allow_unresolved_xmr` Allow unresolved xmr and transform them to blackbox.
- `-resolve_force_xmr_be` Allow force on xmr to be resolved after global-stich phase in protocompiler bottom-up compile flow.
- `-dont_bbox_for_xmr_on_output` Disable blackboxing the module if there are drive XMR on black-box output.
- `-enable_xmr_check <bool>` Enable the XMR connectivity check in WLS (Word Level Synthesis) flow, disabled by default
- `-enable_scalene_xmr_check <bool>` Enable the XMR connectivity check in Scalene Cross Partition Flow. Enabled by default, can be disabled by setting it to false.
- `-disable_rtlname_check <bool>` Disable the rtlname check in WLS (Word Level Synthesis) flow, disabled by default
- `-allow_upward_xmr_inside_interface <bool>` Enable the upward XMR support inside interface.(-Xshwetag=0x8000000)
- `-max_loop_iterations <int>` Specify maximum loop iterations limit.(legacy Compile:MaxLoopIterations=<int>)
- `-allow_implementasflops_on_utf_target <bool>` For utf commands like zforce/zgate/probe_signals if target is memory under this option it will be bit-blasted.
- `-keep_module_order <bool>` Sort module order while processing XMR internally.
- `-set_stack_ctrl_val <int>` Set stack controller value while tracing xtor.
- `-use_vfs <bool>` Enable the use of VFS (virtual file system)
- `-real_number_modeling <bool>` Enable synthesis of real data type with floating point format; default false.
- `-fp_perf_mode <int>` Enable performance optimization on floating point, different value means different levels; default 0.
- `-fp_perf_mode_diag <bool>` Enable diagnostic performance optimization on floating point; default false.
- `-synthesize_shortreal_with_32bits <bool>` Enable synthesis of shortreal in 32-bits floating point format; default false.
- `-enhanced_sv_interface_support <bool>` Enable enhanced sv interface support in WLS (Word Level Synthesis) flow.
- `-enable_xmr_threads <bool>` Enable multi-threaded support in native XMR processing. Enabled by default.
- `-hierarchy_info <path>` Hierarchical path for each module in the design dump directory for bottom up compile
- `-enhanced_var_select_xmr <bool>` avoid module splitting for read XMR with variable select.
- `-disable_bidir_native_xmr <bool>` process bidirect XMR at back-end compile.
- `-handle_complex_vh_xmr <bool>` resolve complex read vh XMRs.
- `-user_override_fpga_prims <bool>` allow xilinx primitives definition by user
- `-consistency_check <bool>` Check the synthesis data structures for consistency.

- `-dwbb_fpga_check_usage <bool>` Check that all DesignWare Building Block components use FPGA (synthesis) models, and not simulation models.
- `-dwbb_fpga_use_library <directory_path>` Use the DWBB FPGA logical libraries at `<directory_path>`.
- `-dwbb_fpga_ <bool>` Use the DWBB FPGA logical libraries that are shipped with VCS. This uses `$VCS_HOME/bin/synopsys_sim.setup`, the lowest priority setup file.
- `-allow_vhdl_as_hwtop <bool>` Allow VHDL as hwtop in UCPC flow.

2.14 system_tasks

Functions

- **system_tasks** args

2.14.1 Detailed Description

Specifies list of hierarchies under which system task/function calls are NOT removed, and are given to synthesis.

Note

All system task/function calls are removed by default except for SVA calls

- `-task #task_name` Specify system task
- `-replace #replace_value` Replace system task with given string
- `-exclude #list_of_hierarchies_to_exclude` Specify list of hierarchies to exclude
- `-remove` SVA Global option to remove all system tasks from SVAs
- `-module #list_of_modules` Specify list of modules under which system task are not removed and are given to synthesis.

One of the following options -task, -remove must be used.

Option -task cannot be used without one of the following options -enable, -replace, -module.

-enable and -module are mutual exclusive, either enabling on all design or on selective modules.

2.15 zcei

Functions

- **zcei** args

2.15.1 Detailed Description

ZCEI transactor options

- `-module {list_of_zcei_transactors_modules}`
- `-instance <xmr path to xtor> (at least one of)[-debug <bool>|-edge_enable <bool>|-disable <xmr to signal> ]`
- `-debug <bool>` Specifies whether visibility must be added inside the transactor or not
- `-edge_enable <bool>` Specifies whether flops initially synchronized by the input zcei clock are updated with an edge enable block around
- `-disable <xmr to signal>` Provides a signal that disables the transactor when it is 0
- `-advanced_synth_command_file <filename>` Where filename is xtor specific global hcsrc commands file name

Note

`-advanced_synth_command_file` must be used with `-module` , and should not be used with `-instance`

2.16 scemi

Functions

- **scemi** args

2.16.1 Detailed Description

SCEMI transactor options

- `-module {list_of_scemi_transactors_modules}`
- `-instance <xmr path to xtor> (at least one of)[-debug <bool>|-edge_enable <bool>|-disable <xmr to signal> ]`
- `-debug <bool>` Specifies whether visibility must be added inside the transactor or not
- `-edge_enable <bool>` Specifies whether flops initially synchronized by the input zcei clock are updated with an edge enable block around
- `-disable <xmr to signal>` Provides a signal that disables the transactor when it is 0
- `-time_stamp [no|uctrl|ctrl]` Timestamp mode: no timestamp/uncontrolled timestamp/controlled timestamp
- `-time_stamp_clk <xmr path to clock>`
- `-advanced_synth_command_file <filename>` Where filename is xtor specific global hsrc commands file name

Note

`-advanced_synth_command_file` must be used with `-module` , and should not be used with `-instance`

2.17 zgate

Functions

- **zgate** args

2.17.1 Detailed Description

```
zgate <MODULE> [-clock_freq [25Mhz|50Mhz|100Mhz]] [-clock_name <string>] [-fnmatch]
[-hier_sep <string>] [-insertion_mode[downstream|global|parallel|serial]]
[-instance_name <string>] [-object_not_found [fatal|warning]] [-reg_init <string>]
[-module <name> -signal <list>] [-rtlname <list>] [-rtlname_file <filename>] [-verbose] [-wire <lis
[-wire_file <filename>]
```

COMMAND DESCRIPTION:

Adds a gate in the design. The module interface must be as follows:

- o input port or port-vector named 'D' or 'I'
- o input port named 'C' for sequential module
- o output port named 'Q' or 'O'. With port-vectors:
- o Input and output must be of equal size
- o Have to match the number of instrumented elements
- o Must be within the same context

Top input ports as well as inout ports cannot be instrumented.

This command will not interrupt a tcl script in case of an error; the number of error(s) is returned instead.

MANDATORY ARGUMENTS:

- **<MODULE>** Specifies the name of the module to be added. In case of conflict between several libraries, the library name can be prepended (library).

* OPTIONAL ARGUMENTS:

- **-clock_freq** [25Mhz|50Mhz|100Mhz] In case of a macro with a 'C' input port, this option defines the clock frequency of the system clock that will be connected to this port.
- **-clock_name** <string> In case of a macro with a 'C' input port, this option defines the wire hierarchical name that will be connected to this port. If rtlname/rtlname_file option is specified, the hierarchical primary clock path must be an RTL path name too. Following special clocks may be specified as well : driverClk, zFilterClk, zDelayClk, compositeClk.
- **-fnmatch** This options applies only to names specified in 'wire/rtlname' and 'wire_file/rtlname_file' options. Only wire or signal names will be matched as fnmatch expressions. This cannot be used when the input and output ports of the zgate module are vectors. Type: switch.
- **-hier_sep** <string> Specifies the hierarchical separator used to parse wire names if specified.
Default value: ..
- **-insertion_mode** [downstream|global|parallel|serial] Specifies the way the module will be instantiated.

- Details about allowed values:

`downstream`: All downstream readers will be driven by the zgate instance. zgate instantiation context depends on port direction when using the `rtlname` option.

`global`: The zgate output will affect the whole hierarchical net on which it is inserted.

`parallel`: Connects the wires to the inputs of the zgate instance.

`serial`: Direction of the wire will be computed and the zgate inserted accordingly. In case of multidriver or tristate buses or equipotential loop, insertion might be impossible.

- `-instance_name <string>` Specifies the instance name to be given to the zgate. In case of conflict it will be used as a prefix of the zgate instance name.
- `-object_not_found [fatal|warning]` Specifies the error policy when objects cannot be found.
- Available policies:
 - `fatal`: Default value. A fatal error is displayed.
 - `warning`: A warning is displayed instead of an error.
- `-reg_init <string>` Specifies the initial value of the inserted gate. Value for vectors may be expressed using Verilog number syntax (e.g {32'hBABEFACE}).
- `-rtlname <list>` Applies the command to a list of hierarchical paths.
- `-rtlname_file <filename>` Applies the command to hierarchical paths listed in a file defined in argument. File path is relative to zCui execution directory.
- `-module <modulename>` Specifies the name of the module under which the command is applied to signals specified by `-signal` option.
- `-signal <list>` Applies the command to a list signals. Signals are relative to module specified with `-module` option.
- `-verbose` Verbosity level.
- `-wire <list>` Applies the command to a list of hierarchical wire patterns.
- `-wire_file <filename>` Applies the command to hierarchical wire patterns listed in a file defined in argument. File path is relative to zCui execution directory.

COMMAND USAGE EXAMPLES:

- Example 1:

```
zgate buf -wire "top.CK_2_CK top.OUT_ins_3mult_gnd_bb_in_io_mod_2_OUT"
-ininsertion_mode parallel
  Adds buffer on several wires in parallel mode.
```

- Example 2:

```
zgate fd -clock_name compositeClk -verbose -rtlname
{top.instance.in[15:0]} -ininsertion_mode downstream -reg_init
{16'b1111101011001110} -object_not_found warning
  Adds runtime programmable registers synchronized on composite clock,
  and initialized with corresponding 16-bit bitfield, to top.instance.in
  16-bit width port vector.
```

- Example 3:

```
zgate fd -wire "top.CK_2_CK top.OUT_ins_3mult_gnd_bb_in_io_mod_2_OUT"
-ininsertion_mode serial -instance_name zgate_
  Adds register on several wires in serial mode.
```

2.18 zinject

Functions

- **zinject** args

2.18.1 Detailed Description

```
zinject [-fnmatch] [-init_value [0|1]] [-mode [dynamic|static]] [-object_not_found [fatal|warning]]
        [-pin <list>] [-pin_file <filename>] [-module <name>] [-signal <list>] [-rtlname <list>] [-rtlname_
        [-type [comb|disconnect_driver]] [-wire <list>] [-wire_file <filename>]
```

COMMAND DESCRIPTION:

Assigns signal to a given value at runtime. Automatically released with an event on its driver. Sequential signals are injectable by default.

OPTIONAL ARGUMENTS:

- `-fnmatch` Indicates that object pathnames can contain wildcards.
- `-init_value [0|1]` Specifies initialization value [0|1] of registers that are used to inject undriven wire or injections performed with driver disconnection. Register's initialization value is set to 0 by default.
Default value: 0.
- `-mode [dynamic|static]` zinject command modes.
- Details about allowed values:
`dynamic`: 'dynamic' mode is recommended for system resource preservation over performance. Default mode for injection.
`static`: 'static' mode is recommended for signals that are forced hundreds of times or more during an emulation via dedicated top ports.
 Default value: dynamic.
- `-object_not_found [fatal|warning]` Specifies the error policy when objects cannot be found.
- Available policies:
`fatal`: Default value. A fatal error is displayed.
`warning`: A warning is displayed instead of an error.
- `-pin <list>` Specifies a list of instance ports. The hierarchical net connected to the port is injected.
- `-pin_file <filename>` Specifies whether the command will be applied on pin objects found in a file defined in argument. File path is relative to the zCui execution directory.
- `-rtlname <list>` Applies the command to a list instance port paths. The hierarchical net connected to the port is injected.
- `-rtlname_file <filename>` Applies the command to RTL objects found in a file defined in argument. File path is relative to zCui execution directory.
- `-module <modulename>` Specifies the name of the module under which the signals specified by `-signal` option are injected.

- **-signal <list>** Applies the command to a list signals. Signals are relative to module specified with **-module** option.
- **-type [comb|disconnect_driver]** Specifies the injection behavior.
- Details about allowed values:
comb:
 In case of injection into a wire with a pre-existing combinational driver, logic will be added to the netlist and will hold the injected value until the next event on the injected wire. Otherwise, injection will be ignored with a warning.
disconnect_driver:
 In case of injection into a wire driven by flop or latch, wire may be disconnected and driven by a new register, initialized to 0 (except when **init_value** option is specified). This option allows you to stick a signal with a runtime-programmable value, and guarantees that the injected signal will no longer be influenced by its original driver.
- **-wire <list>** Applies the command to a list of wires.
- **-wire_file <filename>** Applies the command to wire objects found in a file defined in argument. File path is relative to zCui execution directory.

COMMAND USAGE EXAMPLES:

- ```

```
- Example 1:  
 - `zinject -mode dynamic -rtlname dut.instance.wirename`  
 Injects corresponding wire with a runtime programmable register.
  - Example 2:  
 - `zinject -mode static -rtlname dut.instance.wirename`  
 Injects corresponding wire with a created top port (value), which will be used as driver of the injection register.

## Description

Assigns signal to a given value at runtime. Automatically released with an event on its driver. Sequential signals are injectable by default.

## Synopsis

```
zinject [-fnmatch] [-init_value <0|1>] [-mode <dynamic|static>] [-module <string>] [-object_not_found <fatal|warning>] [-pin <list>] [-pin_file <file>] [-rtlname <list>] [-rtlname_file <file>] [-signal <list>] [-type <comb|disconnect_driver>] [-wire <list>] [-wire_file <file>]
```

## Option(s)

- **-fnmatch** Indicates that object pathnames can contain wildcards..
- **-init\_value <enum>** Specifies initialization value [0|1] of registers that are used to inject undriven wire or injections performed with driver disconnection. Default value: 0..  
 0|1  
 - **0**: initialization value 0.  
 Valid values: 0  
 - **1**: initialization value 1.  
 Valid values: 1
- **-mode <enum>** zinject command modes. Default value: dynamic..  
 dynamic|static

- **dynamic**: This mode is recommended for system resource preservation over performance. Default mode for injection.  
Valid values: `dynamic`
- **static**: This mode is recommended for signals that are forced hundreds of times or more during an emulation via dedicated top ports.  
Valid values: `static`
- **-module <string>** Specifies the name of the module under which the signals specified by -signal option are injected..
- **-object\_not\_found <enum>** Specifies the error policy when objects cannot be found..  
`fatal|warning`
  - **fatal**: Default value. A fatal error is displayed.  
Valid values: `fatal`
  - **warning**: A warning is displayed instead of an error.  
Valid values: `warning`
- **-pin <list>** Specifies a list of instance ports. The hierarchical net connected to the port is injected. This option is deprecated..
- **-pin\_file <file>** Specifies whether the command will be applied on pin objects found in a file defined in argument. File path is relative to the zCui execution directory. This option is deprecated.
- **-rtlname <list>** Applies the command to a list instance port paths. The hierarchical net connected to the port is injected..
- **-rtlname\_file <file>** Applies the command to RTL objects found in a file defined in argument. File path is relative to zCui execution directory..
- **-signal <list>** Applies the command to a list signals. Signals are relative to module specified with -module option..
- **-type <enum>** Specifies the injection behavior..  
`comb|disconnect_driver`
  - **comb**: In case of injection into a wire with a pre-existing combinational driver, logic will be added to the netlist and will hold the injected value until the next event on the injected wire. Otherwise, injection will be ignored with a warning.  
Valid values: `comb`
  - **disconnect\_driver**: In case of injection into a wire driven by flop or latch, wire may be disconnected and driven by a new register, initialized to 0 (except when `init_value` option is specified). This option allows you to stick a signal with a runtime-programmable value, and guarantees that the injected signal will no longer be influenced by its original driver.  
Valid values: `disconnect_driver`
- **-wire <list>** Applies the command to a list of wires. This option is deprecated.
- **-wire\_file <file>** Applies the command to wire objects found in a file defined in argument. File path is relative to zCui execution directory. This option is deprecated.

#### Constraint(s)

- At least one of the options should be used
- -module can only be used with -signal

#### Note(s)

- module relative `zinject` cmds are not passed to `zTopBuild`.

#### Example(s)

- `'zinject -mode dynamic -rtlname dut.instance.wirename'`: Injects corresponding wire with a runtime programmable register.
- `'zinject -mode static -rtlname dut.instance.wirename'`: Injects corresponding wire with a created top port (value), which will be used as driver of the injection register.

## 2.19 profile

### Functions

- **profile** args

#### 2.19.1 Detailed Description

At least one of the options:

- `-compile <bool>` Enable profiled compile.
- `-xtors <bool>`. Enable/disable zTune transactional profiler.
- `-xtors_params {param1=val1, param2=val2, ..., paramN=valN}`. Configure zTune transactional profiler. Valid parameters are:

```
TYPE=ZEMI_GLOBAL|ZEMI_DPI|ZEMI_GLOBAL_DPI|LEGACY
ZEMI_GLOBAL: profile global (sampling/internal/dpi/b2b) clock stopping for ZEMI xtors
ZEMI_DPI: profile per DPI clock stopping for ZEMI xtors
ZEMI_GLOBAL_DPI: profile both global and per DPI clock stopping for ZEMI xtors
LEGACY: legacy profiling (per xtor zceiClockControl)
```

- `-dumpvars_report <bool>` Enable dumping of full \$dumpvars stats in dump\_num\_bits.csv.
- `-dumpports_report <bool>` Enable dumping of full \$dumpports stats in dump\_num\_bits.csv.

## 2.20 memory\_preferences

### Functions

- **memory\_preferences** args

#### 2.20.1 Detailed Description

- [-max\_bram\_blocks <integer>|DEFAULT] Specify maximum numbers of BRAM blocks or restore to default the previously set value.
- [-max\_uram\_blocks <integer>|DEFAULT] Specify maximum numbers of URAM blocks or restore to default the previously set value.
- [-max\_ramlut\_blocks|-max\_lutram\_blocks <integer>|DEFAULT] Specify maximum numbers of RAMLUT blocks or restore to default the previously set value.
- [-ramlut\_to\_bram\_threshold|-lutram\_to\_bram\_threshold <integer>|DEFAULT] Specify ratio (percentage) of RAMLUTs to BRAMs or restore to default the previously set value.
- [-bram\_to\_uram\_threshold <integer>|DEFAULT] Specify ratio (percentage) of BRAMs to URAMs or restore to default the previously set value.
- [-scan\_path <list>] .
- [-advanced\_command\_file #filename] Command file for zMem
- [-advanced\_command] <zMem\_command> Specify advanced command for zMem
- [-zmem\_clock\_frequency 25|50|100|200|AUTO] The valid values are 25, 50, 100, 200(in MHz) or AUTO. AUTO stands for current default.
- [-manage\_dual\_port\_bram\_access <bool>] Enable/disable simple port ramlut multiplexing.

## 2.21 assertion\_synthesis

### Functions

- **assertion\_synthesis** args

#### 2.21.1 Detailed Description

`assertion_synthesis [-ignore (<sva_type>)] [-path <mod_inst_path>] [-tree <top_module_name>] [-module <modules list>=<"">] [-assert <assertion_name>]`

`sva_type` can be: ALL|CONCURRENT|IMMEDIATE|ASSERT|COVER|PACKAGE|PROC|PROC\_LOOP|CONCURRENT\_PROC\_LOOP|IMMEDIATE\_PROC\_LOOP

`+assert #instance_name` Synthesize given SVA instance

`-assert #instance_name` Do not synthesize given SVA instance

`+module #module_name` Synthesize all SVAs in all instances of given module

`-module #module_name` Do not synthesize all SVAs in all instances of given module

`+tree #hierarchy_name` Synthesize all SVAs below given hierarchy

`-tree #hierarchy_name` Do not synthesize all SVAs below given hierarchy

`+module_tree #module_name` Synthesize all SVAs of given module and below hierarchy of that module.

`-module_tree #module_name` Do not synthesize all SVAs of given module and below hierarchy of that module.

`+path #hierarchy_name` Synthesize all SVAs inside given hierarchy instance

`-path #hierarchy_name` Do not synthesize all SVAs inside given hierarchy instance

`-ignore ALL|CONCURRENT|IMMEDIATE|ASSERT|COVER|PACKAGE|PROC|PROC_LOOP|CONCURRENT_PROC_LOOP|IMMEDIATE_PROC_LOOP #assertion_type` Do not synthesize all SVAs of given type

`-enable ALL|ASSERT|COVER` Synthesize all/ASSERT/COVER SVAs in a design

`-verbose <bool>` Dump removed SVAs into report file 'reportSVA.log'.

`-detail` Dump SVA fan-in report when used along with `-verbose true`, otherwise this option has no effect

`-dump_constant` Dump value of elaboration constants in SVA fan-in report when used along with `-verbose true` and `-detail`, otherwise this option has no effect

`-dump_state_count <bool>` Dump SVA state count info. in file 'svaStateCountInfo.log'.

`-cover_max_states <int>` Maximum number of states allowed for a Cover property.

`-enable_ctrl_tasks <bool>` Handle assertion control tasks in the design.

`-unique_if_case <bool>` Enable unique case and unique if transformation to generate violation reports

`-never_fatal` Never use SVA as fatal (do not apply SVA trigger)

`-auto_disable` Disable assertions after first failure. Improves performance but increases logic

`-report_only_failure` Only report SVA failures

`-fire_all_triggers, generate (zsva_trigger == 1'b1)` for static trigger in vcs.dve, ignored if `-never_fatal` used

## 2.22 ztopbuild

### Functions

- **ztopbuild** args

#### 2.22.1 Detailed Description

Specify options for Top-Level build (zTopBuild)

- **-edit\_netlist\_file** #filename Specify filename with zTopBuild netlist edition commands
- **-advanced\_command\_file** #filename Specify advanced command file for zTopBuild
- **-advanced\_command** {<zTopBuild\_command>} Specify any legacy zTopBuild command
- **-advanced\_command\_netlist** {<zTopBuild\_netlist\_edition\_command>} Specify any legacy zTopBuild netlist edition command

### Description

Specify options for Top-Level build (zTopBuild)

### Synopsis

```
ztopbuild [-advanced_command <string>] [-advanced_command_file <file>]
[-advanced_command_netlist <string>] [-edit_netlist_file <file>]
```

### Option(s)

- **-advanced\_command** <string> Specify any legacy zTopBuild command.  
legacy zTopBuild command
- **-advanced\_command\_file** <file> Specify advanced command file for zTopBuild.
- **-advanced\_command\_netlist** <string> Specify any legacy zTopBuild netlist edition command.  
legacy zTopBuild netlist command
- **-edit\_netlist\_file** <file> Specify filename with zTopBuild netlist edition commands.

## 2.23 design

Specify options design option

- `[-enable_consistency_check <bool>]` enable dumping of side files needed for design consistency checker scalene

### Description

Design options

### Synopsis

```
design [-convert_strength <bool>] [-convert_strength_only <bool>] [-convert-
_switches <bool>] [-decl_assignment_check <bool>] [-drive_strength_-
support <bool>] [-enable_consistency_check <bool>] [-exclude_modules
<list>] [-filter_context <ALL|DUT>] [-netlist <bool>] [-netlist_mapping
<FPGA>] [-report <list>] [-severity <ERROR|WARNING>]
```

### Option(s)

- **-convert\_strength** <bool> Enable conversion of strength, MOS/RMOS gates and TRANIF gates.
- **-convert\_strength\_only** <bool> Enable conversion of strength.
- **-convert\_switches** <bool> Enable conversion of MOS/RMOS gates and TRANIF gates.
- **-decl\_assignment\_check** <bool> Enable checking if there are declaration assignments.
- **-drive\_strength\_support** <bool> Resolve undriven/multidriver/tristate signals using drive strength semantics. Default:true.
- **-enable\_consistency\_check** <bool> Enable dump all side files required to design consistency checker scalene.
- **-exclude\_modules** <list> Specifies modules to be excluded from consideration. Currently only for -decl\_assignment\_check..
- **-filter\_context** <ALL|DUT> Specifies whether to consider just the DUT, or ALL. Currently only for -decl\_assignment\_check..
- **-netlist** <bool> Enable netlist specific changes..
- **-netlist\_mapping** <FPGA> Specifies netlist mapping option..
- **-report** <list> Generate the requested reports. Supported reports: xmr and force..
- **-severity** <ERROR|WARNING> Specifies severity, currently only for -decl\_assignment\_check..

### Constraint(s)

- -netlist can only be used with -netlist\_mapping
- -netlist\_mapping can only be used with -netlist
- At least one of the options should be used

### Note(s)

- Option -drive\_strength\_support is deprecated

## 2.24 dpi\_synthesis

### Functions

- **dpi\_synthesis** args

#### 2.24.1 Detailed Description

`dpi_synthesis` [`+/-module` <modules list>] [`+/-dpi` <dpi\_name>] [`-enable` ALL|ALC] [`-ignore` FOR\_LOOP] [`-frequency` <int>] [`-enable_wls` <bool>] [`-offline` <bool>] [`-timestamp` <bool>] [`-global_serialized_opt` <bool>] [`-disable_initial_blocks` <bool>] [`-disable_exports` <bool>] [`-optimize_width` <bool>] [`-perf_mode` <int>]

#### Parameters

| <i>args</i> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|             | <ul style="list-style-type: none"> <li>• <code>+dpi</code> #instance_name Synthesize given DPI function</li> <li>• <code>-dpi</code> #instance_name Do not synthesize given DPI function</li> <li>• <code>+module</code> #module_name Synthesize all DPI calls in all instances of given module</li> <li>• <code>-module</code> #module_name Do not synthesize all DPI calls in all instances of given module</li> <li>• <code>-enable</code> [ALL ALC]</li> </ul> |

#### Note

ALL means Synthesize all DPI calls in a design. ALC means Synthesize DPI calls in always\_latch or always\_comb.

ALC can occur in the same dpi\_synthesis command or by itself in a separate dpi\_synthesis command, with other dpi\_synthesis commands deciding which modules or instances will have their DPI functions synthesized.

- `-ignore for_loop` Do not synthesize all DPI calls within a for loop
- `-frequency` <int> Specify the theoretical DPI frequency (in KHz) to reach at run-time
- `-enable_wls` <bool> Synthesize all DPI calls inside initial blocks and all DPI calls with output ports under ZS4 UC2 flow.
- `-offline` <bool> Support Offline mode only when used along with `-enable_wls` true, otherwise this option has no effect
- `-timestamp` <bool> Support timestamp only when used along with `-enable_wls` true, otherwise this option has no effect
- `-global_serialized_opt` <bool> Support global\_serialized\_opt only when used along with `-enable_wls` true, otherwise this option has no effect
- `-disable_initial_blocks` <bool> Disable initial block support when used along with `-enable_wls` true, otherwise this option has no effect
- `-disable_exports` <bool> Disable export DPI support when used along with `-enable_wls` true, otherwise this option has no effect
- `-optimize_width` <bool> Enable optimisation of input bits for DPI call.
- `-perf_mode` <int> Enable capacity optimisation by 1 register per DPI bit argument.

Duplicated command options are allowed.

## 2.25 defmapping

### Functions

- **defmapping** args

#### 2.25.1 Detailed Description

`defmapping` [-path <mod\_inst\_name|top\_module\_name>] [-use\_fpga <fpga\_name>] [-object\_source [rtl|edif]] [-object\_not\_found [fatal|warning]]

##### Parameters

|             |                                                                                                                                                                         |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>args</i> | <ul style="list-style-type: none"> <li>• <code>-path #hierarchy_name</code> Specify hierarchical path of a design block, could be an instance or top module.</li> </ul> |
|-------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

##### Note

On ZS3 HW configuration, the hierarchical instance cannot be a transactor.

- `-use_fpga #fpga_name` Specify FPGA used for given path; `fpga_name` should be in format: `U[0-31].M[0-5].F[0-18]`
- `-object_source [edif|rtl]` Specifies the source of objects.
  - Available policies:
    - `rtl`: Default value. The object is from Rtl, a fatal error is displayed in VCS if object not found.
    - `edif`: The object is from Edif, sanity check of path would be performed in ZEBU backend instead of VCS.
- `-object_not_found [fatal|warning]` Specifies the mode when invalid path is specified.
  - Available policies:
    - `fatal`: Default value. A fatal error is displayed in VCS if the value of option path is not found.
    - `warning`: Only a warning message is displayed in VCS if the value of option path is not found.

##### Examples:

- `defmapping -path Top.inst1 -use_fpga U0.M0.F0 -object_source edif`

## 2.26 create\_new\_target

### Functions

- **create\_new\_target** args

#### 2.26.1 Detailed Description

- **-open\_block** <string> Open target commands block with specific <name>.
- **-close\_block** <string> Close target commands block with specific <name>.

Target block cannot be opened inside another target block.

You can open and close block with the same name several times.

Commands inside blocks with the same name will be assigned to the same target.

Currently only UTF commands and options which are translated only into BE commands are allowed inside a target block.

#### Description

Target block actions (multiple target feature) Handles creation and closure of a target block

#### Synopsis

```
create_new_target [-close_block <string>] [-open_block <string>]
```

#### Option(s)

- **-close\_block** <string> Closes a target block.  
target name
- **-open\_block** <string> Opens a new target block or continue existing target block.  
target name

#### Constraint(s)

- At least one of the options should be used

## 2.27 pbuild

### Functions

- **pbuild** args

#### 2.27.1 Detailed Description

- `-user_script #filename` Specify filename. Mutually exclusive from all other options.
- `-advanced_command_file #filename` Specify advanced command file for pBuild
- `-advanced_command {<pBuild_command>}` Specify any pBuild command

## 2.28 UCVLbExt

- `library extensions` Specify library extensions for the compilation, separated by '+' mark.

## 2.29 UCVcsVerdiFlowNoFSDB

UCVcsVerdiFlowNoFSDB <true>/<false>

### COMMAND DESCRIPTION:

Aborts loading of the Verdi library in UC2 dumpvars.

### MANDATORY ARGUMENTS:

- `-true` Chose not to load the Verdi library in UC2 dumpvars.
- `-false` Load the Verdi library and generate the FSDB header. Also the default value.

## 2.30 architecture\_file

### Description

Specify target architecture file for compilation

### Synopsis

```
architecture_file -filename <file>
```

### Option(s)

- **-filename** <file> Path to architecture file. **Mandatory option**

### Constraint(s)

- At least one of the options should be used

## 2.31 clock\_boundary\_marker

### Functions

- **clock\_boundary\_marker** args

### 2.31.1 Detailed Description

#### Description

Adds a clock\_boundary\_marker black box module in the design which will stop the clock cone propagation. Black box will be removed after clock tree annotation.

#### Synopsis

```
clock_boundary_marker [-break] [-fnmatch] [-module <string>] [-object_-
not_found <fatal|warning>] [-rtlname <string>] [-rtlname_file <string>]
[-signal <string>] [-verbose]
```

#### Option(s)

- **-break** Replace marker bb with FD if marker bb is in clock cone..
- **-fnmatch** Matches hierarchical names using fnmatch patterns. This option applies only to names specified in 'rtlname'/'signal'. .
- **-module <string>** Module name for clock boundary marker..  
string
- **-object\_not\_found <fatal|warning>** Specify error policy in case of objects is not found..  
policy value
  - **fatal**: In case object cannot be found, a fatal error is emitted. This is the default.  
Valid values: fatal|Fatal|FATAL
  - **warning**: In case object cannot be found, a warning is emitted instead of an error.  
Valid values: warning|Warning|WARNING
- **-rtlname <string>** Specifies the hierarchical signal name(s) where marker has to be inserted..  
string
- **-rtlname\_file <string>** Specifies the hierarchical signal name(s) in a file where marker has to be inserted..  
string
- **-signal <string>** Path of module port used. It should be used only with module option..  
string
- **-verbose** Gives more details about command processing..

#### Constraint(s)

- At least one of the options should be used
- check illegal option combinations

#### Note(s)

- -module and -signal must be given together.
- Either '-rtlname/-rtlname\_file' or '-module <> -signal<>' are supported in one command.

#### Example(s)

- 'clock\_boundary\_marker -rtlname {hw\_top.dt1.myclk2 hw\_top.dt1.myclk1}' => Adds marker on multiple wires/ports in the design.

- 'clock\_boundary\_marker -rtlname {hwtop.t.or\*} -fnmatch' => Adds marker on multiple wires/ports matching the wild card.
- 'clock\_boundary\_marker -rtlname {hwtop.t.or\*} -fnmatch -break' => Adds marker on multiple wires/ports matching the wild card and replace it with FD after clock tree annotation if marker is in clock cone.
- 'clock\_boundary\_marker -module {dut1} -signal {myclk2}' => Adds marker on every instance of given module's signal.

## 2.32 clock\_config

### Description

Define clock related parameters or add new zceiClockPort

### Synopsis

```
clock_config [-accuracy <16|24|32>] [-add_clock_port <string>] [-clock-
_name <string>] [-clock_number <2|4|8|16>] [-driverclk_dyn_opt <bool>]
[-fk_synchronizer <bool>] [-frequency <string>] [-number_of_bufgs <int>]
[-parameters <CLK_PERIOD=string,DEF_RESET=string,DEF_RESETN=string,cclock-
Name=string,clockNum=string>] [-ports <cclock=string,creset=string,cresetn=string>]
[-share_clock_bus <bool>] [-synthesis <bool>]
```

### Option(s)

- **-accuracy <16|24|32>** Specify the accuracy of clock generation (bits).
- **-add\_clock\_port <string>** Specify the type of the clock port to be added.  
Currently, only zceiClockPort is supported
- **-clock\_name <string>** Specify the name of the clock port or the xmr to the clock port instance path.  
masterfrequency|driverfrequency|xclkfrequency|#XMR\_to\_zClockPort\_instance
- **-clock\_number <2|4|8|16>** Specify the number of resources allocated for the Clock Generator.
- **-driverclk\_dyn\_opt <bool>** Dynamically optimized driver clock frequency.
- **-fk\_synchronizer <bool>** Force the use of synchronizer on the RTB to force clocks generated from none zceiClockport to be aligned with zceiClockPort one.
- **-frequency <string>** Specify clock frequency.  
Can be used with 'MHz'/'kHz' units, in format '100 MHz'
- **-number\_of\_bufgs <int>** Specify the number of bufgs available for DUT clock routing in the FPGAs.  
A non-negative integer number.
- **-parameters <CLK\_PERIOD=string,DEF\_RESET=string,DEF\_RESETN=string,cclock-Name=string,clockNum=string>** Specify the parameter values of the clock port if '-add\_clock\_port' exists.
  - **CLK\_PERIOD <string>** Specify the value of parameter 'CLK\_PERIOD' of zceiClockPort.
  - **DEF\_RESET <string>** Specify the value of parameter 'DEF\_RESET' of zceiClockPort.
  - **DEF\_RESETN <string>** Specify the value of parameter 'DEF\_RESETN' of zceiClockPort.
  - **cclockName <string>** Specify the value of parameter 'cclockName' of zceiClockPort.
  - **clockNum <string>** Specify the value of parameter 'clockNum' of zceiClockPort.
- **-ports <cclock=string,creset=string,cresetn=string>** Specify the port connection of the clock port if '-add\_clock\_port' exists.
  - **cclock <string>** Specify the port connection of port 'cclock' of zceiClockPort.  
Must exist when '-add\_clock\_port zceiClockPort' is specified
  - **creset <string>** Specify the port connection of port 'creset' of zceiClockPort.
  - **cresetn <string>** Specify the port connection of port 'cresetn' of zceiClockPort.
- **-share\_clock\_bus <bool>** Use remaining clock bus bits for trigger routing. Only supported for Orion; Default: true.
- **-synthesis <bool>** Used if -clock\_name is Top-down XMR to zClockPort instance.

### Constraint(s)

- At least one of the options should be used
- -share\_clock\_bus cannot be used with any other option

- check illegal option combinations
- the accepted value for `-add_clock_port` is only `zceiClockPort` now

**Note(s)**

- `-clock_name` and `-frequency` are the only options supported in Taurus flow
- `-accuracy 16` is only supported for Orion
- `-share_clock_bus` is only supported for Orion

**Example(s)**

- `clock_config -clock_number 8`
- `clock_config -clock_name masterfrequency -frequency "160MHz"`
- `clock_config -clock_name top.c2 -synthesis true -frequency "450 Mhz"`
- `clock_config -add_clock_port zceiClockPort -clock_name top.dut.clock -ports {cclock=clk}`

## 2.33 clock\_delay

### Description

Enables Verilog clock delays support

### Synopsis

```
clock_delay [-auto_tolerance <bool>] [-clock_wire <string>] [-clockdelayport-
_timescale <string>] [-debug <bool>] [-duty_high <int>] [-duty_low <int>]
[-enable_var_delay_replication <bool>] [-gate_delay <bool>] [-ignore_-
zero_delay <bool>] [-insert_clock_delay_port_in_module <string>] [-instance-
_name <string>] [-module <list>] [-perf_mode <int>] [-phase <int>]
[-reset_duration <int>] [-reset_value <int>] [-reset_wire <string>]
[-sdm_mcp <bool>] [-sdm_mcp_grpl_ref_clocks <bool>] [-sdm_mcp_mode <int>]
[-sdm_mcp_num_ref_clocks <int>] [-tolerance <int>] [-transport_delay_-
depth <int>]
```

### Option(s)

- **-auto\_tolerance <bool>** compute and set max tolerance for clock\_delay.
- **-clock\_wire <string>** clock path.  
clock path
- **-clockdelayport\_timescale <string>** Specify the time scale of all clock delay ports.  
[1|10|100] [s|ms|us|ns|fs|ps]
- **-debug <bool>** Enables debug mode.
- **-duty\_high <int>** Set dutyHigh for clock delay port defined in UTF command.  
A non-negative integer number.
- **-duty\_low <int>** Set dutyLow for clock delay port defined in UTF command.  
A non-negative integer number.
- **-enable\_var\_delay\_replication <bool>** Enable replication with variable delays.
- **-gate\_delay <bool>** enable clock\_delay mode for gate delays.
- **-ignore\_zero\_delay <bool>** ignore processing of #0 delays.
- **-insert\_clock\_delay\_port\_in\_module <string>** Define clockDelayPort in module.  
module name
- **-instance\_name <string>** Specify the instance name of clock delay port.  
clock delay port instance name
- **-module <list>** list of clock delay modules.
- **-perf\_mode <int>** clock\_delay performance modes Only valid for ZS4 onwards. Default Value is 0. 0 - No additional performance optimizations 1 - Module level clock delay replication 2 - Fpga level clock delay replication 3 - No replication, clock delays and \$time consumers moved to terminal fpga 4 - No replication, clock delays moved to terminal fpga. \$time consumers stay on any partitioned fpga.  
A non-negative integer number.
- **-phase <int>** Set phase for clock delay port defined in UTF command.  
A non-negative integer number.
- **-reset\_duration <int>** Set resetDuration for clock delay port defined in UTF command.  
A non-negative integer number.
- **-reset\_value <int>** Set resetValue for clock delay port defined in UTF command.  
A non-negative integer number.
- **-reset\_wire <string>** reset path.  
reset path

- **-sdc\_mcp** <bool> enable clock\_delay based support for SDC MCP computation with clock group based tolerance with edges aligned with the posedge of the fastest clock. '-sdc\_mcp\_mode' option may override the default tolerance mode..
- **-sdc\_mcp\_grp1\_ref\_clocks** <bool> enable group1 clocks as reference clocks in -sdc\_mcp\_mode 2.
- **-sdc\_mcp\_mode** <int> clock\_delay based support for SDC MCP computation Only valid for Z-S4 onwards. 0 - Without any tolerance optimization 1 - Default clock\_delay tolerance optimization 2 - Tolerance with fastest clock period tolerance window 3 - Tolerance with edges aligned with the fastest clock 4 - Tolerance with edges aligned with the posedge of the fastest clock 5 - Clock group based tolerance with edges aligned with the posedge of the fastest clock.  
A non-negative integer number.
- **-sdc\_mcp\_num\_ref\_clocks** <int> enable additional reference clock creation for -sdc\_mcp\_mode 2 .  
A non-negative integer number.
- **-tolerance** <int> Set tolerance for clock\_delay.  
A non-negative integer number.
- **-transport\_delay\_depth** <int> Specify the depth of the transport delay scheduler.  
A positive integer number.

#### Constraint(s)

- -clockdelayport\_timescale is to set in a valid format

## 2.34 clock\_frequency

### Description

Enables clock frequencies support

### Synopsis

```
clock_frequency [-file <string>]
```

### Option(s)

- **-file <string>** Specify name of clock frequency file..  
Specify name of clock frequency file.

## 2.35 cluster\_constraint

### Description

Specify clustering constraints

### Synopsis

```
cluster_constraint [-accept_overflow <bram=int,dsp=int,lut=int,qiwc=int,ramlut=int,reg=int,uram=int>]
[-allow_ovrd_user_max_fill <bool>] [-command <merge_blocks|add_group|defcore>]
[-exclusive <bool>] [-fnmatch <bool>] [-group_name <string>] [-instance
<list>] [-level <core|fpga|both>] [-max_fpga <int>] [-min_fpga <int>]
[-overflow <bram=int,dsp=int,lut=int,qiwc=int,ramlut=int,reg=int,uram=int>]
[-rtlname] [-set_max_fill <bram=int,dsp=int,uram=int,reg=int,ramlut=int,write_port_ip=int,write_port_bit=int,read_port_ip=int,read_port_bit=int,zcei_mess_in=int,zcei_mess_out=int>]
```

### Option(s)

- **-accept\_overflow <bram=int,dsp=int,lut=int,qiwc=int,ramlut=int,reg=int,uram=int>**  
Allows filling rates overflow for different types of logic elements according to specified percentage. Available logic elements are: reg, lut, ramlut, bram, dsp. Default are respectively 5, 10, 10, 0, 0.
  - **bram <int>** bram overflow in %.  
A non-negative integer number.
  - **dsp <int>** dsp overflow in %.  
A non-negative integer number.
  - **lut <int>** lut overflow in %.  
A non-negative integer number.
  - **qiwc <int>** qiwc overflow in %.  
A non-negative integer number.
  - **ramlut <int>** ramlut overflow in %.  
A non-negative integer number.
  - **reg <int>** reg overflow in %.  
A non-negative integer number.
  - **uram <int>** uram overflow in %.  
A non-negative integer number.
- **-allow\_ovrd\_user\_max\_fill <bool>** Allows to AC algorithm override user settings on max\_fill constraints.
- **-command <merge\_blocks|add\_group|defcore>** Specifies cluster\_constraint command.
  - **merge\_blocks:** The auto-clustering will keep the specified blocks on the same fpga. The command can handle multiple statements sharing a common block, in this case all the blocks will be merged. This command handle hierarchy overlapping in the following way: if an instance of the hierarchy is touched by several merge\_block commands, the merge\_block command that references the closest of its ancestor will be selected. Usage: cluster\_constraint -command MERGE\_BLOCKS -instance <list> [-level CORE|FPGA|BOTH] [-fnmatch <bool>]  
Valid values: merge\_blocks|MERGE\_BLOCKS
  - **add\_group:** Defines a logic group of instances that may require the resources of more than one FPGA. One can define multiple groups with several commands. Any instance that does not belong to a group is attached to the default group. It is possible to create a real partitioning of the design by means of group definitions if one declares a group at the top of the design. Thus, there will be no mixing of instances attached to groups and instances attached to the default group. Usage: cluster\_constraint -command ADD\_GROUP -group\_name <string> -instance <list> [-exclusive <bool>] [-fnmatch <bool>] [-max\_fpga <int>] [-min\_fpga <int>] [-overflow reg=<percentage>,lut=<percentage>,ramlut=<percentage>,bram=<percentage>,dsp=<percentage>]  
Valid values: add\_group|ADD\_GROUP
  - **defcore:** Appends a list of instance paths to be handled as a core and processed altogether. Automatic-clustering will use these definitions to find the best partitioning of the design. Usage: cluster\_constraint -command DEFCORE -group\_name <string> -instance <list>

Valid values: defcore|DEFCORE

- **-exclusive** <bool> No blocks belonging to the default group can be mapped on this core, core is isolated. Default is false. The option is used with -command ADD\_GROUP..
- **-fnmatch** <bool> Specifies if the other parameters are interpreted as pattern. The default is false. The option is used with -command MERGE\_BLOCKS and -command ADD\_GROUP..
- **-group\_name** <string> Assigns a name to the group of instances. Mandatory option used with -command ADD\_GROUP and -command DEFCORE..  
Group name
- **-instance** <list> Specifies a list of instance paths to be grouped together. This group must fit on one FPGA. The option could be used with MERGE\_BLOCKS, ADD\_GROUP and DEFCORE -command option..
- **-level** <core|fpga|both> Specifies which tool will process the command core level partitioner (zTopBuild,) fpga level partitioner (zCoreBuild) or both. The default is both. The option could be used with -command MERGE\_BLOCKS and -command ADD\_GROUP..
  - **core**: Core level.  
Valid values: core|CORE
  - **fpga**: Fpga level.  
Valid values: fpga|FPGA
  - **both**: Both core and fpga level.  
Valid values: both|BOTH
- **-max\_fpga** <int> Constraint that specifies the maximum number of FPGAs allowed for this group. The option used with -command ADD\_GROUP..  
A non-negative integer number.
- **-min\_fpga** <int> Reserves at least this number of FPGA for the group. The option used with -command ADD\_GROUP..  
A non-negative integer number.
- **-overflow** <bram=int,dsp=int,lut=int,qiwc=int,ramlut=int,reg=int,uram=int>  
Specifies overflow percentage which is allowed for different types of logic elements. Available logic elements are: reg, lut, ramlut, bram, dsp. Defaults are respectively 5, 10, 10, 0, 0. The option used with -command ADD\_GROUP..
  - **bram** <int> bram overflow in %.  
A non-negative integer number.
  - **dsp** <int> dsp overflow in %.  
A non-negative integer number.
  - **lut** <int> lut overflow in %.  
A non-negative integer number.
  - **qiwc** <int> qiwc overflow in %.  
A non-negative integer number.
  - **ramlut** <int> ramlut overflow in %.  
A non-negative integer number.
  - **reg** <int> reg overflow in %.  
A non-negative integer number.
  - **uram** <int> uram overflow in %.  
A non-negative integer number.
- **-rtlname** Specifies if the input path is interpreted as RTLNAME. The default is false. The option is used with -command ADD\_GROUP..
- **-set\_max\_fill** <bram=int,dsp=int,fwc\_bit=int,fwc\_ip=int,lut=int,qiwc=int,ramlut=int,reg=int,uram=int,write\_port\_bit=int,write\_port\_ip=int,zcei\_mess\_in=int,zcei\_mess\_out=int> Specifies maximum FPGA filling rates for different types of logic elements that consume FPGA hardware resources. Available logic elements are: reg, lut, ramlut, bram, dsp, qiwc, fwc\_bit, fwc\_ip, read\_port\_bit, read\_port\_ip, write\_port\_bit, write\_port\_ip, zcei\_mess\_in, zcei\_mess\_out..

- **bram** <int> bram fillrate in %.  
A non-negative integer number.
- **dsp** <int> dsp fillrate in %.  
A non-negative integer number.
- **fwc\_bit** <int> fwc bit fillrate in %.  
A non-negative integer number.
- **fwc\_ip** <int> fwc ip fillrate in %.  
A non-negative integer number.
- **lut** <int> lut fillrate in %.  
A non-negative integer number.
- **qiwc** <int> qiwc fillrate in %.  
A non-negative integer number.
- **ramlut** <int> ramlut fillrate in %.  
A non-negative integer number.
- **read\_port\_bit** <int> read port bit fillrate in %.  
A non-negative integer number.
- **read\_port\_ip** <int> read port ip fillrate in %.  
A non-negative integer number.
- **reg** <int> reg fillrate in %.  
A non-negative integer number.
- **uram** <int> uram fillrate in %.  
A non-negative integer number.
- **write\_port\_bit** <int> write port bit fillrate in %.  
A non-negative integer number.
- **write\_port\_ip** <int> write port ip fillrate in %.  
A non-negative integer number.
- **zcei\_mess\_in** <int> zcei message in fillrate in %.  
A non-negative integer number.
- **zcei\_mess\_out** <int> zcei message out fillrate in %.  
A non-negative integer number.

#### Constraint(s)

- At least one of the options should be used
- -command can only be used with -instance
- -instance can only be used with -command
- -command DEFCORE option without setting -group\_name option
- -command DEFCORE with at least one on of: -fnmatch, exclusive, -min\_fpga, -max\_fpga and -overflow
- -command MERGE\_BLOCKS with at least one on of: -group\_name, exclusive, -min\_fpga, -max\_fpga and -overflow
- -command ADD\_GROUP option without setting -group\_name option

#### Note(s)

- -set\_max\_fill, -allow\_ovrd\_user\_max\_fill, -accept\_overflow options can be specified without -command option.

#### Example(s)

- cluster\_constraint -set\_max\_fill {reg = 90, lut = 50 }
- cluster\_constraint -set\_max\_fill qiwc=90,bram=70

## 2.36 clustering

### Description

Specify clustering related parameters

### Synopsis

```
clustering [-atomic_module <list>] [-auto_block_selection <bool>] [-core-
_adaptive_filling_rates <bool>] [-core_incremental_partitioning <bool>]
[-core_partitioning_goal <density|performance>] [-core_timing_optimization
<none|post_partitioning|partitioning_driven>] [-input_mapping_file <file>]
[-netlist_file_mode <gzfile|gzstream|stream>] [-output_mapping_file <file>]
[-resource_usage <Safe|Medium|Aggressive>] [-resource_usage_details <bram=string,dsp=
_bit=string,lut=string,lutram=string,qiwc=string,ramlut=string,reg=string,sdt=string,
_bits=string,uram=string>] [-routability_aware <bool>] [-routability-
_aware_params <effort_level=enum,num_threads=string>] [-system_auto_-
core_generation <bool>] [-system_auto_core_generation_advanced_command
<string>] [-system_auto_core_generation_advanced_file <file>] [-system-
_core_definition_file <file>] [-system_incremental_partitioning <bool>]
[-system_timing_optimization <none|post_partitioning|partitioning_driven>]
```

### Option(s)

- **-atomic\_module <list>** (legacy zTopBuild cluster model -atomic\_module=module\_name).
- **-auto\_block\_selection <bool>** Enable/disable Automatic Block Selection. This big grain hierarchy based partitioning model block selection improves the partitioner's speed, memory usage and the quality of result for the designs with a well structured hierarchical tree in its upper levels..
- **-core\_adaptive\_filling\_rates <bool>** Enable adaptive filling rates (legacy ZPF:Dop-CoreAdaptativeFR).
- **-core\_incremental\_partitioning <bool>** Enable incremental partitioning (legacy ZPF:CoreReuseExistingMapping).
- **-core\_partitioning\_goal <density|performance>** Specify core partitioning goal (legacy ZPF:CorePartitioningMode).
  - **density:** Density oriented zCore level partitioner.  
Valid values: `density|DENSITY`
  - **performance:** Performance oriented zCore level partitioner.  
Valid values: `performance|PERFORMANCE`
- **-core\_timing\_optimization <none|post\_partitioning|partitioning\_driven>** Specify core partitioning timing optimization mode (legacy ZPF:PopCoreTimingOptimizationMode).
  - **none:** No timing optimization.  
Valid values: `none|NONE`
  - **post\_partitioning:** Post partitioning timing optimization.  
Valid values: `post_partitioning|POST_PARTITIONING`
  - **partitioning\_driven:** Partitioning driven timing optimization.  
Valid values: `partitioning_driven|PARTITIONING_DRIVEN`
- **-input\_mapping\_file <file>** Specify filename with complete mapping. Mutually exclusive from all other options. (legacy ZPF:ClusterMode AutoRecover,ClusterRecoverMapping).
- **-netlist\_file\_mode <gzfile|gzstream|stream>** Specify modes for driving netlist file from zTopBuild to zTopClustering. .
  - **gzfile:** gzipped netlist file (default)  
Valid values: `gzfile|GZFILE`
  - **gzstream:** gzipped netlist file through a pipe  
Valid values: `gzstream|GZSTREAM`
  - **stream:** netlist file through a pipe  
Valid values: `stream|STREAM`

- **-output\_mapping\_file** <file> Specify output mapping file (legacy ZPF:ZCUI\_set Cluster-AutoOutputMapping).
- **-resource\_usage** <Safe|Medium|Aggressive> Specify resource utilization targets (legacy ZPF:ClusterMode,ClusterUtDspFillRate,ClusterUtLutFillRate,ClusterUtBramFillRate,ClusterUtRegisterFillRate,ClusterUtRamlutFillRate).
  - **Safe:** Low filling rates.  
Valid values: Safe|safe|SAFE
  - **Medium:** Standard filling rates.  
Valid values: Medium|medium|MEDIUM
  - **Aggressive:** High filling rates.  
Valid values: Aggressive|aggressive|AGGRESSIVE
- **-resource\_usage\_details** <bram=string,dsp=string,fwc\_bit=string,lut=string,lutram=\_bits=string,uram=string> Specify manual resource utilization targets. Available resources types are: dsp, bram, lut, reg, ramlut, uram, sdt, sdt\_bits and qiwc(legacy ZPF:ClusterMode,ClusterUtDspFillRate,ClusterUtLutFillRate,ClusterUtBramFillRate,ClusterUtRegisterFillRate,ClusterUtRamlutFillRate,ClusterUtUramFillRate,ClusterUtSdtFillRate,ClusterUtSdtBitsFillRate,ClusterUtQiWCFillRate).
  - **bram** <string> bram fillrate in %.  
bram fillrate
  - **dsp** <string> dsp fillrate in %.  
dsp fillrate
  - **fwc\_bit** <string> fwc bit fillrate in %.  
fwc bit fillrate
  - **lut** <string> lut fillrate in %.  
lut fillrate
  - **lutram** <string> ramlut fillrate in %.  
ramlut fillrate
  - **qiwc** <string> qiwc fillrate in %.  
qiwc fillrate
  - **ramlut** <string> ramlut fillrate in %.  
ramlut fillrate
  - **reg** <string> reg fillrate in %.  
reg fillrate
  - **sdt** <string> sdt fillrate in %.  
sdt fillrate
  - **sdt\_bits** <string> sdt bits fillrate in %.  
sdt bits fillrate
  - **uram** <string> uram fillrate in %.  
uram fillrate
- **-routability\_aware** <bool> Enable/disable routability aware clustering mode..
- **-routability\_aware\_params** <effort\_level=enum,num\_threads=string> Setting advanced routability analysis parameters values. Number of threads can be positive integer value or MAX for maximal available number of threads (default)..
  - **effort\_level** <LOW|MEDIUM|HIGH> Specify routing effort level.
    - \* **LOW:** low effort level  
Valid values: LOW|Low|low
    - \* **MEDIUM:** medium effort level  
Valid values: MEDIUM|Medium|medium
    - \* **HIGH:** high effort level  
Valid values: HIGH|High|high
  - **num\_threads** <string> can be positive integer value or MAX for maximal available number of threads (default)..  
<int> | MAX
- **-system\_auto\_core\_generation** <bool> Enable automatic zcore generation. Please see note (legacy ZPF:ClusterMode,AutoCoreDefinition).

- **-system\_auto\_core\_generation\_advanced\_command** <string> Specify advanced command for zTopClustering.  
zTopClustering command
- **-system\_auto\_core\_generation\_advanced\_file** <file> Specify advanced command file for automatic zcore generation (legacy ZPF:AutoCoreDefinitionFile).
- **-system\_core\_definition\_file** <file> Specify filename with zcore (defcore) definitions (legacy ZPF:ClusterMode,ZTopBuildCoreDefinitionFilename).
- **-system\_incremental\_partitioning** <bool> Enable incremental partitioning (legacy ZPF:CoreReuseExistingMapping).
- **-system\_timing\_optimization** <none|post\_partitioning|partitioning\_driven> Specify system partitioning timing optimization mode (legacy ZPF:AutoCoreTimingOptimizationMode).
  - **none**: No timing optimization.  
Valid values: none|NONE
  - **post\_partitioning**: Post partitioning timing optimization.  
Valid values: post\_partitioning|POST\_PARTITIONING
  - **partitioning\_driven**: Partitioning driven timing optimization.  
Valid values: partitioning\_driven|PARTITIONING\_DRIVEN

**Constraint(s)**

- At least one of the options should be used
- -input\_mapping\_file cannot be used with any of the following options: -system\_auto\_core\_generation, -system\_auto\_core\_generation\_advanced\_file, -system\_auto\_core\_generation\_advanced\_command, -system\_timing\_optimization, -system\_incremental\_partitioning, -system\_core\_definition\_file, -resource\_usage, -resource\_usage\_details, -core\_partitioning\_goal, -core\_adaptive\_filling\_rates, -core\_incremental\_partitioning, -core\_timing\_optimization
- -resource\_usage cannot be used with any of the following options: -resource\_usage\_details

**Note(s)**

- To enable -system\_auto\_core\_generation\_advanced\_file \<file\> please use -system\_auto\_core\_generation t
- -system\_auto\_core\_generation\_advanced\_file \<file\> option is valid when -system\_auto\_core\_generation \<bool\> mode option is enabled, or when design\_size is greater than one module.

**Example(s)**

- clustering -resource\_usage\_details { lut =60, reg= 30, dsp = 50 }
- clustering -resource\_usage\_details lut=60,dsp=50,fvt=75

## 2.37 code\_coverage

### Description

This command is used to enable code coverage. If enabled, the default state is that force/release statements are honored. If disabled, the default state is that force/release statements are ignored.

### Synopsis

```
code_coverage [-line <bool>] [-toggle <bool>]
```

### Option(s)

- **-line** <bool> Enable line coverage..
- **-toggle** <bool> Enable toggle coverage..

### Constraint(s)

- At least one of the options should be used

## 2.38 compile

### Description

This command specifies compile related options.

### Synopsis

```
compile [-adv] [-design_features <file>] [-edge_alignment_mode <int>]
[-fractal <bool>] [-hydra <bool>] [-incremental <bool>] [-incremental_
params <netlist_edition=bool,partition=bool,pnr=bool,synthesis=bool,system_
_route=bool>] [-objective <None|Fast_Turn_Around|Bringup|Performance|
Advance_Performance|Capacity>] [-sdc <file>] [-sem <bool>] [-sem_params
<build_flags=string,mode=enum>]
```

### Option(s)

- **-adv** this is the documentation of -adv option.
- **-design\_features <file>** Specify name of designFeatures file..
- **-edge\_alignment\_mode <int>** clock\_delay based support for SDC MCP computation Only valid for ZS4 onwards. 0 - Without any tolerance optimization 1 - Default clock\_delay tolerance optimization 2 - Tolerance with fastest clock period tolerance window 3 - Tolerance with edges aligned with the fastest clock (deprecated) 4 - Tolerance with edges aligned with the posedge of the fastest clock (deprecated) 5 - Clock group based tolerance with edges aligned with the posedge of the fastest clock (deprecated).  
A non-negative integer number.
- **-fractal <bool>** enable/disable fractal flow.
- **-hydra <bool>** enable/disable hydra flow.
- **-incremental <bool>** enable/disable incremental compile.
- **-incremental\_params <netlist\_edition=bool,partition=bool,pnr=bool,synthesis=bool,\_route=bool>** Specify incremental parameters. Available parameter types are: synthesis, partition, system\_route, pnr, netlist\_edition.
  - **netlist\_edition <bool>** enable/disable advanced incremental netlist mode.
  - **partition <bool>** enable/disable partition incremental compile.
  - **pnr <bool>** enable/disable P&R incremental compile.
  - **synthesis <bool>** enable/disable synthesis incremental compile.
  - **system\_route <bool>** enable/disable system\_route incremental compile.
- **-objective <None|Fast\_Turn\_Around|Bringup|Performance|Advance\_Performance|Capacity>** define compilation objective.  
objective type
- **-sdc <file>** Specify name of sdc file..
- **-sem <bool>** enable/disable sem flow.
- **-sem\_params <build\_flags=string,mode=enum>** Specify SEM flow parameters.
  - **build\_flags <string>** VCS additional build flag.  
flag values
  - **mode <CYCLE|BEHAVIOR>** define SEM mode.  
mode

### Constraint(s)

- At least one of the options should be used
- -sdc can only be used with -objective Advance\_Performance
- -design\_features can only be used with -objective Advance\_Performance
- -edge\_alignment\_mode can only be used with -objective Advance\_Performance

- -adv can only be used with -objective Advance\_Performance
- -sdc cannot be used with any of the following options: -design\_features

**Example(s)**

- compile -incremental true
- compile -incremental true -incremental\_params {synthesis=false,partition=true,system\_route=false,pnr=true,netlist\_edition=true}

## 2.39 coverage

### Description

This command is used to enable and disable coverage collection feature.

### Synopsis

```
coverage [-auto_bin_max <int>] [-enable <bool>]
```

### Option(s)

- **-auto\_bin\_max** <int> Set global auto bin max value..  
A non-negative integer number.
- **-enable** <bool> Enables or disables the tracking in the compilation..

### Constraint(s)

- At least one of the options should be used

## 2.40 debug

### Description

Debug options

### Synopsis

```
debug <CEL_FILE> <LIST_OF_CEL_FILES> <LIST_OF_SIGNALS_FILE> <SIGNALS-
_FILE> <TRACE_BUFFER_CONTROL> [-add_clock_counters <bool>] [-advanced-
_command_file <file>] [-always_accessible <file>] [-dumpports_maxbits
<int>] [-dumpvars_maxbits <int>] [-enable_bram_rw <bool>] [-encryption-
_support <bool>] [-memory_backdoor <all|readmem|none|selected>] [-memory-
_backdoor_instances <list>] [-notifier <asn|uip|false>] [-notifier_-
params <asn_max_bits=int,asn_max_fanins=int,asn_mode=enum,asn_post_-
compilation_cel=file,uip_cel_file_list=file,uip_msg_cnl=enum,uip_opt=enum,uip-
_preserve_bits=int,uip_signal_file_list=file>] [-offline_debug <bool>]
[-offline_debug_params <incl_xtors=bool>] [-rtlname <list>] [-trigger
<bool>] [-trigger_params <cel_file=file,list_of_cel_files=file,list-
_of_signals_file=file,signals_file=file,trace_buffer_control=string>]
[-verdi_db <true|false|old>] [-waveform_reconstruction <bool>] [-waveform-
_reconstruction_params <csa=bool,fsdb=bool,partitions=string,simzilla=enum,simzilla-
_distributed_build=bool,simzilla_forward_saif=file,swave=enum,threads=int>]
```

### Option(s)

- **CEL\_FILE** <file> Specify the CEL (complex event language) file..
- **LIST\_OF\_CEL\_FILES** <file> Specify the CEL (complex event language) file list. The file format of the list is one CEL file path per line..
- **LIST\_OF\_SIGNALS\_FILE** <file> Specify the list of signal files. The file format of the list is one signal file path per line..
- **SIGNALS\_FILE** <file> Specify the SIGNALS file..
- **TRACE\_BUFFER\_CONTROL** <string> enable SABRE flush.  
flush
- **-add\_clock\_counters** <bool> Add clock counters for debugging glitches. (legacy ZPF:Clock-CounterFlag).
- **-advanced\_command\_file** <file> Command file for zDbPostProc (e.g. importing memories). (legacy ZPF:ZDBPostprocCommandFilename).
- **-always\_accessible** <file> File with dynamic probes that should always be accessible (legacy ZPF:DynamicProbeFileName)..
- **-dumpports\_maxbits** <int> Specify max number of bits allowed to be dumped per a single dumpports command..  
A non-negative integer number.
- **-dumpvars\_maxbits** <int> Specify max number of bits allowed to be dumped per a single dumpvars command..  
A non-negative integer number.
- **-enable\_bram\_rw** <bool> Enable BRAM read/write logging. Only required for offline debugging with RLDRAM instantiations. (legacy ZPF:GlobalBramRWFlag).
- **-encryption\_support** <bool> Enable/disable encryption support in RhinoDB..
- **-memory\_backdoor** <all|readmem|none|selected> Add backdoor access for memories in Taurus flow..
  - **all**: add backdoor access to all memory instances in the netlist.  
Valid values: `all`|`ALL`
  - **readmem**: add backdoor access to memory instances related to \$readmem in design  
Valid values: `readmem`|`READMEM`

- **none**: add backdoor access to all memory instances in the netlist.  
Valid values: `none|NONE`
- **selected**: add backdoor access to memory instances specified by '-memory\_backdoor\_instances' option.  
Valid values: `selected|SELECTED`
- **-memory\_backdoor\_instances <list>** Specify memory instances for backdoor access enabling (when "-memory\_backdoor selected" option is used). Wildcard name is supported. Can not accept escape name. If you really want to use escape name, use underscore instead of backslash. For example, use '123mem' instead of 'Smem'.
- **-notifier <asn|uip|false>** Enable/Disable the notifier and debug dumping..
  - **asn**: The Notifier engine integrated. The list of signal are applied to the Notifier engine with the ECO flow.  
Valid values: `asn|ASN`
  - **uip**: The Notifier engine is integrated with a 1st list of signals. They are provided with a text file or with an FSM described with CEL language. After compilation, the list of signals can be updated thanks to the \$dumpvars task call definitions applied at compile time. Below UIP parameters are available.  
Valid values: `uip|UIP`
  - **false**: resets all previous setting of notifier and its params
- **-notifier\_params <asn\_max\_bits=int,asn\_max\_fanins=int,asn\_mode=enum,asn\_post\_compilation\_cel=file,uip\_cel\_file\_list=file,uip\_msg\_cnl=enum,uip\_opt=enum,uip\_preserve\_bits=int,uip\_signal\_file\_list=file>** notifier params {param1=val1, param2=val2, ..., paramN=valN>}.
  - **ASN\_MAX\_BITS <int>** Sets the bus width to specified size..  
A non-negative integer number.
  - **ASN\_MAX\_FANINS <int>** Enables to set the maximal fanin size Notifier Engines..  
A non-negative integer number.
  - **ASN\_MODE <pipeline|accurate>** Defines the optimization target notifier collector tree..
    - \* **pipeline**: Reduce as much as possible frequency penalty. I.e try to avoid congested areas.  
Valid values: `pipeline|PIPELINE`
    - \* **accurate**: Reduce the maximal depth of the tree, i.e optimize the delay of notifier tree.  
Valid values: `accurate|ACCURATE`
  - **ASN\_POST\_COMPILATION\_CEL <file>** Enables user to set a list of signals he wants to track already when launching zebu..
  - **UIP\_CEL\_FILE\_LIST <file>** Specify the CEL (complex event language) file list for notifier.-  
The file format of the list is one CEL file path per line..
  - **UIP\_MSG\_CNL <1|2|4>** Specify the message pair number for notifier. Default is 1..
  - **UIP\_OPT <capacity|balanced|performance>** Specify the optimization strategy for debug dump..
    - \* **capacity**: allocate the largest number of debug resources.  
Valid values: `capacity|CAPACITY`
    - \* **balanced**: allocate the resource to balanced performance and capacity (default)  
Valid values: `balanced|BALANCED`
    - \* **performance**: allocate the resource for best performance.  
Valid values: `performance|PERFORMANCE`
  - **UIP\_PRESERVE\_BITS <int>** Preserve the debug resources for ECO..  
A positive integer number.
  - **UIP\_SIGNAL\_FILE\_LIST <file>** Specify the list of signal files for notifier. The file format of the list is one signal file path per line. And the format of signal file is one signal per line..
- **-offline\_debug <bool>** Insert sniffer infrastructure for Post-Run Debug. (legacy ZPF:UseOfflineDebugFlag).
- **-offline\_debug\_params <incl\_xtors=bool>** offline debug params (comma separated list of name=value pairs).
  - **INCL\_XTORS <bool>** Enable/disable the new zPRD. (when offline\_debug is enabled).

- **-rtlname** <list> Explicitly specify which top ports to provide debug visibility for..
- **-trigger** <bool> Enables FSM Based HW Trigger Technology. To be used with -trigger\_params..
- **-trigger\_params** <cel\_file=file,list\_of\_cel\_files=file,list\_of\_signals\_file=file,signals\_file=file,trace\_buffer\_control=string> Signals used by FSM Based HW Trigger..
  - **CEL\_FILE** <file> Specify the CEL (complex event language) file..
  - **LIST\_OF\_CEL\_FILES** <file> Specify the CEL (complex event language) file list.The file format of the list is one CEL file path per line..
  - **LIST\_OF\_SIGNALS\_FILE** <file> Specify the list of signal files.The file format of the list is one signal file path per line..
  - **SIGNALS\_FILE** <file> Specify the SIGNALS file..
  - **TRACE\_BUFFER\_CONTROL** <string> enable SABRE flush.  
flush
- **-verdi\_db** <true|false|old> Control VCS/Verdi flow..
  - **true**: Enable Post elab verdi flow  
Valid values: true|TRUE|1|True|t|T|y|Y|yes|YES
  - **false**: Disable verdi flow  
Valid values: false|FALSE|0|False|f|F|n|N|no|NO
  - **old**: Enable vcs verdi flow  
Valid values: old|OLD
- **-waveform\_reconstruction** <bool> Enable/disable all waveform reconstruction parameters..
- **-waveform\_reconstruction\_params** <csa=bool,fsdb=bool,partitions=string,simzilla=en\_ \_distributed\_build=bool,simzilla\_forward\_saif=file,swave=enum,threads=int>  
Enable/disable a series of debug parameters.
  - **CSA** <bool> .
  - **FSDB** <bool> .
  - **PARTITIONS** <string> number of graph partitions.  
integer value or auto
  - **SIMZILLA** <t|f|SEQUENTIAL> .  
ENABLE|DISABLE|SEQUENTIAL
    - \* **t**: Enable SIMZILLA  
Valid values: t|enabled|Enabled|ENABLED|true|True|TRUE
    - \* **f**: Disable SIMZILLA  
Valid values: f|disabled|Disabled|DISABLED|false|False|FALSE
    - \* **SEQUENTIAL**: Enable Sequential SIMZILLA  
Valid values: SEQUENTIAL|sequential|Sequential
  - **SIMZILLA\_DISTRIBUTED\_BUILD** <bool> .
  - **SIMZILLA\_FORWARD\_SAIF** <file> .
  - **SWAVE** <enabled|disabled|gls|ENABLE\_SEQ> .
    - \* **enabled**: Enable SWave  
Valid values: enabled|Enabled|ENABLED|true|True|TRUE
    - \* **disabled**: Disable SWave  
Valid values: disabled|Disabled|DISABLED|false|False|FALSE
    - \* **gls**: Enable SWave in GLS mode  
Valid values: gls|Gls|GLS
    - \* **ENABLE\_SEQ**: Enable Sequential SWave  
Valid values: ENABLE\_SEQ|enable\_seq|enable\_sequential
  - **THREADS** <int> number of threads for simzilla.  
A positive integer number.

#### Constraint(s)

- At least one of the options should be used

- -all cannot be used with any of the following options: -offline\_debug, -verdi\_db
- -rtlname option must be specified with -enable\_hwtop\_ports\_access select
- -memory\_backdoor\_instances cannot include escaped names
- -add\_clock\_counters is deprecated command and not supported now.

**Note(s)**

- -offline\_debug will also implicitly enable csa and debug -set\_dyn\_probes\_on\_dut\_ios t commands.
- ASN and UIP are mutual exclusive
- ASN(UIP) parameters can be used only when ASN(UIP) notifier is set.

**Example(s)**

- debug -notifier UIP -notifier\_params {UIP\_SIGNAL\_FILE\_LIST=./signal.lst, UIP\_OPT=CAPACITY, UIP\_CEL\_FILE\_LIST=cel.lst, UIP\_MSG\_CNL=1, UIP\_PRESERVE\_BITS=128}
- debug -notifier ASN -notifier\_params {ASN\_MODE=PIPELINE, ASN\_POST\_COMPILATION\_CEL=cel.-txt, ASN\_MAX\_FANINS=500}

## 2.41 design\_size

### Description

Specify number of modules to be used

### Synopsis

```
design_size [-max_number_of_modules <float>] [-mode <AUTO|FULL>] [-number-
_of_modules <float>] [-resource_margin <float>] [-resource_margin_-
details <bram=float,dsp=float,fwc_bit=float,fwc_ip=float,lut=float,lut6=float,qiwc-
_bit=float,ramlut=float,read_ports_bits=float,read_ports_ips=float,reg=float,uram=flo
_ports_bits=float,write_ports_ips=float,zc_trace_bits=float,zcei_mess-
_in=float,zcei_mess_out=float,zrm_ports=float,zrm_slots=float,zview_-
bits=float>]
```

### Option(s)

- **-max\_number\_of\_modules <float>** Represents the maximum number of modules can be used in case of AUTO mode. Should be a positive integer, 0.25 or 0.5. When compiling for ZS4, ZS5 HW it is possible to specify 0.5 to map the design on a half-module. When compiling for ZS5 HW it is possible to specify 0.25 to map the design on a quarter-module. .  
A positive float number.
- **-mode <AUTO|FULL>** Specifies design\_size configuration mode.
  - **AUTO**: computes the minimal number of boards required  
Valid values: AUTO|Auto|auto
  - **FULL**: full target  
Valid values: FULL|Full|full
- **-number\_of\_modules <float>** Specifies maximum number of modules to be used. Should be a positive integer, 0.25 or 0.5. When compiling for ZS4, ZS5 HW it is possible to specify 0.5 to map the design on a half-module. When compiling for ZS5 HW it is possible to specify 0.25 to map the design on a quarter-module. .  
A positive float number.
- **-resource\_margin <float>** Specifies margin for design size estimation .  
A positive float number.
- **-resource\_margin\_details <bram=float,dsp=float,fwc\_bit=float,fwc\_ip=float,lut=float,lut6=float,qiwc\_bit=float,ramlut=float,read\_ports\_bits=float,read\_ports\_ips=float,reg=float,uram=float,write\_ports\_bits=float,write\_ports\_ips=float,zc\_trace\_bits=float,zcei\_mess\_in=float,zcei\_mess\_out=float,zrm\_ports=float,zrm\_slots=float,zview\_bits=float>** Specify margin for design size estimation for the specified resource type Available resources types are: lut, lut6, bram, reg, dsp, ramlut, fwc\_bit, fwc\_ip, qiwc\_read\_port\_bits, write\_port\_bits, zcei\_mess\_out, zc\_trace\_bits, zview\_bits, zrm\_slots, zrm\_ports..
  - **bram <float>** bram fillrate in %.  
A positive float number.
  - **dsp <float>** dsp fillrate in %.  
A positive float number.
  - **fwc\_bit <float>** fwc\_bit fillrate in %.  
A positive float number.
  - **fwc\_ip <float>** ramlut fillrate in %.  
A positive float number.
  - **lut <float>** lut fillrate in %.  
A positive float number.
  - **lut6 <float>** lut6 fillrate in %.  
A positive float number.

- **qiwc\_bit** <float> qiwc\_bit fillrate in %.  
A positive float number.
- **ramlut** <float> ramlut fillrate in %.  
A positive float number.
- **read\_ports\_bits** <float> read\_port\_bits fillrate in %.  
A positive float number.
- **read\_ports\_ips** <float> read\_port\_ips fillrate in %.  
A positive float number.
- **reg** <float> reg fillrate in %.  
A positive float number.
- **uram** <float> bram fillrate in %.  
A positive float number.
- **write\_ports\_bits** <float> write\_port\_bits fillrate in %.  
A positive float number.
- **write\_ports\_ips** <float> write\_port\_ips fillrate in %.  
A positive float number.
- **zc\_trace\_bits** <float> zc\_trace\_bits fillrate in %.  
A positive float number.
- **zcei\_mess\_in** <float> zcei\_mess\_in fillrate in %.  
A positive float number.
- **zcei\_mess\_out** <float> zcei\_mess\_out fillrate in %.  
A positive float number.
- **zrm\_ports** <float> zrm\_ports fillrate in %.  
A positive float number.
- **zrm\_slots** <float> zrm\_slots fillrate in %.  
A positive float number.
- **zview\_bits** <float> zview\_bits fillrate in %.  
A positive float number.

#### Constraint(s)

- -number\_of\_modules cannot be used with any of the following options: -mode
- -number\_of\_modules cannot be used with any of the following options: -max\_number\_of\_modules
- At least one of the options should be used

#### Note(s)

- -mode option could not be used with -number\_of\_modules option.
- -mode AUTO does not imply the automatic core partitioning. It must be enabled explicitly by 'clustering -system\_auto\_core\_generation true'

#### Example(s)

- design\_size -resource\_margin\_details {lut=20}
- design\_size -resource\_margin 15

## 2.42 disk\_space\_checker

### Description

Disk space checker options

### Synopsis

```
disk_space_checker -enable <bool> [-command <string>] [-constants <core=float,elab=
_fpga=float,synth=float,system=float>] [-constants_mode <AUTO|MANUAL>]
[-email_address <string>] [-email_address_batch <string>] [-email_
notification <NONE|BATCH|GUI|ALL>] [-log_notification <NONE|BATCH|GUI|ALL>]
[-warning_box_notification <bool>]
```

### Option(s)

- **-command <string>** Shell command which will be executed in case there is no more disk space.  
Shell command to be executed
- **-constants <core=float,elab=float,pp\_fpga=float,synth=float,system=float>**  
Relevant only in MANUAL constants mode. Assign a positive floating point value to a compilation step constant. Constant value can be: elab, synth, system, core or pp\_fpga.
  - **core <float>** core.  
A non-negative float number.
  - **elab <float>** elab.  
A non-negative float number.
  - **pp\_fpga <float>** pp\_fpga.  
A non-negative float number.
  - **synth <float>** synth.  
A non-negative float number.
  - **system <float>** system.  
A non-negative float number.
- **-constants\_mode <AUTO|MANUAL>** in AUTO mode -constants options shouldn't be provided.  
Default: AUTO.
- **-email\_address <string>** Indicates the email address used to inform the user about the missing disk space. Required if -email\_notification==BATCH|ALL.  
Email address used to inform the user about
- **-email\_address\_batch <string>** Indicates the email address used to inform the user about the missing disk space. Required if -email\_notification==BATCH|ALL.  
Email address used to inform the user about
- **-email\_notification <NONE|BATCH|GUI|ALL>** Enables the notification by email about the missing disk space in batch/gui or both modes, NONE disables it. Default: NONE.
- **-enable <bool>** Activates or deactivates disk space checker functionality. This option is mandatory.. **Mandatory option**
- **-log\_notification <NONE|BATCH|GUI|ALL>** Enables the notification in log file about the missing disk space in batch/gui or both modes, NONE disables it. Default: NONE.
- **-warning\_box\_notification <bool>** Activates a warning box to notify the user about the missing disk space. Default: false..

### Constraint(s)

- At least one of the options should be used

### Note(s)

- elab, synth: Common to all backends. Can not be used inside create\_new\_target blocks.
- system, core, pp\_fpga: backend settings applied on current target context.

## 2.43 eco

### Description

Specify eco options

### Synopsis

```
eco [-advanced_command <string>] [-advanced_command_file <file>] [-change-
_not_feasible <DEFAULT|FATAL|WARNING>] [-depth <int>] [-disable_alias]
[-disable_constant_prop] [-disable_remove_loadless] [-enable <bool>]
[-force_add_equiv_signals <bool>] [-frontend_protect <bool>] [-ignore-
_limit] [-no_flatten] [-recompile_on_error <NEVER|ON_OVERWRITE_ONLY>]
[-report_driver] [-reserve_design_change <int>] [-reserve_fan_inout <int>]
[-reserve_force <int>] [-reserve_force_assign <int>] [-reserve_fwc <int>]
[-reserve_monitor <int>] [-reserve_ports] [-reserve_qiwc <int>] [-reserve-
_scope <list>] [-reserve_signal_file <file>] [-reserve_trigger <int>]
[-reserve_xtor <int>] [-skip_checking <bool>] [-valueset_fwc <string>]
[-valueset_qiwc <string>]
```

### Option(s)

- **-advanced\_command <string>** Specify any zECO command.  
zECO command
- **-advanced\_command\_file <file>** Specify advanced command file for zECO.
- **-change\_not\_feasible <DEFAULT|FATAL|WARNING>** Set the behavior when signal change cannot be handled in ECO compilation..
- **-depth <int>** Set the depth for reserve scope.  
A non-negative integer number.
- **-disable\_alias** Disable optimization for alias..
- **-disable\_constant\_prop** Disable constant propagation..
- **-disable\_remove\_loadless** Disable remove loadless..
- **-enable <bool>** ignore previous eco commands..
- **-force\_add\_equiv\_signals <bool>** Enable/disable force adding equivalent signals..
- **-frontend\_protect <bool>** Full scope dump\_any zebu protection in front end..
- **-ignore\_limit** Ignore resource limitation check..
- **-no\_flatten** Disable flatten in zNetgen..
- **-recompile\_on\_error <NEVER|ON\_OVERWRITE\_ONLY>** Set the recompile on error mode..
- **-report\_driver** Enable fanin driver debug dump..
- **-reserve\_design\_change <int>** Reserve design change number for ECO.  
A non-negative integer number.
- **-reserve\_fan\_inout <int>** Reserve fan inout for ECO..  
A non-negative integer number.
- **-reserve\_force <int>** Reserve force bit number in each chip..  
A non-negative integer number.
- **-reserve\_force\_assign <int>** Reserve force assign number for ECO.  
A non-negative integer number.
- **-reserve\_fwc <int>** Reserve fwc IP bit number in each chip..  
A non-negative integer number.
- **-reserve\_monitor <int>** Reserve monitors in each chip..  
A non-negative integer number.
- **-reserve\_ports** Reserve all ports for the design..

- **-reserve\_qiwc** <int> Reserve qiwc bit number in each chip..  
A non-negative integer number.
- **-reserve\_scope** <list> Specify the list of modules that will reserve resources for ECO compilations.
- **-reserve\_signal\_file** <file> Specify signal file to be protected for ECO.
- **-reserve\_trigger** <int> Reserve trigger number for ECO..  
A non-negative integer number.
- **-reserve\_xtor** <int> Reserve xtor size for xtor based force assign..  
A non-negative integer number.
- **-skip\_checking** <bool> Enable/disable skip checking..
- **-valueset\_fw** <string> Specify the fw valueset name for ECO..  
Default is ZECO\_PRESERVE\_FW.
- **-valueset\_qiwc** <string> Specify the qiwc valueset name for ECO..  
Default is ZECO\_PRESERVE\_QIWC.

#### Constraint(s)

- check illegal option combinations

## 2.44 encrypted\_mod

### Description

This command is now deprecated.

### Synopsis

```
encrypted_mod [-delete_readback <bool>] [-encrypt_v2 <bool>] [-mangle
<bool>]
```

### Option(s)

- **-delete\_readback** <bool> Enables encryption flow deletion of readback database.
- **-encrypt\_v2** <bool> Enables encryption version 2 flow.
- **-mangle** <bool> Enables encryption flow based on mangle.

### Constraint(s)

- At least one of the options should be used

## 2.45 enhanced\_write\_access

### Description

set the write access configuration with given scope information. Selected signals will be protected as forceable widely.

### Synopsis

```
enhanced_write_access -application <Force|Eco> [-depth <int>] [-instance
<string>] [-module <string>] [-type <list>]
```

### Option(s)

- **-application <Force|Eco>** Indicates the method to keep write\_access on signals.  
type {Force | Eco} **Mandatory option**
  - **Force:** 'Force' Indicates that the selected signals will be forceable for write accessing. Should be set with -type {low\_cost port sequential all}  
Valid values: Force|force|FORCE
  - **Eco:** 'Eco' Indicates that writability will not be kept by inserting additional IP for forceability. Keep the writability as ECOableShould be set with -type {constant undriven unused alias}  
Valid values: Eco|eco|ECO
- **-depth <int>** Specifies writability depth..  
An integer number.
- **-instance <string>** Specifies the name of the instance under which the signals are forced. .  
string
- **-module <string>** Specifies the name of the module under which the signals are forced. .  
string
  - **warning:** A warning is displayed instead of an error.  
Valid values: warning|WARNING|Warning
  - **fatal:** A fatal error is displayed.  
Valid values: fatal|FATAL|Fatal
- **-type <list>** specified the protection target and method.

### Constraint(s)

- At least one of the options should be used
- option -module cannot be used with option -uniquify

## 2.46 fault\_emulation

### Description

This command indicates the fault emulation database and the campaign name.

### Synopsis

```
fault_emulation [-campaign <string>] [-dut_path <string>] [-fdb_name
<string>] [-fdb_server <string>] [-object_not_found <string>]
```

### Option(s)

- **-campaign <string>** Specify the campaign name..  
Specify the campaign name.
- **-dut\_path <string>** Specify the DUT path..  
Specify the DUT path.
- **-fdb\_name <string>** Specify the fault emulation database name..  
Specify the fault emulation database name.
- **-fdb\_server <string>** Specify the fault emulation database server..  
Specify the fault emulation database server.
- **-object\_not\_found <string>** Specify the message severity for not found objects. Default is fatal..  
Specify the message severity for not found objects. Default is fatal.

### Constraint(s)

- At least one of the options should be used

### Example(s)

- `fault_emulation -fdb_server <host name>:80 -fdb_name FDB1 -campaign fc_1`

## 2.47 force

### Description

Force wires and ports

### Synopsis

```
force <mode> [-assign_as_edit <bool>] [-disconnect] [-fnmatch] [-global-
Verbose <bool>] [-hier_sep <string>] [-local_driver_check] [-module
<string>] [-net <string>] [-no_power] [-object_not_found <fatal|warning>]
[-only_if <list>] [-out <string>] [-pin <string>] [-pin_input <string>]
[-pin_output <string>] [-reg_init <string>] [-rtlname <string>] [-rtlname-
_input <string>] [-rtlname_output <string>] [-source_dve <string>] [-source-
_net <string>] [-source_pin <string>] [-source_rtlname <string>] [-value
<string>] [-verbose]
```

### Option(s)

- **mode** <assign|config|default|undriven> Force command modes.  
mode value **Mandatory option**
  - **assign**: 'assign' mode forces the value of a pin or a net to be equal to some other pin, net or constant. In this mode the older driver(s) of the net get disconnected but for tristate buses  
Valid values: assign|Assign|ASSIGN
  - **config**: 'config' mode is used to process all commands together or one by one  
Valid values: config|Config|CONFIG
  - **default**: 'default' mode is used to allow change of the global verbosity level  
Valid values: default|Default|DEFAULT
  - **undriven**: 'undriven' mode is used to set the value of undriven nets  
Valid values: undriven|Undriven|UNDRIVEN
- **-assign\_as\_edit** <bool> If value is 'yes', process commands one by one. if the value is 'no', process all commands together. This option is deprecated..
- **-disconnect** Disconnect ALL drivers of the source (net or pin) before force processing. This option is unactivated by default. By default we perform automatic disconnection except when both source and target objects are tristate or when source object is not driven. This option is deprecated..
- **-fnmatch** Matches hierarchical names using fnmatch patterns.
- **-globalVerbose** <bool> Report disconnected/connected DRIVERS for ALL 'force commands'.
- **-hier\_sep** <string> Specifies hierachy separator used in hierarchical names.  
string
- **-local\_driver\_check** Only local connection of target wires are explored to perform necessary disconnection.
- **-module** <string> Module name for module relative force assign. Supported switches for this mode are -rtlname/-rtlname\_input/rtlname\_output/-pin/-pin\_input/-pin\_output/-net for target signals, -source\_rtlname/-source\_pin/-source\_net for source signals, -value [0/1/REG], -reg\_init and -disconnect. No hierarchical names are allowed for target/source signals, and -fnmatch is only allowed in -value. -fnmatch would result in fatal error if used with target/source signals. In case of module name conflict between several libraries, the library name can be prepended (library name name)..  
string
- **-net** <string> Path of net to force. This options is deprecated.  
string
- **-no\_power** Propagate constant throughout power domains.
- **-object\_not\_found** <fatal|warning> Specify error policy in case of objects cannot be forced.  
policy value

- **fatal**: In case object cannot be found, a fatal error is emitted. This is the default  
Valid values: `fatal|Fatal|FATAL`
- **warning**: In case object cannot be found, a warning is emitted instead of an error  
Valid values: `warning|Warning|WARNING`
- **-only\_if <list>** Specifies driver types to remove when 'value' option is set to REG. This option is deprecated.
- **-out <string>** Choose log output {logfile / file / report}. This option is deprecated.  
string
- **-pin <string>** Specifies a path of instance port. The hierarchical net connected to the port is forced. This options is deprecated.  
string
- **-pin\_input <string>** Specifies a path of instance port. The signal previously connected to the pin is disconnected, then the pin is forced. The signal down the hierarchy of pin is forced. This options is deprecated.  
string
- **-pin\_output <string>** Specifies a path of instance port. The signal up the hierarchy of the pin is forced. This options is deprecated.  
string
- **-reg\_init <string>** Init of REG [0|1]. This option is used only when option value equals REG.  
string
- **-rtlname <string>** Specifies a RTL path of instance port. The hierarchical net connected to the port is forced.  
string
- **-rtlname\_input <string>** Specifies a RTL path of instance port. The signal previously connected to the pin is disconnected, then the pin is forced. The signal down the hierarchy of pin is forced.  
string
- **-rtlname\_output <string>** Specifies a RTL path of instance port. The signal up the hierarchy of the pin is forced.  
string
- **-source\_dve <string>** DVE signal name used as source in force assign. This option is deprecated.  
string
- **-source\_net <string>** Path of net used as source in force assign. This options is deprecated.  
string
- **-source\_pin <string>** Path of instance port used as source in force assign. This options is deprecated.  
string
- **-source\_rtlname <string>** RTL Path of instance port or net used as source in force assign.  
string
- **-value <string>** Value of force nets/pins[0|1|REG].  
string
- **-verbose** Report disconnected/connected DRIVERS for ONLY THIS 'force command'.

#### Constraint(s)

- At least one of the options should be used

#### Example(s)

- Assigning the value of dut.a.b.wirename to dut.c.d.wirename (original drivers of dut.c.d.wirename are always disconnected) force assign -source\_net dut.a.b -net dut.c.d -disconnect

- Assigning binary value 1 to all 16 bits (equivalent to {16'b1111111111111111}) of dut.instance.in port vector force assign -pin {dut.instance.in[15:0]} -value 1
- Assigning decimal value 8 (equivalent to {16'b0000000000001000}) to dut.instance.in port vector force assign -pin {dut.instance.in[15:0]} -value 8
- Assigning 16 bits binary value FACE, to dut.instance.in port vector, with runtime programmable registers, initialized with corresponding bitfield force assign -pin {dut.instance.in[15:0]} -value REG -reg\_init {16'b1111101011001110}
- Assigning 32 bits hexadecimal value BABEFACE, to dut.instance.in port vector force assign -pin {dut.instance.in[31:0]} -value {32'hBABEFACE}
- Assigning binary value 1 to the least significant bit of dut.instance.in port vector and all other bits are forced to 0 force assign -pin {dut.instance.in[31:16]} -value 'b1
- force default force default -globalVerbose yes
- Find undriven nets and ports in the design and force them with a runtime programmable register. The number of forced objects (result returned by the command) is written on standard output puts "undriven count : [force undriven -value REG -reg\_init 1]"
- Module relative force assign: (1) Multidimensional indexing: force assign -module mod -rtlname y -source\_rtlname x[0][0] -disconnect (2) Wildcard matching: force assign -module mod\* -rtlname sig\* -value 1'b1 -fnmatch (3) Vector assignment: force assign -module mod -rtlname sig\_vec -value REG -reg\_init 8'b1110\_0001 (4) Inside generate block: force assign -module mod -net genblk[0].sig -source\_pin in (5) List of target signals with -net/-pin: force assign -module mod -net {x y z} -source\_net w Note that -rtlname and the source side cannot accept a list of signals. Both will trigger an error. (6) Part selection: force assign -module mod -rtlname x[1:0] -source\_net y[1:0] Note that there should be only one part select per signal (e.g. x[1:0][1:0] is not allowed), and the part select should be at the end of the signal name (e.g. x[1:0].y is not allowed).

## 2.48 fpga

### Description

Specify fpga PAR options

### Synopsis

```
fpga [-advanced_command_file <file>] [-complexity_predictor <bool>]
[-complexity_predictor_bundle_size <int>] [-debug_policy <Disabled|Enabled|Expert>] [-file_policy <Regular|Compress|Suppress>] [-first_par_policy <Original|Winner|Both|BothAnyway>] [-inter_die_tdm <bool>] [-inter_die_tdm_params <strategy=enum,time_budgets=bool>] [-local_disk_path <string>] [-max_retry <int>] [-multi_process <Single_Host|Multi_Host|None>] [-multi_process_activation_percent_limit <int>] [-multi_process_params <incl_parff=bool>] [-name <list>] [-param <string>] [-parff <t|f|MULTI_STAGE>] [-parff_multi_stage <bool>] [-parff_winner_strategy <Early|Regular|BestTiming>] [-use_local_disk <bool>] [-user_complexity_threshold <int>]
```

### Option(s)

- **-advanced\_command\_file <file>** Specify advanced command file to specify PARFF settings (legacy ZPF:UseFpgaMultiLaunch).
- **-complexity\_predictor <bool>** Enables or disables FPGA complexity predictor. Default value: false (legacy ZPF:ZCUI\_set FpgaComplexityPredictor).
- **-complexity\_predictor\_bundle\_size <int>** Number of FPGAs in each complexity predictor bundle. Default value is 8 (legacy ZPF:ZCUI\_set NumberOfFpgaPerBundle).  
A positive integer number.
- **-debug\_policy <Disabled|Enabled|Expert>** Specify debug policy (legacy ZPF:ISE-DebugPolicy).
  - **Disabled:** Improve disk space regarding P&R information  
Valid values: Disabled|DISABLED|disabled
  - **Enabled:** Enable some Vivado reports  
Valid values: Enabled|ENABLED|enabled
  - **Expert:** Launch specific PARFF strategies files for design analysis  
Valid values: Expert|EXPERT|expert
- **-file\_policy <Regular|Compress|Suppress>** Specify file policy (legacy ZPF:ISEFile-Policy).
- **-first\_par\_policy <Original|Winner|Both|BothAnyway>** Specify first FPGA P&R policy. Only valid with incremental zPar "zpar -incremental". (legacy ZPF:ISEFirstPARPolicy).
- **-inter\_die\_tdm <bool>** enable PDM.
- **-inter\_die\_tdm\_params <strategy=enum,time\_budgets=bool>** {strategy=PDM\_DIRECT|PDM\_PARFF}.
  - **strategy <Pdm\_Direct|Pdm\_Parff|Sc\_Tdm\_Direct|Sc\_Tdm\_Parff>** Specify PDM strategy.
    - \* **Pdm\_Direct:** PDM will be enabled even for Original compilation. User will define which PARFF stages that they will also like to introduce PDM based on the PARFF file  
Valid values: Pdm\_Direct|PDM\_DIRECT|pdm\_direct
    - \* **Pdm\_Parff:** User will define which PARFF stages that they will like to introduce PDM based on the PARFF file  
Valid values: Pdm\_Parff|PDM\_PARFF|pdm\_parff
    - \* **Sc\_Tdm\_Direct:** SC TDM will be enabled even for Original compilation. User will define which PARFF stages that they will also like to introduce SC TDM based on the PARFF file  
Valid values: Sc\_Tdm\_Direct|SC\_TDM\_DIRECT|sc\_tdm\_direct

- \* **Sc\_Tdm\_Parff**: User will define which PARFF stages that they will like to introduce SC TDM based on the PARFF file  
Valid values: `Sc_Tdm_Parff|SC_TDM_PARFF|sc_tdm_parff`
- **time\_budgets** <bool> Enable budgeting based PDM.
- **-local\_disk\_path** <string> (legacy ZPF:ZCUI\_set FpgaLocalDiskComplationDir).  
local disk compilation path
- **-max\_retry** <int> specify the max number of retry jobs.  
A non-negative integer number.
- **-multi\_process** <Single\_Host|Multi\_Host|None> Define multi process policy.
- **-multi\_process\_activation\_percent\_limit** <int> Percentage of FPGA job enabling the MPF feature.  
A non-negative integer number.
- **-multi\_process\_params** <incl\_parff=bool> set multi process handling options.
  - **INCL\_PARFF** <bool> let PARFF strategies also use the multiprocessing parameters.
- **-name** <list> list of fpga names.
- **-param** <string> fpga params.  
{par\_name1=some\_value1, par\_name2=some\_value2,...}
- **-parff** <t|f|MULTI\_STAGE> Enables or disables parallel automatic recompilation of failing FPGAs. Default value is true.
- **-parff\_multi\_stage** <bool> Enables or disables parallel automatic recompilation of failing FPGAs - multi stage mode. Default value is true.
- **-parff\_winner\_strategy** <Early|Regular|BestTiming> Specify parff winner detection strategy. (legacy ZPF:ParffWinnerStrategy).
- **-use\_local\_disk** <bool> (legacy ZPF:ZCUI\_set UseFpgaLocalDiskCompilation).
- **-user\_complexity\_threshold** <int> Threshold compile time in hours to classify FPGA as Complex. The Default value is 4 (legacy ZPF:ZCUI\_set FpgaUserComplexityThreshold).  
An integer number.

#### Constraint(s)

- At least one of the options should be used
- -param can only be used with -name
- -name can only be used with -param

#### Note(s)

- -parff\_multi\_stage t also implicitly enables -parff. -parff f -parff\_multi\_stage t is not a legal combination.
- if -use\_local\_disk is set to true then dir path must be supplied using -local\_disk\_path

#### Example(s)

- -param par1=val1,par1=val2 or -param {par1=val1, par2=val2}

## 2.49 fusa

### Description

This command is now deprecated. Please use `fault_emulation` instead.

### Synopsis

```
fusa [-campaign <string>] [-dut_path <string>] [-fdb_name <string>]
[-fdb_server <string>] [-object_not_found <string>]
```

Deprecated command: To be obsoleted. #Please use `fault_emulation` instead.

### Option(s)

- **-campaign <string>** Specify the campaign name..  
Specify the campaign name.
- **-dut\_path <string>** Specify the DUT path..  
Specify the DUT path.
- **-fdb\_name <string>** Specify the fault emulation database name..  
Specify the fault emulation database name.
- **-fdb\_server <string>** Specify the fault emulation database server..  
Specify the fault emulation database server.
- **-object\_not\_found <string>** Specify the message severity for not found objects. Default is fatal..  
Specify the message severity for not found objects. Default is fatal.

### Constraint(s)

- At least one of the options should be used

## 2.50 global\_db

### Synopsis

```
global_db [-merge_start_fpga_percent <int>] [-number_of_threads <int>]
```

### Option(s)

- **-merge\_start\_fpga\_percent <int>** In order to shorten compile time (wall clock), this option allows to start the global DB merger already after a set percentage of passing (P&R) FPGAs..  
A non-negative integer number.
- **-number\_of\_threads <int>** Specify the number of threads can be used by the DB\_Global.  
A non-negative integer number.

### Constraint(s)

- At least one of the options should be used

## 2.51 grid\_cmd

### Description

Specify GRID/LSF commands for compilation

### Synopsis

```
grid_cmd [-clear_all] [-compute_unit <int>] [-cores <int>] [-delete
<string>] [-local] [-mem_limit <string>] [-njobs <int>] [-queue <string>]
[-submit <string>]
```

### Option(s)

- **-clear\_all** Clears out all previous grid\_cmd settings..
- **-compute\_unit <int>** Compute unit for compute cost metric.  
A non-negative integer number.
- **-cores <int>** Number of cores.  
A non-negative integer number.
- **-delete <string>** Job deletion string (if not present, kill -9 will be used).  
job\_deletion\_string
- **-local** Run jobs in this queue locally (mutually exclusive to -submit and -delete options).
- **-mem\_limit <string>** Specify the memory limit to be used.
- **-njobs <int>** Specify the number of parallel jobs..  
A non-negative integer number.
- **-queue <string>** Specify queue name (if not present this command is referring to all previously defined queues).  
queue\_name
- **-submit <string>** Job submission string.  
job\_submission\_string

### Constraint(s)

- At least one of the options should be used
- -clear\_all cannot be used with any other option
- -local cannot be used with any of the following options: -delete, -submit

## 2.52 grid\_engine

### Description

Relaunch jobs in case of grid failure

### Synopsis

```
grid_engine [-clear <all>] [-name <string>] [-property <string>] [-value
<string>]
```

### Option(s)

- **-clear** <all> Clears out all previous grid\_engine declarations..
- **-name** <string> Specify engine name.  
RETRY\_UPON\_FAILURE
- **-property** <string> Specify property name..  
property\_string
- **-value** <string> Specify property value as string..  
value\_string

### Constraint(s)

- At least one of the options should be used
- -clear cannot be used with any other option
- -implementation cannot be used with any other option
- At least one of the following options should be used: -implementation|-name|-property|-clear
- -property can only be used with -value
- -value can only be used with -property

### Note(s)

- Only one property can be declared per command.

### Example(s)

- grid\_engine -name RETRY\_UPON\_FAILURE -property numberOfRetries -value 5
- grid\_engine -name RETRY\_UPON\_FAILURE -property retryExpression -value {error 131}
- grid\_engine -name RETRY\_UPON\_FAILURE -property invertedRetryExpression -value {^a, b or c}
- grid\_engine -name RETRY\_UPON\_FAILURE -property delay -value 10
- grid\_engine -property gridvendor -value "SGE"
- grid\_engine -clear

## 2.53 grid\_task\_association

### Description

Specify associations of Zebu tasks and GRID/LSF queues

### Synopsis

```
grid_task_association [-bundle_mode <bool>] [-clear_all] [-queue <string>]
[-reserve_after <string>] [-task <string>] [-threads <int>]
```

### Option(s)

- **-bundle\_mode <bool>** The tasks will run as dynamic bundles and share the same grid slots.
- **-clear\_all** Clears out all previous grid\_task\_association settings..
- **-queue <string>** Specify queue name (if not present this command is referring to all previously defined queues).  
queue\_name
- **-reserve\_after <string>** Specify task class name task class triggering new slots.
- **-task <string>** Specify task name .  
task\_name
- **-threads <int>** Specify number of threads for the given task.  
A positive integer number.

### Constraint(s)

- -clear\_all cannot be used with any other option
- At least one of the options should be used

## 2.54 load\_edif

### Description

Relates an edif file to a module or vhdl unit

### Synopsis

```
load_edif [-filename <file>] [-module <string>]
```

### Option(s)

- **-filename** <**file**> Specify EDIF file.
- **-module** <**string**> Specify module or entity name.  
module or entity name

### Constraint(s)

- At least one of the options should be used

### Example(s)

- load\_edif -module my\_entity -filename my\_mod.edf

## 2.55 memories

### Description

Memory inference options

### Synopsis

```
memories [-algorithm <DIRECT|TWO_PASS>] [-convert_reset <bool>] [-drop_write_only <bool>] [-exclude <list>] [-flops] [-inline_readmem] [-instance <list>] [-make_sync_writes <false|posedge|negedge|dual|driver|COMPOSITE>] [-make_sync_writes_specific <FALSE|DRIVER|DUAL|COMPOSITE>] [-module <list>] [-profile <bool>] [-set_default_type <LUT|BRAM|URAM|ZRM|READ_AFTER_WRITE|READ_BEFORE_WRITE|NO_READ_ON_WRITE|OPTIMIZE_CAPACITY|OPTIMIZE_PERFORMANCE|WORD_LENGTH>] [-signal <list>] [-sparse_params <num_pages=int,on_chip_promotion=enum,page_size=int,verbose=bool,words_per_page=int>] [-type <LUT|BRAM|URAM|ZRM|READ_AFTER_WRITE|READ_BEFORE_WRITE|NO_READ_ON_WRITE|OPTIMIZE_CAPACITY|OPTIMIZE_PERFORMANCE|WORD_LENGTH|ZMEM_CLOCK_FREQUENCY|ZMEM_TARGET_FREQUENCY|REPLICATE|BRAM_DUAL_PORT_MULTIPLICED|AUTO|SPARSE>] [-under_instance <list>] [-zmem] [-zmem_port_threshold <int>] [-zmem_scripts <list>] [-zmem_size_threshold <int>] [-zrm_latency_encoding <ONEHOT|BINARY>]
```

### Option(s)

- **-algorithm** <DIRECT|TWO\_PASS> Specify algorithm.
- **-convert\_reset** <bool> Convert reset logic, so there will be fewer memory ports, increasing the chance that the array can be implemented as a memory Global option. Default is false for UC Phase 1, and true for WLS (Word Level Synthesis). (legacy zFast: Compile:ConvertMemReset).
- **-drop\_write\_only** <bool> Drop write-only memories. Global option. (legacy zFast:DropWriteOnlyMemories).
- **-exclude** <list> list of excluded instances.
- **-flops** Specify memories to be implemented as flops.
- **-inline\_readmem** .
- **-instance** <list> list of instances.
- **-make\_sync\_writes** <false|posedge|negedge|dual|driver|COMPOSITE> Converts asynchronous writes to synchronous writes (legacy zFast:makesyncwrites).
- **-make\_sync\_writes\_specific** <FALSE|DRIVER|DUAL|COMPOSITE> make sync writes specific.
- **-module** <list> list of modules.
- **-profile** <bool> Enable/disable profile.
- **-set\_default\_type** <LUT|BRAM|URAM|ZRM|READ\_AFTER\_WRITE|READ\_BEFORE\_WRITE|NO\_READ\_ON\_WRITE|OPTIMIZE\_CAPACITY|OPTIMIZE\_PERFORMANCE|WORD\_LENGTH> Specify default type.
- **-signal** <list> list of signals.
- **-sparse\_params** <num\_pages=int,on\_chip\_promotion=enum,page\_size=int,verbose=bool,words\_per\_page=int> Enable/disable a series of sparse\_memoryAvailable params are : size\_of\_page, words\_per\_page, num\_of\_page.
  - **num\_pages** <int> number of graph pages.  
A positive integer number.
  - **on\_chip\_promotion** <auto|true|false> .
  - **page\_size** <int> size of page.  
A positive integer number.
  - **verbose** <bool> verbose.

- **- words\_per\_page <int>** number of words per page.  
A positive integer number.
- **-type <LUT|BRAM|URAM|ZRM|READ\_AFTER\_WRITE|READ\_BEFORE\_WRITE|NO\_READ\_ON\_WRITE|OPTIMIZE\_CAPACITY|OPTIMIZE\_PERFORMANCE|WORD\_LENGTH|ZMEM\_CLOCK\_FREQUENCY|ZMEM\_TARGET\_FREQUENCY|REPLICATE|BRAM\_DUAL\_PORT\_MULTIPLEXED|AUTO|SPARSE>** Specify type.
- **-under\_instance <list>** list of under instances.
- **-zmem** Specify memories to be implemented as zMems.
- **-zmem\_port\_threshold <int>** Specify port threshold for flop vs. zMem memories for automatic memory inference (legacy zFast:MemoryPortThreshold).  
A positive integer number.
- **-zmem\_scripts <list>** Specify list of zMem scripts.
- **-zmem\_size\_threshold <int>** Specify size threshold for flop vs. zMem memories for automatic memory inference (legacy zFast:MemorySizeThreshold).  
A positive integer number.
- **-zrm\_latency\_encoding <ONEHOT|BINARY>** Latency encoding type.

#### Constraint(s)

- -exclude can only be used with -inline\_readmem
- -flops cannot be used with any of the following options: -zmem, -set\_default\_type, -type, -make\_sync\_writes\_specific
- -flops: -inline\_readmem is not compatible with -algorithm TWO\_PASS
- -flops: using -algorithm implies also using -instance
- -inline\_readmem can only be used with -flops
- -algorithm can only be used with -flops
- -zmem: if no -instance use -set\_default\_type; -make\_sync\_writes\_specific cannot be used without -instance; -inline\_readmem is illegal; cannot use -make\_sync\_writes\_specific with -type
- REPLICATE is the only -type that can be used without a -zmem
- -type, when used without -zmem, must be used with -instance
- -sparse\_params, must be used with -type SPARSE
- -module cannot be used with any of the following options: -instance
- -set\_default\_type can only be used with -zmem
- -make\_sync\_writes\_specific can only be used with -zmem
- when -type or -set\_default\_type value is WORD\_LENGTH, or -type is ZMEM\_CLOCK\_FREQUENCY, ZMEM\_TARGET\_FREQUENCY, REPLICATE, or BRAM\_DUAL\_PORT\_MULTIPLEXED, -instance value shall be an even-length list of pairs {inst1 num1 ...}
- '-make\_sync\_writes INDRIVER' is obsolete
- '-make\_sync\_writes INVDRIVER' is obsolete
- '-make\_sync\_writes\_specific DRIVER\_NEGEDGE' is obsolete
- At least one of the options should be used

#### Note(s)

- Usage for -flops option: -flops -instance #instances [-inline\_readmem [-exclude #instances]] [-algorithm DIRECT|TWO\_PASS] using -inline\_readmem (legacy zfast: +InlineReadmem) also implies implement as flops (legacy zfast: +Compile:ImplementAsFlops) for the specific modules or instances
- memories -flops -algorithm TWO\_PASS -instance {module\_name.variable\_name} is same as legacy zfast:Compile:ForceFlops=(module\_name.variable\_name)
- memories -flops -algorithm DIRECT ... is same as memories -flops ... (i.e. -algorithm DIRECT is the default)

- Usage for -zmem option: -zmem -instance #instances [-type LUT|BRAM|ZRM|READ\_AFTER\_WRITE|READ\_BEFORE\_WRITE|NO\_READ\_ON\_WRITE|OPTIMIZE\_CAPACITY|OPTIMIZE\_PERFORMANCE|WORD\_LENGTH|ZMEM\_CLOCK\_FREQUENCY|ZMEM\_TARGET\_FREQUENCY|REPLICATE|BRAM\_DUAL\_PORT\_MULTIPLEXED|AUTO] Or [-set\_default\_type LUT|BRAM|ZRM|READ\_AFTER\_WRITE|READ\_BEFORE\_WRITE|NO\_READ\_ON\_WRITE|OPTIMIZE\_CAPACITY|OPTIMIZE\_PERFORMANCE|WORD\_LENGTH] Or [-make\_sync\_writes\_specific FALSE|DRIVER|DUAL|COMPOSITE]
- when using -zmem in combination with -type WORD\_LENGTH or -type REPLICATE, -instance <list>: <list> should have the following format {instance1 <int> instance2 <int> ...} where <int> is the word length or replication factor per instance.
- when using -zmem in combination with -type ZMEM\_CLOCK\_FREQUENCY, -instance <list>: <list> should have the following format {instance1 <int> instance2 <int> ...} where <int> is 25, 50, 100, or 200.
- when using -zmem in combination with -type ZMEM\_TARGET\_FREQUENCY, -instance <list>: <list> should have the following format {instance1 <int> instance2 <int> ...} where <int> is target frequency for zMem in kHz [1-200000].
- when using -zmem in combination with -type BRAM\_DUAL\_PORT\_MULTIPLEXED, -instance <list>: <list> should have the following format {instance1 <int> instance2 <int> ...} where <int> is 0 for false, or 1 for true.
- using -zmem in combination with -make\_sync\_writes\_specific (zfast: +MakeSyncWrites:<...>) also implies implement as memory (legacy zfast: +Compile:ImplementAsMemory) for the specific instances
- -type must be used with -zmem, except for '-type REPLICATE', which may be used either with or without -zmem, and, like all other -type values, may not be used with -flops.
- When -type is used without -zmem, it must be used with -instance. Due to no -zmem, the arrays specified by -instance are not required to be implemented as zMemes. -type will only apply to those arrays which are implemented as zMems (if any), and has no effect on those arrays which are implemented as flops.
- using -inline\_readmem (legacy zfast: +InlineReadmem) also implies implement as flops (legacy zfast: +Compile:ImplementAsFlops) for the specific instances

## 2.56 modular\_compile

### Description

Specifies modular compile flow for list of modules.

### Synopsis

```
modular_compile [-module <list>]
```

### Option(s)

- **-module <list>** Specifies the modules..

### Constraint(s)

- At least one of the options should be used

### Note(s)

- Usage: modular\_compile -module <list> modular\_compile -module module\_name modular\_compile -module {module1 module2 module3 ....}

## 2.57 move\_initial\_to\_sw

### Description

Move initial blocks to SW

### Synopsis

```
move_initial_to_sw [-enable <bool>] [-exclude <list>]
```

### Option(s)

- **-enable** <bool> Specify whether initial block move to SW flow is enabled or not..
- **-exclude** <list> Modules which should be excluded for initial block move.

### Example(s)

- `move_initial_to_sw -enable true`
- `move_initial_to_sw -enable true -exclude {module1 module2}`

## 2.58 nfs\_settings

### Description

Specify options for NFS file handling

### Synopsis

```
nfs_settings [-access_level <0|1|2|3|4|5>] [-delay <int>] [-max_retries
<int>]
```

### Option(s)

- **-access\_level** <0|1|2|3|4|5> Specify access level for retries.
- **-delay** <int> Specify delay interval (ms) for retries.  
A non-negative integer number.
- **-max\_retries** <int> Specify maximum number of NFS retries.  
A non-negative integer number.

### Constraint(s)

- At least one of the options should be used

## 2.59 performance

### Description

Specify performance options

### Synopsis

```
performance [-clock_cone_mcp <bool>] [-clock_opt <bool>] [-clock_opt2
<dpi_call=bool>] [-dynamic_emu_stop_counter <int>] [-dynamic_emulation
<dpi=bool,export_dpi_first=bool,loop=bool,var_clock_delays=bool,vlog_
_force=bool,zforce=bool,zgate=bool>] [-fetch_mode <bool>] [-safe_mode
<bool>] [-safe_mode_params <clock_latch=bool,memory=bool,set_reset=bool,skew-
_time=bool>] [-zemi3_fm_mcp_opt <bool>] [-zemi3_fm_opt <bool>]
```

### Option(s)

- **-clock\_cone\_mcp <bool>** Enables/Disables fetch mode clock cone mcp.
- **-clock\_opt <bool>** Optimize away the rarely used negedge (or posedge) of fastest design clock to improve throughput of the design.
- **-clock\_opt2 <dpi\_call=bool>** Optimize away the rarely used negedge (or posedge) of fastest design clock to improve throughput of the design.
  - **dpi\_call <bool>** Values brought in by the DPI calls are consumed by the design in POS cycle.
- **-dynamic\_emu\_stop\_counter <int>** Specify number of driverClk cycles to stop during run-time to ensure dynamic emulation behavior.  
A non-negative integer number.
- **-dynamic\_emulation <dpi=bool,export\_dpi\_first=bool,loop=bool,var\_
clock\_delays=bool,vlog\_force=bool,zforce=bool,zgate=bool>** Enables MCP evaluation to ignore certain rarely used paths and ensure correctness at runtime by holding driverClk when those paths are active.
  - **dpi <bool>** Enables dynamic emulation on dpi paths.
  - **export\_dpi\_first <bool>** Enables dynamic emulation on paths from export DPI first event.
  - **loop <bool>** To introduce a "dynamic stoppage" feature when the value of break register changes in consecutive cycle.
  - **var\_clock\_delays <bool>** Enables the variable delay .
  - **vlog\_force <bool>** Enables dynamic emulation on verilog force paths.
  - **zforce <bool>** Enables dynamic emulation on zforce paths.
  - **zgate <bool>** Enables dynamic emulation on zgate paths.
- **-fetch\_mode <bool>** Enables/Disables fetch mode.
- **-safe\_mode <bool>** Enables/Disables all safe mode performance parameters except clock\_latch.
- **-safe\_mode\_params <clock\_latch=bool,memory=bool,set\_reset=bool,skew\_
time=bool>** Specifies explicitly safe mode performance parameters.
  - **clock\_latch <bool>** .
  - **memory <bool>** .
  - **set\_reset <bool>** .
  - **skew\_time <bool>** .
- **-zemi3\_fm\_mcp\_opt <bool>** Make use of MCP analysis based on instrumentation generated by "-zemi3\_fm\_opt option".
- **-zemi3\_fm\_opt <bool>** Optimizes LUT usage and throughput in fetch mode.

### Note(s)

•

## 2.60 post\_compile\_command

### Description

post compilation command for zCui

### Synopsis

```
post_compile_command <script> [-on_error <IGNORE|EXIT>]
```

### Option(s)

- **script** <string> script commands to be executed after compilation.  
script
- **-on\_error** <enum> enable/disable exit error verification, disabled by default.  
IGNORE|EXIT
  - **EXIT**: exit on error verification enabled  
Valid values: EXIT|Exit|exit
  - **IGNORE**: exit on error verification disabled  
Valid values: IGNORE|Ignore|ignore

### Constraint(s)

- At least one of the options should be used

## 2.61 pre\_compile\_command

### Description

pre compilation command for zCui

### Synopsis

```
pre_compile_command <script> [-on_error <IGNORE|EXIT>]
```

### Option(s)

- **script** <string> script commands to be executed before compilation.  
script
- **-on\_error** <enum> enable/disable exit error verification, disabled by default.  
IGNORE|EXIT
  - **EXIT**: exit on error verification enabled  
Valid values: EXIT|Exit|exit
  - **IGNORE**: exit on error verification disabled  
Valid values: IGNORE|Ignore|ignore

### Constraint(s)

- At least one of the options should be used

## 2.62 probe\_config

### Description

Configure static probes

### Synopsis

```
probe_config [-action <create|delete>] [-bank <string>] [-clock <string>]
[-clock_path <string>] [-edge <pos|neg|both>] [-type <sram_trace>]
```

### Option(s)

- **-action** <create|delete> Create..  
create
- **-bank** <string> Specify bank name..  
bank name
- **-clock** <string> Specify sampling clock..  
clock name
- **-clock\_path** <string> Specify user defined sampling clock signal path for converged mode only..  
signal name
- **-edge** <pos|neg|both> Specify sampling edge..
- **-type** <sram\_trace> Specify probe type..  
sram\_trace

### Constraint(s)

- At least one of the options should be used

### Note(s)

- -action, -bank, -edge options are not supported in converged mode. Converged mode specific options-  
: -clock composite\_clk | driver\_clk | user\_defined : Specify clock in converged mode only. -clock\_path  
signal\_name Specify user defined sampling clock signal path for converged mode only. For user\_defined  
clock -clock\_path option should be specified.
- Probe configuration is only required for "probe\_signals -type sram\_trace" support.

## 2.63 probe\_signals

### Description

Adds debug visibility to signals

### Synopsis

```
probe_signals [-bank <string>] [-clock_name <string>] [-depth <int>]
[-fnmatch] [-group <string>] [-hier_sep <string>] [-init_value <string>]
[-instance <string>] [-instance_file <file>] [-label <string>] [-module
<string>] [-module_file <file>] [-object_not_found <string>] [-port
<list>] [-port_file <file>] [-regexp <string>] [-rtlname <list>] [-rtlname-
_file <file>] [-type <dynamic|static|sram_trace|read_port|write_port>]
[-wire <list>] [-wire_file <file>]
```

### Option(s)

- **-bank <string>** Specify bank number used for static probes with 'sram\_trace' type..  
Specify bank number used for static probes with 'sram\_trace' type.
- **-clock\_name <string>** Only supported by the UCPC product. Specify sampling clock. This option is deprecated..  
Only supported by the UCPC product. Specify sampling clock. This option is deprecated.
- **-depth <int>** Specify depth of search for given context determined by module/instance..  
An integer number.
- **-fnmatch** Specify that fnmatch pattern is used to match names..
- **-group <string>** Specify group for probes..  
Specify group for probes.
- **-hier\_sep <string>** Specify hierarchical separator..  
char
- **-init\_value <string>** Specify initial value for write\_port probes..  
Specify initial value for write\_port probes.
- **-instance <string>** Specify name of instance to be probed.  
Specify name of instance to be probed
- **-instance\_file <file>** Specify name of file with instance to be probed. This option is deprecated..
- **-label <string>** Specify label for read\_port|write\_port probes. This option is deprecated..  
Specify label for read\_port|write\_port probes.
- **-module <string>** Specify name of module to be probed.  
Specify name of module to be probed
- **-module\_file <file>** Specify name of file with module to be probed. This option is deprecated..
- **-object\_not\_found <string>** Specify level of severity message in case object is not found.  
warning|fatal
- **-port <list>** Specify list of ports to be probed. This option is deprecated..
- **-port\_file <file>** Specify name of file with ports to be probed. This option is deprecated..
- **-regexp <string>** Specify regex pattern used to match names..  
regex\_pattern
- **-rtlname <list>** Specify list with signals to be probed..
- **-rtlname\_file <file>** Specify name of file with signals to be probed..
- **-type <dynamic|static|sram\_trace|read\_port|write\_port>** Specify probe type..
- **-wire <list>** Specify list of wires to be probed. This option is deprecated..
- **-wire\_file <file>** Specify name of file with wires to be probed. This option is deprecated..

### Constraint(s)

- At least one of the options should be used

## 2.64 read\_classification

### Description

Classification reader

### Synopsis

```
read_classification <classifFileNoOpt> [-edif <bool>] [-file <file>]
```

### Option(s)

- **classifFileNoOpt** <file> Specify name of classification file..
- **-edif** <bool> Enable EDIF output.
- **-file** <file> Specify name of classification file..

### Constraint(s)

- classifFileNoOpt cannot be used with any of the following options: -file
- At least one of the following options should be used: classifFileNoOpt|-file

## 2.65 read\_sdc

### Description

SDC reader

### Synopsis

```
read_sdc <fileWithoutArgument> [-designFeatures <file>] [-edif <bool>]
[-file <file>] [-instance <string>] [-noClockGen <bool>] [-resilient
<bool>]
```

### Option(s)

- **fileWithoutArgument** <file> Specify name of sdc file..
- **-designFeatures** <file> Specify name of designFeatures file..
- **-edif** <bool> when edif option is set all sources names will be resolved at zTopBuild stage.
- **-file** <file> Specify name of sdc file..
- **-instance** <string> Specify instance name for which sdc file is provided..  
Specify instance name for which sdc file is provided.
- **-noClockGen** <bool> when noClockGen option is set all constraints in this file no rtl clock generation will occur.
- **-resilient** <bool> Enable SDC handling with resilient flow.

### Constraint(s)

- At least one of the options should be used
- fileWithoutArgument cannot be used with any of the following options: -file
- -designFeatures cannot be used with any of the following options: fileWithoutArgument, -file
- At least one of the following options should be used: fileWithoutArgument|-file|-designFeatures

## 2.66 reg\_init

### Description

Initializes registers and latches.

### Synopsis

```
reg_init [-default_value <0|1>] [-enable_wls <bool>] [-exclude_modules
<list>] [-filter_context <ALL|DUT>] [-fnmatch] [-inst_path <string>]
[-value <0|1>]
```

### Option(s)

- **-default\_value** <0|1> Specifies the initialization value of uninitialized registers and latches. Available values are '0' or '1'. Default value: 0..
- **-enable\_wls** <bool> When true, enables synthesis support for 'reg\_init -default\_value'. Default is false..
- **-exclude\_modules** <list> Specifies modules to be excluded from consideration. Currently only for '-default\_value <0|1> -enable\_wls true'..
- **-filter\_context** <ALL|DUT> Specifies whether to consider just the DUT, or ALL. Currently only for '-default\_value <0|1> -enable\_wls true'. Default is DUT..
- **-fnmatch** Indicates that -inst\_path and -exclude\_modules can contain wildcards. Type: switch..
- **-inst\_path** <string> Specifies the path of the register or latch you would like to initialize. Will override previous initialization values..  
string
- **-value** <0|1> Specifies the initialization value of the register or latch given by -inst\_path. Available values are '0' or '1'. Default value: 0..

### Constraint(s)

- -inst\_path can only be used with -value
- At least one of the options should be used
- option '-enable\_wls true' cannot be used with option -inst\_path or with option -value.

## 2.67 run\_manager

### Description

Specify run manager options

### Synopsis

```
run_manager [-debug <bool>] [-number_of_instances <int>]
```

### Option(s)

- **-debug** <bool> Specifies whether visibility must be added inside the run manager instances.
- **-number\_of\_instances** <int> Specify the number of run manager instances. This should be a positive number.  
A non-negative integer number.

### Constraint(s)

- At least one of the options should be used

## 2.68 scheduling

### Description

Specify scheduling options

### Synopsis

```
scheduling [-memory <ALL|CLOCK_CONE|DISABLE>] [-verbose <bool>]
```

### Option(s)

- **-memory** <**ALL**|**CLOCK\_CONE**|**DISABLE**> Enables or disables scheduling of memories. Default value is ALL.
- **-verbose** <**bool**> Enables verbose logging. Default value is false.

## 2.69 set\_bus\_interface

### Functions

- **set\_bus\_interface** args

### 2.69.1 Detailed Description

#### Description

Adds a set\_bus\_interface black box module in the design which will mark the port where downstream tool will add register. Black box will be removed after adding registers.

#### Synopsis

```
set_bus_interface [-clock <string>] [-latency <int>] [-module <string>]
[-port <list>]
```

#### Option(s)

- **-clock <string>** clock which will drive the register.  
string
- **-latency <int>** Set latency between two ip blocks..  
A non-negative integer number.
- **-module <string>** Module name for IP.  
string
- **-port <list>** Name of port where tool will add register..

#### Constraint(s)

- At least one of the options should be used
- check illegal option combinations

#### Note(s)

- -name, -module and -port are mandatory arguments.
- clock and max latency are optional arguments default latency is 2.

#### Example(s)

- 'set\_bus\_interface -clock {top.ACLK} -module {axi\_m} -port {DATA}' => Add 2 registers in front of port DATA, drive them using clock top.ACLK
- 'set\_bus\_interface -clock {top.ACLK} -module {axi\_m} -port {data\_snd, data\_rcv} max\_latency {1}' => Add 1 register in front of port data\_snd and data\_rcv, drive them using top.ACLK

## 2.70 set\_dualedge

### Description

set dual edge

### Synopsis

```
set_dualedge [-instance <string>]
```

Deprecated command: To be obsoleted. #

### Option(s)

- **-instance** <**string**> instance path.  
string representing instance path

### Constraint(s)

- At least one of the options should be used

### Note(s)

- set value "yes" to defparam 'dualedge' of a given C\_COSIM instance.

## 2.71 set\_false\_path\_from

### Description

This command is used to specify false path constraints

### Synopsis

```
set_false_path_from [-case_sens <bool>] [-ins <string>] [-match <string>]
[-match_alias <string>] [-match_ins <string>] [-match_port <string>]
[-match_wire <string>] [-port <string>] [-verbose <bool>]
```

### Option(s)

- **-case\_sens <bool>** Matching pattern is case sensitive.
- **-ins <string>** Specifies the name of the FPGA instance in zCoreBuild generated EDIF file.  
Name of the FPGA instance
- **-match <string>** Matching pattern using the form Ins\_pattern/Port\_pattern.  
Matching pattern using the form Ins\_pattern/Port\_pattern
- **-match\_alias <string>** Matching alias name.  
Matching alias name
- **-match\_ins <string>** Matching pattern for FPGA instance name.  
Matching pattern for FPGA instance name
- **-match\_port <string>** Matching port name pattern.  
Matching port name pattern
- **-match\_wire <string>** Matching wire name.  
Matching wire name
- **-port <string>** Specifies FPGA port name.  
FPGA port name
- **-verbose <bool>** Displays the name of false paths found.

### Constraint(s)

- At least one of the options should be used

### Note(s)

- This command is used to specify false path constraints. All paths starting from this port should be considered as false paths. This port is defined with the '-port=<Port\_name>' argument. If it is not a primary port then the '-ins=<Ins\_name>' argument must be provided to specify the name of the FPGA instance of the port. Port name and instance name can be specified directly with the '-port=<Port\_name>' and '-ins=<Ins\_name>' arguments or with wildcards with '-match\_port=<Pattern\_port>' and '-match\_ins=<Pattern\_ins>'. zTime supports Unix wildcards. Alternatively the '-match' option can be used to specify a match pattern like Ins\_pattern/Port\_pattern.

## 2.72 set\_false\_path\_to

### Description

This command is used to specify false path constraints.

### Synopsis

```
set_false_path_to [-case_sens <bool>] [-ins <string>] [-match <string>]
[-match_alias <string>] [-match_ins <string>] [-match_port <string>]
[-match_wire <string>] [-port <string>] [-verbose <bool>]
```

### Option(s)

- **-case\_sens <bool>** Matching pattern is case sensitive.
- **-ins <string>** Specifies the name of the FPGA instance in zCoreBuild generated EDIF file.  
Name of the FPGA instance
- **-match <string>** Matching pattern using the form Ins\_pattern/Port\_pattern.  
Matching pattern using the form Ins\_pattern/Port\_pattern
- **-match\_alias <string>** Matching alias name.  
Matching alias name
- **-match\_ins <string>** Matching pattern for FPGA instance name.  
Matching pattern for FPGA instance name
- **-match\_port <string>** Matching port name pattern.  
Matching port name pattern
- **-match\_wire <string>** Matching wire name.  
Matching wire name
- **-port <string>** Specifies FPGA port name.  
FPGA port name
- **-verbose <bool>** Displays the name of false paths found.

### Constraint(s)

- At least one of the options should be used

### Note(s)

- This command is used to specify false path constraints. All paths ending on this port should be considered as false paths. This port is defined with the '-port=<Port\_name>' argument. If it is not a primary port, i.e. not in IF FPGA, then the '-ins=<Ins\_name>' argument must be provided to specify the name of the FPGA instance of the port. Port name and instance name can be specified directly with the '-port=<Port\_name>' and '-ins=<Ins\_name>' arguments or with wildcards with '-match\_port=<Pattern\_port>' and '-match\_ins=<Pattern\_ins>'. This command supports Unix wildcards. Alternatively the '-match' option can be used to specify a match pattern like Ins\_pattern/Port\_pattern.

## 2.73 set\_hwtop

### Description

Specify hardware top module name

### Synopsis

```
set_hwtop [-module <string>]
```

### Option(s)

- **-module** **<string>** Top-level module.  
top module name

### Constraint(s)

- At least one of the options should be used

### Example(s)

- `set_hwtop -module top_module_name`

## 2.74 set\_perf\_flow

### Description

Define perf flow

### Synopsis

```
set_perf_flow <mode> [-patch <string>] [-patch_list <list>]
```

### Option(s)

- **mode** <string> perf flow mode.
- **-patch** <string> patch name.
- **-patch\_list** <list> list of allowed patch names.

## 2.75 set\_power\_profile

### Synopsis

```
set_power_profile [-bucket_file <file>] [-udp_width <16|32>] [-uip_if_-
fx_pairs <0|1|2|4>]
```

Deprecated command: To be obsoleted. #Please use power\_profile instead.

### Option(s)

- **-bucket\_file <file>** Specify the bucket file output by Spyglass..
- **-udp\_width <16|32>** Specify the UDP bit width..  
Default is 16 which means each UDP bit width is 16k bits.
- **-uip\_if\_fx\_pairs <0|1|2|4>** Specify message pair number for conditions, 0|1|2|4. .  
Default is 0 which means enabling UIP without conditions. 1 means enable UIP with 1 message pair for conditions and so on.

### Constraint(s)

- At least one of the options should be used

### Example(s)

- `set_power_profile -uip_if_fx_pairs 1 -udp_width 32 -bucket_file {./bucket.txt}`

## 2.76 set\_sw\_control\_signal

### Description

Specify a signal as software-accessible. Equivalent to creating a DVE register.

### Synopsis

```
set_sw_control_signal -hdl_path <string> [-name <string>] [-sync <bool>]
```

### Option(s)

- **-hdl\_path** <string> Specify the hierarchical path for the software control signal.  
hierarchical path **Mandatory option**
- **-name** <string> Specify the name of software control signal name.  
signal name
- **-sync** <bool> Specify whether the software control signal is synchronized with system clock.

### Constraint(s)

- At least one of the options should be used

### Note(s)

- -sync option is supported only in nodve mode.

### Example(s)

- `set_sw_control_signal -name my_reg -hdl_path hw_top.ctrl_reg -sync true`

## 2.77 set\_trigger

### Description

Specify a static trigger based on signal full path string

### Synopsis

```
set_trigger -hdl_path <string> -name <string>
```

### Option(s)

- **-hdl\_path <string>** Specify the hierarchical path for the static trigger signal.  
hierarchical path **Mandatory option**
- **-name <string>** Specify the static trigger name.  
trigger name **Mandatory option**

### Example(s)

- `set_trigger -name my_trig -hdl_path {hw_top.trig_sig}`

## 2.78 set\_zcei\_partition

### Description

for zTopBuild to specify FPGA for zcei partition

### Synopsis

```
set_zcei_partition [-use_fpga <string>]
```

### Option(s)

- **-use\_fpga** **<string>** Specify the target FPGA for zcei partition.  
FPGA name

### Constraint(s)

- At least one of the options should be used

## 2.79 signature

### Description

Accelerate debug of not functional emulation against a functional reference

### Synopsis

```
signature [-fpga_stability_check <bool>]
```

### Option(s)

- **-fpga\_stability\_check <bool>** Generates a trigger when an fpga output is not stable.

## 2.80 simxl

### Description

This command indicates the operation mode in time coupled simulation acceleration with HDL testbench. If it is set to false or if the command is not specified, acceleration behaves as legacy HDL\_COSIM mode (time decoupled). User can also specify the mode of execution either congruent or fast mode. Congruent mode is by default. However, the 'mode' option has no effect if the 'enable' option is set to false.

### Synopsis

```
simxl [-enable <bool>] [-mode <congruent|fast>]
```

### Option(s)

- **-enable** <bool> Specify whether the simxl flow is enabled or not..
- **-mode** <congruent|fast> Specify whether the execution mode is congruent or fast. Default is congruent..

### Constraint(s)

- At least one of the following options should be used: -enable

### Example(s)

- `simxl -enable TRUE -mode FAST`

## 2.81 simxl\_allow\_concurrent

### Description

Allow concurrent calls of SW->HW tasks/functions

### Synopsis

```
simxl_allow_concurrent -module <string> -tf <list> [-count <int>]
```

### Option(s)

- **-count** <int> Number of concurrent calls of task/function allowed..  
A positive integer number.
- **-module** <string> Specify the name of the target module..  
Specify the name of the target module. **Mandatory option**
- **-tf** <list> Specify the SW->HW tasks/functions that can be called concurrently.. **Mandatory option**

### Example(s)

- `simxl_allow_concurrent -module subdut -tf {TASK1 FUNC1} -count 3`

## 2.82 simxl\_enable\_zdpi

### Description

Enable zDPI in DUT modules in SIMXL flow

### Synopsis

```
simxl_enable_zdpi [-all <bool>] [-module <list>]
```

### Option(s)

- **-all** <bool> Specify if zDPI needs to be enabled for all modules..
- **-module** <list> Specify modules for which zDPI needs to be enabled.

### Constraint(s)

- At least one of the options should be used

### Example(s)

- `simxl_enable_zdpi -all`
- `simxl_enable_zdpi -module {module1 module2}`

## 2.83 simxl\_move\_tf\_to\_tb

### Description

In SimXL mode, we target to provide the same or similar emulation environment as simulator and we encourage user to use task and function to exchange data between TB and DUV for performance. By moving those tasks and functions to TB, we can resolve performance or HW resource issues or provide solutions for those unsupported cases.

### Synopsis

```
simxl_move_tf_to_tb [-module <string>] [-target <list>] [-tf <list>]
```

### Option(s)

- **-module <string>** Specify the name of the target module..  
Specify the name of the target module.
- **-target <list>** Specify which tasks/functions need be moved from DUT to TB. All the task/function definition in DUT will be moved to TB..
- **-tf <list>** Specify which tasks/functions need be moved from DUT to TB. All the task/function definition in DUT will be moved to TB..

### Constraint(s)

- At least one of the options should be used

### Example(s)

- `simxl_move_tf_to_tb -module subdut -tf {TASK1 FUNC1}`

## 2.84 simxl\_move\_to\_tb

### Description

There may be some behavior code in some modules which are instantiated in DUT (HW). This command indicates all such module instantiations in DUT will be moved to TB. User can also specify some instances being moved to TB if necessary. At least one option should be specified.

### Synopsis

```
simxl_move_to_tb [-instance <list>] [-module <list>]
```

### Option(s)

- **-instance <list>** Specify some extra hierarchy instances in DUT need be moved to TB..
- **-module <list>** Specify which modules need be moved from DUT to TB. All the module instantiations in DUT will be moved to TB..

### Constraint(s)

- At least one of the options should be used

### Example(s)

- `simxl_move_to_tb -module {TOP1 TOP2} -instance {TOP3.dut.I1 TOP3.dut.I2}`

## 2.85 simxl\_set\_hwtop

### Description

Specify hardware top module or instance names in SimXL flow. Multiple top modules and full hierarchy instances can be set as hardware top.

### Synopsis

```
simxl_set_hwtop [-instance <list>] [-module <list>]
```

### Option(s)

- **-instance <list>** Specify full hierarchy instances to be moved to HW..
- **-module <list>** Specify top-level modules to be moved to HW..

### Constraint(s)

- At least one of the options should be used

### Example(s)

- `simxl_set_hwtop -module {TOP1 TOP2} -instance {TOP3.dut.I1 TOP3.dut.I2}`

## 2.86 simxl\_xlbram

### Description

XLBRAM can help on improving the data throughput in SimXL by providing low-level DPI functions to access memory. This UTF command is for replacing the specified memory in user code with pre-defined XLBRAM modules.

### Synopsis

```
simxl_xlbram [-enable <bool>] [-instance <list>]
```

### Option(s)

- **-enable** <bool> Specify if XLBRAM replacement needs to be enabled.
- **-instance** <list> Specify the path of memory instances to be replaced.

### Constraint(s)

- At least one of the options should be used

### Example(s)

- `simxl_xlbram -enable -instance {tb.dut.a.mem tb.dut.b.mem}`

## 2.87 smart\_zice\_config

### Description

Smart Z-Ice options

### Synopsis

```
smart_zice_config [-connector <HE10>] [-vcc <1v5|1v8|2v5|3v3>]
```

### Option(s)

- **-connector** <HE10> Enable HE10 adapter.
- **-vcc** <1v5|1v8|2v5|3v3> Value of VCC used for Smart Z-Ice.

### Constraint(s)

- At least one of the following options should be used: -vcc|-connector

### Example(s)

- smart\_zice\_config -vcc 1v8
- smart\_zice\_config -vcc 3V3 -connector HE10

## 2.88 synthesis\_preferences

### Description

Specify synthesis preferences

### Synopsis

```
synthesis_preferences [-bundle_mode <string>] [-enable_fmcheck_debug
<bool>] [-enable_wls <bool>] [-log_files <ALWAYS|NEVER|DEFAULT>]
```

### Option(s)

- **-bundle\_mode** <string> Specify bundle mode for synthesis.  
UNITARY | SIZE=<int> | CARDINAL=<int>
- **-enable\_fmcheck\_debug** <bool> Enabling/Disabling the dumping of encrypted db.
- **-enable\_wls** <bool> Enable the WLS (Word Level Synthesis) flow.
- **-log\_files** <ALWAYS|NEVER|DEFAULT> Whether to create log files in zcui.work/design/synth-  
\_Default\_RTL\_Group/synthesis\_log\_dir.
  - **ALWAYS**: Do create the log files.  
Valid values: ALWAYS|always|Always
  - **NEVER**: Do not create the log files.  
Valid values: NEVER|never|Never
  - **DEFAULT**: The default, currently, is to create the log files.  
Valid values: DEFAULT|default|Default

### Constraint(s)

- At least one of the options should be used

## 2.89 timing\_analysis

### Description

Specify timing analysis options

### Synopsis

```
timing_analysis [-advanced_async_set_reset_analysis <bool>] [-advanced-
_command <string>] [-checker_command_file <file>] [-constraints_input-
_file <file>] [-delay_min_zfilter_skew <int>] [-fpga_advanced_command
<string>] [-fpga_advanced_command_file <file>] [-on_error <FATAL|WARNI-
NG>] [-post_fpga <SKEW|FILTER|NONE|BACK_ANNOTATED>] [-post_fpga_placement
<bool>] [-pre_fpga <LINEAR_DELAY_MODEL|CONSTANTS_BASE_DELAY_MODEL>] [-report-
_input_file <file>] [-title <string>] [-use_hdl_names <bool>]
```

### Option(s)

- **-advanced\_async\_set\_reset\_analysis <bool>** Enables advanced asynchronous set/reset analysis mode (adds asynchronous path into account at multi-cycle path analysis). Default value: False..
- **-advanced\_command <string>** Specify any legacy zTime command.  
ztime\_command
- **-checker\_command\_file <file>** Specify user filename.
- **-constraints\_input\_file <file>** Specify filename with zTime constraints.
- **-delay\_min\_zfilter\_skew <int>** Specify the minimum delay (in [ps]) between filter to skew.  
Default: 40000..  
A non-negative integer number.
- **-fpga\_advanced\_command <string>** Command for zFpgaTiming.  
zFpgaTiming\_command
- **-fpga\_advanced\_command\_file <file>** Command for zFpgaTiming.
- **-on\_error <FATAL|WARNING>** indicate if zTiNa execution status should be taken into account by zCui or not. WARNING is the default value.
- **-post\_fpga <SKEW|FILTER|NONE|BACK\_ANNOTATED>** .
  - **FULL**: Enable the full timing analysis based on the timing  
Valid values: FULL|Full|full
  - **SKEW**: Enable the skew time update computation  
Valid values: SKEW|Skew|skew
  - **FILTER**: Enable the filter time update computation  
Valid values: FILTER|Filter|filter
  - **NONE**: Disable the timing analysis  
Valid values: NONE|None|none
  - **BACK\_ANNOTATED**: Enable SDF timing analysis and disable SKEW and FILTER timing analysis  
Valid values: BACK\_ANNOTATED|Back\_Annotated|back\_annotated
- **-post\_fpga\_placement <bool>** Enable/disable the place timing analysis after the ZTIME\_ANALYSE\_PLACE phase. Default: false..
- **-pre\_fpga <LINEAR\_DELAY\_MODEL|CONSTANTS\_BASE\_DELAY\_MODEL>** enables lut/fanout based delay model in zTime pre fpga. CONSTANTS\_BASE\_DELAY\_MODEL is the default value which disables the pre fpga timing analysis..
- **-report\_input\_file <file>** Specify filename with ztime report commands.
- **-title <string>** Specify the task name of zTiNa.  
ztina\_task\_name
- **-use\_hdl\_names <bool>** Enable/disable the usage of HDL names in zTime report. Default: false..

**Constraint(s)**

- At least one of the options should be used
- -on\_error can only be used with -checker\_command\_file
- -title can only be used with -checker\_command\_file

## 2.90 timing\_constraint

### Description

Specify timing constraints for zTime

### Synopsis

```
timing_constraint [-command <SET_FALSE_PATH>] [-from_match <string>]
[-from_match_alias <string>] [-global_match <string>] [-to_match <string>]
[-to_match_alias <string>] [-verbose <bool>]
```

### Option(s)

- **-command <SET\_FALSE\_PATH>** Specify false path constraints and port/instance/alias name(currently wildcard patterns are based, case sensitive) to or from which all paths should be considered as false paths is taken in argument..
- **-from\_match <string>** Matching pattern of ports/instances from which all paths should be considered as false paths.  
pattern
- **-from\_match\_alias <string>** Matching pattern of aliases from which all paths should be considered as false paths.  
pattern
- **-global\_match <string>** Matching pattern of aliases and wires on which all paths should be considered as false paths.  
pattern
- **-to\_match <string>** Matching pattern of ports/instances to which all paths should be considered as false paths.  
pattern
- **-to\_match\_alias <string>** Matching pattern of aliases to which all paths should be considered as false paths.  
pattern
- **-verbose <bool>** Display the name of false paths found.

### Constraint(s)

- At least one of the options should be used

## 2.91 transparent\_latch

### Description

Specify transparent latches

### Synopsis

```
transparent_latch [-fnmatch] [-module <string>] [-object_not_found <fatal|warning>]
[-rtlname <string>] [-signal <string>]
```

### Option(s)

- **-fnmatch** Matches hierarchical names using fnmatch patterns.
- **-module <string>** Module name for transparent latch.  
string
- **-object\_not\_found <fatal|warning>** Specify error policy in case of objects cannot be forced.  
policy value
  - **fatal**: In case object cannot be found, a fatal error is emitted. This is the default  
Valid values: `fatal|Fatal|FATAL`
  - **warning**: In case object cannot be found, a warning is emitted instead of an error  
Valid values: `warning|Warning|WARNING`
- **-rtlname <string>** Specifies a the hierarchical signal name of transparent latch .  
string
- **-signal <string>** Path of module port used .  
string

### Constraint(s)

- At least one of the options should be used

## 2.92 tristate

### Description

Specifies the processings on tristate buses.

### Synopsis

```
tristate <buses> [-conflict <wand|wor>] [-dynamic] [-exclude <string>]
[-fnmatch] [-keeper] [-pulldown] [-pullup] [-rtlname]
```

### Option(s)

- **buses** <list> List of hierarchical paths to tristate buses. These paths may contain '\*' to match several buses if you use the -fnmatch option.. **Mandatory option**
- **-conflict** <wand|wor> Specifies the behavior of the bus if several tristate drivers are simultaneously enabled.If this option is not specified, the most appropriate conflict resolution for each bus will be used.Details about allowed values:wand: In case of multiple drivers, the option specifies that the resolution function will be AND.wor: In case of multiple drivers, the option specifies that the resolution function will be OR..
- **-dynamic** .
- **-exclude** <string> Excludes buses whose names match the string defined in argument. This string can contain wildcards when the fnmatch option is used..  
string value
- **-fnmatch** Indicates that bus names can contain wildcards..
- **-keeper** Specifies if the bus must be implemented as a keeper..
  - **warning**: A warning is displayed instead of an error  
Valid values: warning|WARNING|Warning
  - **fatal**: A fatal error is displayed  
Valid values: fatal|FATAL|Fatal
- **-pulldown** Specifies if the bus must be implemented as a pulldown..
- **-pullup** Specifies if the bus must be implemented as a pullup..
- **-rtlname** Specifies whether the given list of hierarchical paths to tristate buses must be considered as signal paths..

### Constraint(s)

- At least one of the options should be used

### Example(s)

- Example 1:tristate -pulldown {top.ins0.busz top.ins\*.busz\*} -exclude busz1 -fnmatchSpecifies tristate bus resolution as pulldown for buses whose hierarchical names are matching 'top.ins0.busz top.ins\*.busz\*' wildcardbut whose names are not 'busz1'.
- Example 2: tristate -pullup {top.ins0.busz top.ins1.busz} -conflict worSpecifies tristate bus resolution as pullup for buses top.ins0.busz andtop.ins1.busz. If several drivers of this bus are enabledaltogether, the resolution function will be OR for these signals.

## 2.93 vcs\_exec\_command

### Description

Specify VCS Command line

### Synopsis

```
vcs_exec_command <script>
```

### Option(s)

- **script** <list> vcs\_command\_line\_or\_script VCS command. **Mandatory option**

### Constraint(s)

- At least one of the options should be used

## 2.94 verilog\_force\_release

### Description

This command is used to enable or disable support for Verilog force/release statements. If enabled (enable/enable-wls/wire\_semantic), the default state is that force/release statements are honored. If disabled, the default state is that force/release statements are ignored.

### Synopsis

```
verilog_force_release [-create_task <bool>] [-enable <bool>] [-enable-
_wls <bool>] [-enable_zprop <bool>] [-enhanced_wls <bool>] [-exclude
<list>] [-include <list>] [-local_driver_check <bool>] [-treat_as_wire
<bool>] [-wire_semantic <bool>]
```

### Option(s)

- **-create\_task <bool>** create sys-task for synthesis Verilog force/release support..
- **-enable <bool>** Sets the default state to enabled/disabled for backend support for Verilog force/release statements..
- **-enable\_wls <bool>** Enable synthesis support for Verilog force statements, -enable -include and -exclude are ignored in this mode..
- **-enable\_zprop <bool>** Enable z value propagation for synthesis Verilog force/release support..
- **-enhanced\_wls <bool>** Enable enhanced synthesis support for Verilog force statements, -enable -include and -exclude are ignored in this mode..
- **-exclude <list>** Disables support for force/release statements in these modules, regardless of the default state..
- **-include <list>** Enables support for force/release statements in these modules, regardless of the default state..
- **-local\_driver\_check <bool>** Do not trace driver of the forced signals and only local connection of target wires are forced when set to true..
- **-treat\_as\_wire <bool>** Also trace driver of the signals with type reg | logic | bit when set to true..
- **-wire\_semantic <bool>** Enable wire semantic flow for verilog force release implementation..

### Constraint(s)

- At least one of the options should be used

## 2.95 wire\_resolution

### Description

1. Specifies multidriver resolution (WAND/WOR) for list of signals.
2. Defines default multidriver resolution for nets on which no specific resolution has been defined, default tristate bus resolution, default undriven nets behaviour.

### Synopsis

```
wire_resolution [-conflict <wand|wor>] [-default_conflict <wand|wor>]
[-default_tristate <pullup|pulldown|keeper>] [-default_undriven <0|1>]
[-default_xmr_conflict <wand|wor|xmr>] [-exclude <string>] [-fnmatch
<bool>] [-tristate <pullup|pulldown|keeper>] [-wires <list>]
```

### Option(s)

- **-conflict** <wand|wor> Specifies the default behaviour of the bus in case several drivers are simultaneously enabled..
- **-default\_conflict** <wand|wor> Will be applied on all wires which don't have specific -conflict configuration..
- **-default\_tristate** <pullup|pulldown|keeper> Defines default resolution function when no tristate driver is active on the bus..
- **-default\_undriven** <0|1> Sets the default value for undriven nets..
- **-default\_xmr\_conflict** <wand|wor|xmr> Only supported by the UCPC product. Will be applied on all XMRs. The XMR value means that the XMR wins..
- **-exclude** <string> Exclude wires whose names are matching pattern string given in argument.. string value
- **-fnmatch** <bool> Indicates that bus names can contain wildcards..
- **-tristate** <pullup|pulldown|keeper> Specifies the default tristate behaviour of the bus in case no drivers are enabled..
- **-wires** <list> Specifies the wires to be resolved..

### Constraint(s)

- At least one of the options should be used
- -default\_conflict cannot be used with any of the following options: -conflict, -wires, -fnmatch, -exclude
- -default\_xmr\_conflict cannot be used with any of the following options: -conflict, -wires, -fnmatch, -exclude
- -default\_undriven cannot be used with any of the following options: -conflict, -wires, -fnmatch, -exclude
- -default\_tristate cannot be used with any of the following options: -conflict, -wires, -fnmatch, -exclude
- -conflict cannot be used with any of the following options: -default\_conflict, -default\_undriven, -default\_tristate
- -wires cannot be used with any of the following options: -default\_conflict, -default\_undriven, -default\_tristate
- -fnmatch cannot be used with any of the following options: -default\_conflict, -default\_undriven, -default\_tristate
- -exclude cannot be used with any of the following options: -default\_conflict, -default\_undriven, -default\_tristate
- -conflict can only be used with -wires
- -fnmatch can only be used with -wires
- -exclude can only be used with -wires

## Note(s)

- usage 1: wire\_resolution -conflict <WAND|WOR> -wires <list> [-fnmatch <bool>] [-exclude <pattern>]
- usage 2: wire\_resolution [-default\_conflict WOR|WAND] [-default\_undriven 0|1] [-default\_tristate PULLUP|PULLDOWN|KEEPER]
- Mix of two usage not allowed

## 2.96 xtors

### Description

Marks instance as Xtor

### Synopsis

```
xtors [-add <list>] [-add_xtor_path <string>] [-mapping_script <string>]
[-type <string>] [-use_zebu_ip_root <bool>]
```

### Option(s)

- **-add <list>** Specify the instance(s) which will be marked as Xtor..
- **-add\_xtor\_path <string>** Add path to transactor (legacy ZPF:"ZCUI\_add -driver\_path").  
string
- **-mapping\_script <string>** Specify transactor defmapping (legacy ZPF:XtorMappingFile-Name).  
string
- **-type <string>** Specify the type of Xtor (now support only zemi3)..  
string
- **-use\_zebu\_ip\_root <bool>** Use ZEBU\_IP\_ROOT to locate transactors (legacy ZPF:UseZebu-IpRoot).

### Constraint(s)

- -add can only be used with -type
- -type can only be used with -add
- At least one of the options should be used

## 2.97 zemi3

### Description

ZEMI3 transactor options

### Synopsis

```
zemi3 [-advanced_synth_command_file <file>] [-all_tf_args_are_automatic
<bool>] [-allow_mixed_design_clocks <bool>] [-allow_streaming <bool>]
[-async_reset_init <bool>] [-auto_flush_interval <int>] [-buffered_-
out_ports_default_width <int>] [-buffered_out_ports_disable <bool>]
[-buffered_out_ports_regardless <bool>] [-call_final_import <bool>]
[-debug <bool>] [-debug_zemi_behav <bool>] [-default_cclk <string>]
[-delayed_dpi <bool>] [-deterministic_dpi <bool>] [-disable <string>]
[-disable_all_xtors <bool>] [-do_profiling <bool>] [-do_verbose_profiling
<bool>] [-edge_enable <bool>] [-emit_initial <bool>] [-exports_excluded-
_from_back_to_back <list>] [-external_controlled_clocks <list>] [-fm-
dpi_mcp <int>] [-global_serialized_opt <bool>] [-hard_max_in_port_-
width <int>] [-hard_max_out_port_width <int>] [-heartbeat_period <int>]
[-instance <list>] [-max_clocked_loop_iterations <int>] [-max_in_port-
_width <int>] [-max_loop_iterations <int>] [-max_out_port_width <int>]
[-merge_multiple_comb_writers <bool>] [-module <list>] [-optimize_dpi_-
constant_args <bool>] [-port_optimization_mode <0|1|2|3>] [-port_width_-
slop_percent <int>] [-profile_counters_width <int>] [-protolink <bool>]
[-reduce_control_paths <bool>] [-resolve_events_globally <bool>] [-sample-
_clock_mode <bool>] [-serialize_all_imports <bool>] [-start_in_back-
_to_back <bool>] [-streaming_dpi_opt <bool>] [-streaming_is_default
<bool>] [-support_dollar_display <bool>] [-task_streaming_is_default
<bool>] [-timestamp <bool>] [-transform_latch <bool>] [-treat_warnings-
_as_errors <bool>]
```

### Option(s)

- **-advanced\_synth\_command\_file <file>** filename is xtor specific global hcsrc commands file name.
- **-all\_tf\_args\_are\_automatic <bool>** all\_tf\_args\_are\_automatic.
- **-allow\_mixed\_design\_clocks <bool>** allow\_mixed\_design\_clocks.
- **-allow\_streaming <bool>** allow streaming.
- **-async\_reset\_init <bool>** -async\_reset\_init is a global option.
- **-auto\_flush\_interval <int>** .  
A positive integer number.
- **-buffered\_out\_ports\_default\_width <int>** .  
A positive integer number.
- **-buffered\_out\_ports\_disable <bool>** buffered\_out\_ports\_disable.
- **-buffered\_out\_ports\_regardless <bool>** buffered\_out\_ports\_regardless.
- **-call\_final\_import <bool>** -call\_final\_import is a global option.
- **-debug <bool>** Specifies whether visibility must be added inside the transactor or not.
- **-debug\_zemi\_behav <bool>** debug\_zemi\_behav.
- **-default\_cclk <string>** xmr to signal that is connected to zceiClockport.  
xmr to signal that is connected to zceiClockport
- **-delayed\_dpi <bool>** -delayed\_dpi is a global option.
- **-deterministic\_dpi <bool>** -deterministic\_dpi is a global option.
- **-disable <string>** Provides a signal that disables the transactor when it is 0.  
xmr to signal

- **-disable\_all\_xtors** <bool> -disable\_all\_xtors is a global option which provides a disable signal for each transactor instance.
- **-do\_profiling** <bool> do profiling.
- **-do\_verbose\_profiling** <bool> do verbose profiling.
- **-edge\_enable** <bool> Specifies whether flops initially synchronized by the input zcei clock are updated with an edge enable block around.
- **-emit\_initial** <bool> emit\_initial.
- **-exports\_excluded\_from\_back\_to\_back** <list> list\_of\_export\_DPI\_excluded\_from\_back\_to\_back\_mode.
- **-external\_controlled\_clocks** <list> list\_of\_external\_controlled\_clocks.
- **-fm\_dpi\_mcp** <int> -fm\_dpi\_mcp is a global option.  
A positive integer number.
- **-global\_serialized\_opt** <bool> -global\_serialized\_opt is a global option.
- **-hard\_max\_in\_port\_width** <int> .  
A positive integer number.
- **-hard\_max\_out\_port\_width** <int> .  
A positive integer number.
- **-heartbeat\_period** <int> -heartbeat\_period is a global option.  
A positive integer number.
- **-instance** <list> list of instances.
- **-max\_clocked\_loop\_iterations** <int> .  
A positive integer number.
- **-max\_in\_port\_width** <int> .  
A positive integer number.
- **-max\_loop\_iterations** <int> -max\_loop\_iterations can be used with or without -module.  
A positive integer number.
- **-max\_out\_port\_width** <int> .  
A positive integer number.
- **-merge\_multiple\_comb\_writers** <bool> merge\_multiple\_comb\_writers.
- **-module** <list> list of transactors modules.
- **-optimize\_dpi\_constant\_args** <bool> -optimize\_dpi\_constant\_args is a global option.
- **-port\_optimization\_mode** <0|1|2|3> Specify port optimization mode.  
0|1|2|3 0 for AREA\_OPT, 1 for THROUGHPUT\_OPT, 2 for LATENCY\_OPT, 3 for DEFAULT\_OPT
- **-port\_width\_slop\_percent** <int> .  
A positive integer number.
- **-profile\_counters\_width** <int> .  
A positive integer number.
- **-protolink** <bool> protolinks.
- **-reduce\_control\_paths** <bool> reduce control paths.
- **-resolve\_events\_globally** <bool> -resolve\_events\_globally is a global option.
- **-sample\_clock\_mode** <bool> -sample\_clock\_mode is a global option.
- **-serialize\_all\_imports** <bool> serialize\_all\_imports.
- **-start\_in\_back\_to\_back** <bool> start in back to back.
- **-streaming\_dpi\_opt** <bool> -streaming\_dpi\_opt.
- **-streaming\_is\_default** <bool> streaming is default.
- **-support\_dollar\_display** <bool> support \$DISPLAY.
- **-task\_streaming\_is\_default** <bool> task\_streaming\_is\_default.

- **-timestamp** <bool> -timestamp can be used with or without -module.
- **-transform\_latch** <bool> transform\_latch.
- **-treat\_warnings\_as\_errors** <bool> -treat\_warnings\_as\_errors is a global option.

#### Constraint(s)

- -emit\_initial can only be used with -module
- global options cannot be combined with -instance|-module options
- -advanced\_synth\_command\_file can only be used with -module
- check for mandatory combinations of options
- -advanced\_synth\_command\_file cannot be used with any of the following options: -instance
- -instance cannot be used with any of the following options: -allow\_streaming, -start\_in\_back\_to\_back, -support\_dollar\_display, -reduce\_control\_paths, -do\_profiling, -do\_verbose\_profiling, -streaming\_is\_default, -task\_streaming\_is\_default, -emit\_initial, -serialize\_all\_imports, -hard\_max\_out\_port\_width, -hard\_max\_in\_port\_width, -auto\_flush\_interval, -buffered\_out\_ports\_disable, -buffered\_out\_ports\_default\_width, -buffered\_out\_ports\_regardless, -port\_optimization\_mode, -profile\_counters\_width, -max\_out\_port\_width, -max\_in\_port\_width, -port\_width\_slop\_percent, -all\_tf\_args\_are\_automatic, -external\_controlled\_clocks, -allow\_mixed\_design\_clocks, -exports\_excluded\_from\_back\_to\_back, -max\_loop\_iterations, -max\_clocked\_loop\_iterations, -merge\_multiple\_comb\_writers, -transform\_latch, -debug\_zemi\_behav, -default\_cclk, -sample\_clock\_mode, -timestamp, -protolink, -sample\_clock\_opt, -rt\_diag, -rt\_config, -resolve\_events\_globally, -async\_reset\_init, -optimize\_dpi\_constant\_args, -treat\_warnings\_as\_errors, -fm\_dpi\_mcp, -call\_final\_import, -global\_serialized\_opt, -delayed\_dpi, -deterministic\_dpi, -heartbeat\_period, -streaming\_dpi\_opt, -enforce\_streaming
- -sample\_clock\_mode cannot be used with any of the following options: -sample\_clock\_opt
- At least one of the options should be used
- Option '-instance' should be used with at least one of the following options: -debug|-edge\_enable|-disable
- -instance cannot be used with any of the following options: -advanced\_synth\_command\_file

#### Note(s)

- -advanced\_synth\_command\_file must be used with -module, and should not be used with -instance
- Usage for -instance option: -instance is only valid with -debug|-edge\_enable|-disable, other options cannot be used with -instance
- -default\_cclk </tt> can be used either with -module in which case it refers to specific modules, or without -module in which case it refers to all xtors defined until now.

## 2.98 zemi4

### Description

ZEMI4 transactor options

### Synopsis

```
zemi4 [-advanced_synth_command_file <file>] [-all_tf_args_are_automatic
<bool>] [-allow_mixed_design_clocks <bool>] [-allow_streaming <bool>]
[-async_reset_init <bool>] [-auto_flush_interval <int>] [-buffered_-
out_ports_default_width <int>] [-buffered_out_ports_disable <bool>]
[-buffered_out_ports_regardless <bool>] [-call_final_import <bool>]
[-debug <bool>] [-debug_zemi_behav <bool>] [-default_cclk <string>]
[-delayed_dpi <bool>] [-deterministic_dpi <bool>] [-disable <string>]
[-disable_all_xtors <bool>] [-do_profiling <bool>] [-do_verbose_profiling
<bool>] [-edge_enable <bool>] [-emit_initial <bool>] [-exports_excluded-
_from_back_to_back <list>] [-external_controlled_clocks <list>] [-fm-
_dpi_mcp <int>] [-global_serialized_opt <bool>] [-hard_max_in_port_-
width <int>] [-hard_max_out_port_width <int>] [-heartbeat_period <int>]
[-instance <list>] [-max_clocked_loop_iterations <int>] [-max_in_port_-
width <int>] [-max_loop_iterations <int>] [-max_out_port_width <int>]
[-merge_multiple_comb_writers <bool>] [-module <list>] [-optimize_dpi_-
constant_args <bool>] [-port_optimization_mode <0|1|2|3>] [-port_width_-
slop_percent <int>] [-profile_counters_width <int>] [-protolink <bool>]
[-reduce_control_paths <bool>] [-resolve_events_globally <bool>] [-sample-
_clock_mode <bool>] [-serialize_all_imports <bool>] [-start_in_back-
_to_back <bool>] [-streaming_dpi_opt <bool>] [-streaming_is_default
<bool>] [-support_dollar_display <bool>] [-task_streaming_is_default
<bool>] [-timestamp <bool>] [-transform_latch <bool>] [-treat_warnings-
_as_errors <bool>]
```

Deprecated command: To be obsoleted. #Please use simxl instead.

### Option(s)

- **-advanced\_synth\_command\_file <file>** filename is xtor specific global hcsrc commands file name.
- **-all\_tf\_args\_are\_automatic <bool>** all\_tf\_args\_are\_automatic.
- **-allow\_mixed\_design\_clocks <bool>** allow\_mixed\_design\_clocks.
- **-allow\_streaming <bool>** allow streaming.
- **-async\_reset\_init <bool>** -async\_reset\_init is a global option.
- **-auto\_flush\_interval <int>** .  
A positive integer number.
- **-buffered\_out\_ports\_default\_width <int>** .  
A positive integer number.
- **-buffered\_out\_ports\_disable <bool>** buffered\_out\_ports\_disable.
- **-buffered\_out\_ports\_regardless <bool>** buffered\_out\_ports\_regardless.
- **-call\_final\_import <bool>** -call\_final\_import is a global option.
- **-debug <bool>** Specifies whether visibility must be added inside the transactor or not.
- **-debug\_zemi\_behav <bool>** debug\_zemi\_behav.
- **-default\_cclk <string>** xmr to signal that is connected to zceiClockport.  
xmr to signal that is connected to zceiClockport
- **-delayed\_dpi <bool>** -delayed\_dpi is a global option.

- **-deterministic\_dpi** <bool> -deterministic\_dpi is a global option.
- **-disable** <string> Provides a signal that disables the transactor when it is 0.  
xmr to signal
- **-disable\_all\_xtors** <bool> -disable\_all\_xtors is a global option which provides a disable signal for each transactor instance.
- **-do\_profiling** <bool> do profiling.
- **-do\_verbose\_profiling** <bool> do verbose profiling.
- **-edge\_enable** <bool> Specifies whether flops initially synchronized by the input zcei clock are updated with an edge enable block around.
- **-emit\_initial** <bool> emit\_initial.
- **-exports\_excluded\_from\_back\_to\_back** <list> list\_of\_export\_DPI\_excluded\_from\_-back\_to\_back\_mode.
- **-external\_controlled\_clocks** <list> list\_of\_external\_controlled\_clocks.
- **-fm\_dpi\_mcp** <int> -fm\_dpi\_mcp is a global option.  
A positive integer number.
- **-global\_serialized\_opt** <bool> -global\_serialized\_opt is a global option.
- **-hard\_max\_in\_port\_width** <int> .  
A positive integer number.
- **-hard\_max\_out\_port\_width** <int> .  
A positive integer number.
- **-heartbeat\_period** <int> -heartbeat\_period is a global option.  
A positive integer number.
- **-instance** <list> list of instances.
- **-max\_clocked\_loop\_iterations** <int> .  
A positive integer number.
- **-max\_in\_port\_width** <int> .  
A positive integer number.
- **-max\_loop\_iterations** <int> -max\_loop\_iterations can be used with or without -module.  
A positive integer number.
- **-max\_out\_port\_width** <int> .  
A positive integer number.
- **-merge\_multiple\_comb\_writers** <bool> merge\_multiple\_comb\_writers.
- **-module** <list> list of transactors modules.
- **-optimize\_dpi\_constant\_args** <bool> -optimize\_dpi\_constant\_args is a global option.
- **-port\_optimization\_mode** <0|1|2|3> Specify port optimization mode.  
0|1|2|3 0 for AREA\_OPT, 1 for THROUGHPUT\_OPT, 2 for LATENCY\_OPT, 3 for DEFAULT\_OPT
- **-port\_width\_slop\_percent** <int> .  
A positive integer number.
- **-profile\_counters\_width** <int> .  
A positive integer number.
- **-protolink** <bool> protolinks.
- **-reduce\_control\_paths** <bool> reduce control paths.
- **-resolve\_events\_globally** <bool> -resolve\_events\_globally is a global option.
- **-sample\_clock\_mode** <bool> -sample\_clock\_mode is a global option.
- **-serialize\_all\_imports** <bool> serialize\_all\_imports.
- **-start\_in\_back\_to\_back** <bool> start in back to back.
- **-streaming\_dpi\_opt** <bool> -streaming\_dpi\_opt.

- **-streaming\_is\_default** <bool> streaming is default.
- **-support\_dollar\_display** <bool> support \$DISPLAY.
- **-task\_streaming\_is\_default** <bool> task\_streaming\_is\_default.
- **-timestamp** <bool> -timestamp can be used with or without -module.
- **-transform\_latch** <bool> transform\_latch.
- **-treat\_warnings\_as\_errors** <bool> -treat\_warnings\_as\_errors is a global option.

#### Constraint(s)

- -emit\_initial can only be used with -module
- global options cannot be combined with -instance|-module options
- -advanced\_synth\_command\_file can only be used with -module
- check for mandatory combinations of options
- -advanced\_synth\_command\_file cannot be used with any of the following options: -instance
- -instance cannot be used with any of the following options: -allow\_streaming, -start\_in\_back\_to\_back, -support\_dollar\_display, -reduce\_control\_paths, -do\_profiling, -do\_verbose\_profiling, -streaming\_is\_default, -task\_streaming\_is\_default, -emit\_initial, -serialize\_all\_imports, -hard\_max\_out\_port\_width, -hard\_max\_in\_port\_width, -auto\_flush\_interval, -buffered\_out\_ports\_disable, -buffered\_out\_ports\_default\_width, -buffered\_out\_ports\_regardless, -port\_optimization\_mode, -profile\_counters\_width, -max\_out\_port\_width, -max\_in\_port\_width, -port\_width\_slop\_percent, -all\_tf\_args\_are\_automatic, -external\_controlled\_clocks, -allow\_mixed\_design\_clocks, -exports\_excluded\_from\_back\_to\_back, -max\_loop\_iterations, -max\_clocked\_loop\_iterations, -merge\_multiple\_comb\_writers, -transform\_latch, -debug\_zemi\_behav, -default\_cclk, -sample\_clock\_mode, -timestamp, -protolink, -sample\_clock\_opt, -rt\_diag, -rt\_config, -resolve\_events\_globally, -async\_reset\_init, -optimize\_dpi\_constant\_args, -treat\_warnings\_as\_errors, -fm\_dpi\_mcp, -call\_final\_import, -global\_serialized\_opt, -delayed\_dpi, -deterministic\_dpi, -heartbeat\_period, -streaming\_dpi\_opt, -enforce\_streaming
- -sample\_clock\_mode cannot be used with any of the following options: -sample\_clock\_opt
- At least one of the options should be used

#### Note(s)

- -advanced\_synth\_command\_file must be used with -module, and should not be used with -instance
- Usage for -instance option: -instance is only valid with -debug|-edge\_enable|-disable, other options cannot be used with -instance
- -default\_cclk </tt> can be used either with -module in which case it refers to specific modules, or without -module in which case it refers to all xtors defined until now.

## 2.99 zforce

### Description

Forces and maintains a signal to a given value at runtime until its release. After its release, the original design value is passed.

### Synopsis

```
zforce [-access <all|seq|ports>] [-depth <int>] [-detect_val_change]
[-enable_wls <bool>] [-fnmatch] [-instance <string>] [-mode <dynamic|static>]
[-module <string>] [-object_not_found <warning|fatal>] [-pin <list>]
[-pin_file <file>] [-pin_only] [-rtlname <list>] [-rtlname_file <file>]
[-signal <list>] [-sync <bool>] [-sync_enable] [-type <global|downstream>]
[-uniquify] [-wire <list>] [-wire_file <file>]
```

### Option(s)

- **-access** <all|seq|ports> depth access type.
  - **all**: Protection type all.  
Valid values: all|ALL|All
  - **seq**: Protection type seq.  
Valid values: seq|SEQ|Seq
  - **ports**: Protection type ports.  
Valid values: ports|PORTS|Ports
- **-depth** <int> Specifies forcibility depth..  
An integer number.
- **-detect\_val\_change** Detects any value change on the forced net..
- **-enable\_wls** <bool> Enable synthesis support for zforce.
- **-fnmatch** Indicates that object pathnames can contain wildcards.
- **-instance** <string> Specifies the name of the instance under which the signals are forced. .  
string
- **-mode** <dynamic|static> zforce command modes.
  - **dynamic**: 'DYNAMIC' mode is recommended for system resource preservation over performance.  
Default mode for force.  
Valid values: dynamic|DYNAMIC|Dynamic
  - **static**: 'STATIC' mode is recommended for signals that are forced hundreds of times or more during an emulation via dedicated top ports.  
Valid values: static|STATIC|Static
- **-module** <string> Specifies the name of the module under which the signals specified by -signal option are forced. .  
string
- **-object\_not\_found** <warning|fatal> Specifies the error policy when objects cannot be found.
  - **warning**: A warning is displayed instead of an error.  
Valid values: warning|WARNING|Warning
  - **fatal**: A fatal error is displayed.  
Valid values: fatal|FATAL|Fatal
- **-pin** <list> Specifies a list of instance ports. The hierarchical net connected to the port is forced.  
This options is deprecated.
- **-pin\_file** <file> Specifies whether the command will be applied on pin objects found in a file defined in argument. File path is relative to the zCui execution directory. This options is deprecated.
- **-pin\_only** Forces the net connected to the specified instance port only. Don't force any equivalent nets..

- **-rtlname** <list> Applies the command to a list instance port paths. The hierarchical net connected to the port is forced..
- **-rtlname\_file** <file> Applies the command to RTL objects found in a file defined in argument. File path is relative to zCui execution directory.
- **-signal** <list> Applies the command to a list signals. Signals are relative to module specified with -module option..
- **-sync** <bool> Prevents force from potential glitches at the cost of a register.
- **-sync\_enable** Prevents force from potential glitches at the cost of a register.
- **-type** <global|downstream> Indicates the way the dynamic force need to be inserted.
  - **global**: 'GLOBAL' Default value. Indicates that the dynamic force will apply to all the readers of the hierarchical wire.  
Valid values: global|GLOBAL|Global
  - **downstream**: 'DOWNSTREAM' Indicates that the dynamic force will be applied to the readers downstream. The wire is implicitly oriented and the net may not be multidriven or tristate.  
Valid values: downstream|DOWNSTREAM|Downstream
- **-uniquify** Uniquify the module if zforce is targeted on instance-based signals..
- **-wire** <list> Applies the command to wire objects. This options is deprecated.
- **-wire\_file** <file> Applies the command to wire objects found in a file defined in argument. File path is relative to zCui execution directory.

#### Constraint(s)

- At least one of the options should be used
- option -module cannot be used with option -uniquify

## 2.100 zpar

### Description

System-level place and route (zPar) options

### Synopsis

```
zpar [-advanced_command <string>] [-advanced_command_file <file>] [-direct-
_placement <bool>] [-effort <No|Low|Medium|High>] [-incremental <bool>]
[-max_parallel_jobs <int>] [-placement_number_of_jobs <int>]
```

### Option(s)

- **-advanced\_command <string>** Command for zPar.  
legacy zPar command
- **-advanced\_command\_file <file>** Command file for zPar.
- **-direct\_placement <bool>** Use direct placement.
- **-effort <No|Low|Medium|High>** Specify effort level.
- **-incremental <bool>** Use zPar incremental mode.
- **-max\_parallel\_jobs <int>** Maximum number of zpar jobs allowed(Default:0, parallelism is disabled).  
A non-negative integer number.
- **-placement\_number\_of\_jobs <int>** Number of jobs for placement(Default:0, take default value from zpar).  
A non-negative integer number.

### Constraint(s)

- At least one of the options should be used

### Note(s)

- Higher number can potentially increase performance, but consume more computing resources.
- If value for -placement\_number\_of\_jobs is not 0, -max\_parallel\_jobs with none zero value must also be defined

# Index

architecture\_file, [42](#)  
assertion\_synthesis, [33](#)  
  
blackbox, [5](#)  
  
clock\_boundary\_marker, [43](#)  
clock\_config, [45](#)  
clock\_delay, [47](#)  
clock\_frequency, [49](#)  
clocks, [10](#)  
cluster\_constraint, [50](#)  
clustering, [53](#)  
code\_coverage, [56](#)  
compile, [57](#)  
converged\_flow, [7](#)  
coverage, [59](#)  
create\_new\_target, [38](#)  
  
debug, [60](#)  
defmapping, [37](#)  
design, [35](#)  
design\_size, [64](#)  
disk\_space\_checker, [66](#)  
dpi\_synthesis, [36](#)  
  
eco, [67](#)  
encrypted\_mod, [69](#)  
enhanced\_write\_access, [70](#)  
  
fault\_emulation, [71](#)  
force, [72](#)  
fpga, [75](#)  
fusa, [77](#)  
  
global\_db, [78](#)  
grid\_cmd, [79](#)  
grid\_engine, [80](#)  
grid\_task\_association, [81](#)  
  
keep\_rtl\_read, [11](#)  
keep\_rtl\_write, [12](#)  
  
load\_edif, [82](#)  
  
memories, [83](#)  
memory\_preferences, [32](#)  
modular\_compile, [86](#)  
move\_initial\_to\_sw, [87](#)  
  
nfs\_settings, [88](#)  
  
optimization, [13](#)  
  
pbuilt, [39](#)  
performance, [89](#)  
post\_compile\_command, [90](#)  
power\_profile, [16](#)  
pre\_compile\_command, [91](#)  
probe\_config, [92](#)  
probe\_signals, [93](#)  
profile, [31](#)  
  
read\_classification, [94](#)  
read\_sdc, [95](#)  
reg\_init, [96](#)  
run\_manager, [97](#)  
  
scemi, [25](#)  
scheduling, [98](#)  
set\_app\_var, [8](#)  
set\_bus\_interface, [99](#)  
set\_dualedge, [100](#)  
set\_false\_path\_from, [101](#)  
set\_false\_path\_to, [102](#)  
set\_global\_false\_path, [17](#)  
set\_hwtop, [103](#)  
set\_perf\_flow, [104](#)  
set\_power\_estimation, [18](#)  
set\_power\_profile, [105](#)  
set\_sram\_trace, [9](#)  
set\_sw\_control\_signal, [106](#)  
set\_trigger, [107](#)  
set\_zcei\_partition, [108](#)  
signature, [109](#)  
simxl, [110](#)  
simxl\_allow\_concurrent, [111](#)  
simxl\_enable\_zdpi, [112](#)  
simxl\_move\_tf\_to\_tb, [113](#)  
simxl\_move\_to\_tb, [114](#)  
simxl\_set\_hwtop, [115](#)  
simxl\_xlbram, [116](#)  
smart\_zice\_config, [117](#)  
synplifypro\_synthesis, [19](#)  
synthesis, [20](#)  
synthesis\_preferences, [118](#)  
system\_tasks, [23](#)  
  
timing\_analysis, [119](#)  
timing\_constraint, [121](#)  
transparent\_latch, [122](#)  
tristate, [123](#)  
  
UCVLibExt, [40](#)

UCVcsVerdiFlowNoFSDB, [41](#)

vcs\_exec\_command, [124](#)

verilog\_force\_release, [125](#)

wire\_resolution, [126](#)

xtors, [128](#)

zcei, [24](#)

zemi3, [129](#)

zemi4, [132](#)

zforce, [135](#)

zgate, [26](#)

zinject, [28](#)

zpar, [137](#)

ztopbuild, [34](#)